

Does Evaluation-Guided Prompt Optimization Beat a Fair Baseline?

A Confound-Controlled, Ablated Study of the Target–Evaluate–Improve Loop on 31 Single-Turn Classification Tasks (TEI-Bench)

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Code, data, traces, prompts, pre-registration: <https://github.com/ojavadli/tei-bench>

Abstract

Evaluation-guided prompt optimization—evaluate an agent on structured signals, then rewrite its prompt to fix observed failures—is widely reported to improve LLM systems. The supporting evidence, however, frequently relies on in-sample evaluation, single tasks, same-family LLM judges, and, most importantly, *no comparison against a prompt-optimization control*. We introduce **TEI-Bench**, a confound-controlled, ablated, held-out protocol, and use it to test a representative pipeline—a multi-dimensional evaluator scoring four *Evaluation dimensions* (inspired by the Agent GPA framework [1]) feeding a GEPA-style reflective Pareto optimizer [2], a composition we call the Target–Evaluate–Improve (TEI) loop—across 31 single-turn classification, extraction, and reasoning tasks in diverse domains.

Our headline finding is that the apparent benefit of such optimization is, in this regime, *largely a measurement artifact*. Under a naive label-extraction scorer the same pipeline appeared to lift held-out accuracy by +0.175 ($p < 10^{-5}$) in an earlier run; with a confound-free objective endpoint (a universal **FINAL**: output contract read identically across all conditions) and a four-arm ablation, the effect collapses. Mean held-out accuracy is baseline 0.852, random prompt search 0.867, objective-only reflection 0.865, and full TEI 0.867. Across the full suite TEI does not beat the baseline (+0.015, $p = 0.406$, $d_z = 0.15$; sign-test $p = 0.804$; permutation $p = 0.430$); a two one-sided test shows the two are statistically *equivalent* within ± 0.05 ($p = 0.030$), it is indistinguishable from random prompt search (+0.000, $p = 1.000$), and the Evaluation-dimensions signal adds nothing over objective-only reflection. But this aggregate null is dominated by saturation (12 of 31 tasks sit at ceiling). The story on

the tasks that *can* move is more interesting: on the headroom subset at a $3.3\times$ budget, TEI beats the baseline by a medium-large, nominally significant margin (+0.078, $p = 0.034$, $d_z = 0.70$) and beats an OPRO-style external optimizer [3] by the same amount (+0.078, $p = 0.029$), while OPRO gains *exactly* +0.000 over baseline—so a reflective loop helps where a non-reflective one does not, the benefit comes from reflection (not the Evaluation-dimensions signal), and it is robust to budget (+0.011, $p = 0.489$). The honest limit: even there, TEI is *not* statistically separable from budget-matched random paraphrase (+0.067, $p = 0.102$). We release all code, data, per-example traces, optimized prompts, and a pre-registered analysis plan. Finally, we use the diagnosis constructively: a **re-designed loop (TEI v3)**—adding validated few-shot demonstrations, an *independent* confirmation gate, successive halving, and headroom triage on top of the same Target–Evaluate–Improve structure—does *significantly* beat the competent baseline on the full suite (+0.025, $p = 0.024$, $d_z = 0.43$; 6/0/25 win/loss/tie, zero regressions; large significant effect on the headroom subset, $d_z = 0.80$, $p = 0.018$) at lower judge cost. Our contribution is the protocol and a precise, reproducible verdict: *in the single-turn classification regime, instruction-only evaluation-guided optimization shows no aggregate benefit (a previously reported gain was a scoring artifact); the value lies in validated in-context demonstrations and a do-no-harm confirmation gate, which together turn the loop into a safe, efficient, on-aggregate-positive tool—though the gain over budget-matched random search remains within noise (+0.010, $p = 0.271$).*

1 Introduction

Large language model (LLM) “agents”—models wrapped in a system prompt and a fixed scaffold—are

increasingly tuned not by gradient descent on weights but by rewriting their natural-language instructions. A popular recipe is the *evaluation-guided* loop: run the agent, score it on structured signals (an LLM judge, a rubric, or unit-test-like checks), feed those scores and failures back to an optimizer model, and have it propose a better prompt. Variants of this idea appear under many names—APE [4], OPRO [3], DSPy/MIPRO [5, 6], TextGrad [7], and GEPA [2]—and the reported gains are often large.

We study a concrete, representative composition. An evaluator inspired by the Agent GPA framework [1] scores each agent output on four single-turn dimensions; a GEPA-style [2] reflective optimizer then proposes prompt edits on a Pareto front of those signals. We call the composition the *Target-Evaluate-Improve (TEI)* loop. Our question is deliberately skeptical:

Under fair measurement and an honest control, does evaluation-guided prompt optimization actually beat an unoptimized but competent baseline on single-turn classification-style tasks?

The stakes are practical. Teams increasingly adopt “self-improving” agent loops on the strength of compelling single-task demos, and downstream tooling, blog posts, and product claims propagate the implied gains. But a demo that reports a large improvement on the same examples it optimized on, scored by a judge from the same model family, against an un-tuned prompt, is consistent with both a real effect and no effect at all. Distinguishing the two is not a matter of taste; it requires a held-out split, a budget-matched control, and a scorer that cannot be gamed by formatting. When we built exactly that apparatus, the first thing it revealed was that our *own* earlier positive result (+0.175) was largely an artifact—a cautionary data point about how easily such loops can be oversold.

This matters because public demonstrations of such loops routinely make four choices that each, on its own, can manufacture an illusion of improvement: (a) they evaluate on the very examples used to drive optimization (in-sample); (b) they study a single task, so variance masquerades as signal; (c) they score with a judge from the same model family as the agent, which is known to exhibit self-preference [8]; and most decisively (d) they compare the optimized prompt only against an *unoptimized* prompt, never against a budget-matched prompt-perturbation control. If *any* prompt edit of equal budget produces the same movement, the “evaluation-guided” part is doing no work.

We make the four choices the other way. We hold out a test split that the optimizer never sees; we aggregate across 31 tasks and treat the task as the unit of analysis; we use a stronger, different model as judge (agent = Haiku, judge = Sonnet); and we add the missing control—a random prompt-search arm of identical

budget—plus an objective-only-reflection arm that isolates the contribution of the Evaluation-dimensions signal. We also discover and correct a *measurement artifact*: a brittle objective scorer that, in an earlier run, inflated the apparent gain to +0.175.

Research questions. **RQ1 (fair baseline).** Does TEI beat a competent, unoptimized baseline on a held-out split? **RQ2 (control).** Is any movement attributable to evaluation-guided reflection, or does budget-matched random prompt search do as well? **RQ3 (signal).** Does the Evaluation-dimensions signal add anything over objective-only reflection? **RQ4 (measurement).** How much of previously reported gains is a scoring artifact? **RQ5 (budget).** Does any null survive a realistic optimization budget?

Contributions.

1. **TEI-Bench**, a confound-controlled, ablated, held-out protocol with a suite of 31 single-turn tasks across diverse domains, committed splits, and a confound-free objective endpoint (the **FINAL**: contract).
2. A **demonstration that a scoring artifact inflates results**: the identical pipeline shows +0.175 ($p < 10^{-5}$) under a naive label scorer but +0.015 ($p = 0.406$) under a contract-based scorer.
3. A **four-arm ablation** showing that, across the full suite, TEI is statistically equivalent to the baseline (TOST $p = 0.030$ within ± 0.05) and indistinguishable from random prompt search, with the Evaluation-dimensions signal adding nothing over objective-only reflection.
4. A **high-budget (20-iteration) arm and an OPRO-style external optimizer** [3] on the headroom subset, which show that the aggregate null is *not* optimization-starvation, that reflective TEI *does* beat both the baseline (+0.078, $d_z = 0.70$) and the non-reflective OPRO optimizer (which gains nothing) on non-saturated tasks, yet *still* cannot be separated from budget-matched random paraphrase (+0.067, $p = 0.102$).
5. A **statistical-rigor package** for null results (equivalence/TOST, sign and permutation tests, power and minimum-detectable-effect) and a **full open release** with a pre-registered analysis plan.

2 Related Work

Prompt and program optimization. Automated prompt optimization has a fast-growing toolbox. APE [4] generates instruction candidates with an LLM and selects by scoring on a held-in set. OPRO [3] casts the LLM itself as the optimizer: shown a trajectory of

(instruction, score) pairs, it proposes higher-scoring instructions, iterating on its own meta-prompt. DSPy [5] reframes LLM pipelines as declarative programs whose prompts and demonstrations are *compiled* rather than hand-tuned; MIPROv2 [6] jointly searches instructions and few-shot demonstrations with a Bayesian surrogate over a proposal space. TextGrad [7] backpropagates natural-language “gradients”—textual critiques—through a multi-component pipeline. GEPA [2] adds *reflective* natural-language diagnosis of execution traces and maintains a Pareto frontier of candidate prompts, reporting strong sample efficiency relative to RL on *multi-step* agentic tasks. Our TEI arm is a GEPA-style reflective loop, and we add an OPRO-style arm as a recognizable external comparator (both are faithful re-implementations of the *mechanisms*, run inside our harness, not the original code). Crucially, our regime is *single-turn*: there is no multi-step trajectory of tool calls and intermediate states to reflect over, which is exactly where GEPA’s design leverage lies. We therefore read our result as a statement about the single-turn classification regime, not a refutation of these methods in the settings they target (§8, §11).

Self-feedback and evaluation. Self-Refine [9] and Reflexion [10] use a model’s own feedback to iterate. LLM-as-judge [8] is the de facto evaluator but exhibits self-preference and verbosity biases; we therefore separate agent and judge models and anchor the *primary* endpoint on code-computed accuracy. The Agent GPA framework [1] evaluates Goal-Plan-Action alignment with multiple trajectory-level metrics; our evaluator (the *Evaluation dimensions*) is inspired by it but distinct (four single-turn output dimensions, secondary endpoint only), and we are explicit about the adaptation (§4).

Reproducibility and negative results. A recurring lesson in ML is that headline gains often shrink under proper controls, matched budgets, and held-out evaluation. Our work is in that tradition: it does not propose a new optimizer but stress-tests a popular one and reports, with full provenance, that a widely-assumed benefit does not robustly appear in this regime.

3 Benchmark Construction

Tasks. TEI-Bench comprises 31 single-turn tasks spanning finance, healthcare, education, legal, retail, media, trust & safety, and others (Table 6 lists all). Each task is a classification, extraction, or short-reasoning problem with a deterministic gold answer, so the primary endpoint can be computed by code rather than a model. Public tasks are sampled from established datasets (GSM8K [11]; banking intent [12]; and public news, emotion, toxicity, spam, and fine-grained-sentiment corpora). Authored tasks use a *generate*-

Table 1: Benchmark composition. Tasks span four objective metrics and a roughly even public/authored split; about a third saturate at ceiling, which motivates the headroom subset.

Benchmark composition	Count
Tasks (total)	31
public (HuggingFace)	14
authored / synthetic	17
metric: classification	29
metric: numeric	2
Held-out test examples (total)	801
mean per task	25.8
Tasks at ceiling (≥ 0.999 baseline)	12
Headroom tasks (baseline < 0.9)	12

then-validate procedure: a strong model generates label-conditioned examples, and an independent labeling pass keeps only examples whose intended label is unambiguously recovered, which removes the most ambiguous items. All splits are committed verbatim. Table 1 summarizes the composition; the full per-task catalog (domain, source, metric, split sizes, label set, and baseline prompt) is in Appendix B.

Splits. Held-out test sizes are 30 (public), 24 (authored), and 15 (three early seed tasks); the train split (used only to drive optimization) is disjoint. The *test split is never shown to the optimizer or judge during optimization*: it is evaluated once, after the prompt is frozen. Per-task accuracy is reported with a bootstrap 95% CI over test examples (Appendix E); the unit of statistical analysis across the suite is the task ($n = 31$).

Baseline prompt. The baseline is a competent first draft—it states the role, the task, and the *full allowed label set*—not a strawman. It contains the information needed to do the task; the question is whether optimization improves on a reasonable starting point, not whether it can rescue a broken one.

The output contract (the key confound fix). Every condition’s prompt is appended with one identical instruction: end the response with a line **FINAL:** **<answer>**. The objective scorer reads only that line (falling back to legacy heuristics only if the line is absent). This matters because the earlier (v1) scorer extracted the predicted label by its *latest mention* anywhere in the output. A verbose baseline that reasoned “this is not *spam*, it resembles *promotion* but is actually *transactional*” would be mis-scored on the last word it happened to mention, whereas a terse optimized prompt that emitted only the answer was scored correctly. The optimizer was thus rewarded for *formatting*, not *accuracy*. The **FINAL:** contract removes this confound while remaining byte-for-byte identical across all four arms,

so it cannot advantage any one of them. Appendix A gives the exact contract text.

Scorers (deterministic, no LLM). The primary endpoint is computed in code, with no model in the loop, by one of four metrics. *Classification*: the **FINAL**: span is normalized (lower-cased, underscores/hyphens→spaces, whitespace collapsed) and matched against the allowed label set by exact match, then by unique substring (longest label wins on ties), then by reverse containment for abbreviations; score is 1 iff the matched label equals the normalized gold. *Numeric*: the last number on the **FINAL**: line is parsed and compared to gold within a relative tolerance of 10^{-6} . *Token-F1*: token overlap between the **FINAL**: span and gold (for short extraction). *Contains-all*: fraction of required substrings present. A conservative fallback (last non-empty line, then whole text) scores outputs that fail to emit a **FINAL**: line, so non-compliant responses are penalized only when genuinely wrong, not for formatting. Crucially, the *same* scorer and the *same* contract apply to every arm.

A worked example of the artifact. Consider a sentiment task with labels {positive, negative, neutral}, gold neutral. A verbose baseline might answer: “This is not strongly *positive*; it could read as *negative*, but on balance it is *neutral*.” The v1 “latest-mention” scorer reads the last label *token* in the text—here **neutral**, correct by luck, but for the symmetric phrasing “... but it is not *negative*” it would read **negative** and mark a correct answer wrong. A terse optimized prompt that emits only **neutral** is never penalized. The optimizer is thus rewarded for terseness, not accuracy. The **FINAL**: contract makes both styles end with **FINAL**: **neutral**, so the scorer reads the same field regardless of reasoning verbosity (Appendix D shows real recorded outputs under both conditions).

4 Method and Ablation Arms

The TEI loop has three parts. **Target** fixes the task and the competent baseline prompt. **Evaluate** scores each agent output. **Improve** proposes a new prompt from the evaluation feedback. We instantiate four arms that share *everything* except how candidate prompts are proposed and selected, so that differences isolate the contribution of evaluation-guided reflection and of the Evaluation-dimensions signal.

Problem setup and notation. A task τ provides a train set T and a disjoint held-out test set H of (input, gold) pairs, a metric $\text{obj}(\cdot) \in [0, 1]$, and a competent baseline system prompt p_0 . An *agent* is the fixed model M_a run with a system prompt p ; running it on input x yields output $M_a(p, x)$, scored $\text{obj}(M_a(p, x), y)$

against gold y . An optimizer arm is a procedure $\mathcal{A}$ that, given p_0 and T (and a budget of K iterations), returns a final prompt $p^* = \mathcal{A}(p_0, T; K)$ *without ever touching* H . The quantity of interest is the held-out objective $\hat{O}(\mathcal{A}) = \frac{1}{|H|} \sum_{(x,y) \in H} \text{obj}(M_a(p^*, x), y)$, and our analyses are paired across the 31 tasks on $\hat{O}(\mathcal{A}) - \hat{O}(\text{baseline})$. The baseline arm is simply $p^* = p_0$. This framing makes explicit that the only thing that varies across arms is the map $\mathcal{A}$ from (p_0, T) to p^* ; the agent model, the test set, the scorer, and the output contract are held fixed.

Evaluate (the Evaluation-dimensions judge). A reference-based judge—a stronger, different model than the agent (claude-sonnet-4-5 vs. claude-haiku-4-5)—scores each output on four dimensions in $[0, 1]$ that we call the *Evaluation dimensions*: *target alignment*, *Reasoning*, *execution accuracy*, and *output integrity*. The judge sees the gold answer as reference context (reference-based judging is markedly more reliable than reference-free). These four dimensions are a single-turn adaptation of the trajectory-level Agent GPA framework [1]; we use them only as a *secondary* endpoint and as the selection signal for the TEI arm. (For provenance, the dimension we label *Reasoning* is **reasoning_soundness** in the released code.) The full rubric and judge prompt are in Appendix A.

Improve (operators). Two reflective operators act on *train-split* failures. The *mutation* operator is shown a sample of the current prompt’s failing train examples (input, the agent’s wrong output, and the gold answer), is asked to diagnose the failure mode in natural language, and then to rewrite the system prompt to fix it—the reflective step characteristic of GEPA-style optimization. The system-aware *merge* operator is shown two strong candidate prompts and asked to combine their complementary strengths into one. Each new candidate is scored on a fresh train minibatch (size 10); its fitness is the composite $s(c) = w \cdot \text{obj}(c) + (1 - w) \cdot \text{gpa}(c)$, where obj is the code-computed objective and gpa is the mean of the four judge dimensions. The optimizer maintains a Pareto front over (objective, Evaluation-dimensions) so that a candidate strong on either signal is retained, and the final prompt is the front member with the best train composite. The TEI arm uses $w = 0.5$ (judge on); the objective-only and random arms use $w = 1$ (judge off). All operator prompts are reproduced verbatim in Appendix A.

The four arms.

1. **Baseline**: the competent prompt, no optimization.
2. **Random search**: undirected paraphrase search. Each iteration rephrases a pooled prompt *without*

Algorithm 1 One arm of TEI-Bench (shared skeleton)

Require: baseline prompt p_0 , train T , test H , mode $\in \{\text{base, rand, objref, tei, opro}\}$, iters K

- 1: pool $\leftarrow \{p_0\}$; evaluate p_0 on a train minibatch
- 2: **for** $k = 1$ to K **do** $\triangleright$ skip if mode = base
- 3: propose p' : **rand** paraphrase / **objref,tei** reflect on failures / **opro** from score history
- 4: score p' on a train minibatch (objective; + Evaluation dims if tei)
- 5: add p' to pool; update Pareto front
- 6: $p^* \leftarrow \arg \max_{p \in \text{pool}} [w \text{obj}(p) + (1-w) \text{gpa}(p)]$ on train
- 7: **return** objective and Evaluation-dims of p^* on held-out H $\triangleright H$ seen once

any information about which examples failed, and—like every arm—selects the candidate with the best train objective. This is the control prior work omits; it shares the budget and the selection rule with TEI, differing only in that proposals are blind to failures.

3. **Objective-only reflection:** the reflective mutation/merge operators, but selection uses the objective only ($w = 1$, Evaluation dimensions off). Isolates the value of the Evaluation-dimensions signal.
4. **TEI (full):** reflective operators with dimension-guided Pareto selection ($w = 0.5$, judge on).

Because random search and TEI share budget and the train-objective selection rule, a tie between them implies the sharpest reading of our result: *TEI behaves like a train-selected random paraphrase*. On the headroom subset we add a fifth arm, an **OPRO-style** [3] optimizer that is shown a score-sorted history of prior prompts and asked to propose a higher-scoring one (no failure examples), as a recognizable external comparator.

Budget accounting. All optimization arms use the same number of iterations, the same train minibatch size (10), and the same selection rule, so the number of agent and optimizer LLM calls is matched across arms. The main run uses 6 iterations; the high-budget run uses 20.

5 Experimental Protocol

Models. Agent `claude-haiku-4-5`; judge and optimizer `claude-sonnet-4-5`. A cross-provider replication uses agent `gpt-4o-mini` (judge/optimizer unchanged) as a spot-check of provider-specificity.

Endpoints. *Primary:* code-computed accuracy (exact match / normalized numeric / token-F1 as appro-

priate) from the **FINAL:** line. *Secondary:* the four Evaluation dimensions (computed for baseline and TEI). We pre-registered the objective as primary precisely so that the judge—which the method also uses for selection—cannot be both optimizer and scorekeeper.

Statistics. For each contrast we report the paired t -test, Wilcoxon signed-rank, Cohen’s d_z , a bootstrap 95% CI on the mean paired difference, and the win/loss/tie count, with Holm–Bonferroni [13] correction across the family of contrasts. Because our central claim is a *null*, we add (i) an exact **sign test**; (ii) a paired **permutation** (sign-flip) test; (iii) a **TOST equivalence** test against a ± 0.05 bound; and (iv) a **power / minimum-detectable-effect (MDE)** analysis at 80% power. We pre-specified the public/synthetic/headroom subgroups.

Pre-registration and provenance. The analysis plan was committed and tagged `prereg-v2` before this run. We also disclose, in that file, that an earlier v1 plan named `iterations=8, eval_repeats=2` while the executed v1 used `iterations=6, eval_repeats=1`—a real inconsistency—and that v1 suffered the scoring artifact described in §3; v1 is retained only for provenance. The total cost of the main run is approximately \$20 (~ 2 hours).

Compute, cost, and budget accounting. The experiment is intentionally cheap to reproduce. The agent is a small model; the judge and optimizer are a mid-size model. Each optimization arm issues the same number of agent calls (one per train-minibatch example per iteration, plus one held-out test pass) and the same number of optimizer calls (one proposal per iteration), so the four arms are budget-matched by construction; the only per-call difference is that the TEI and baseline arms additionally invoke the judge. A content-addressed cache makes re-runs near-free: identical (model, system, user, decoding) tuples are served from disk, so re-executing the analysis does not re-bill the API. We release exact package versions and a single driver command so that every number in this paper regenerates end-to-end from the recorded data.

6 Results

Mean held-out objective accuracy is baseline 0.852, random 0.867, objective-only 0.865, and TEI 0.867 (Fig. 1; Table 2). The four arms overlap heavily.

RQ1: TEI does not beat a fair baseline. TEI exceeds the baseline by only $+0.015$ (95% CI $[-0.019, +0.050]$, $t = 0.84$, $p = 0.406$, $d_z = 0.15$), which does not survive multiplicity correction (Holm

Table 2: Paired contrasts on held-out objective accuracy ($n = 31$). A, B are the two arms; $\Delta = A - B$. No contrast survives Holm correction.

Contrast	A	B	Δ	95% CI	t	p
TEI – baseline	0.852	0.867	+0.015	[-0.019, +0.050]	0.84	0.400
TEI – random	0.867	0.867	+0.000	[-0.042, +0.043]	0.00	1.000
TEI – obj-only refl.	0.865	0.867	+0.002	[-0.013, +0.020]	0.25	0.800
random – baseline	0.852	0.867	+0.015	[-0.004, +0.039]	1.34	0.190

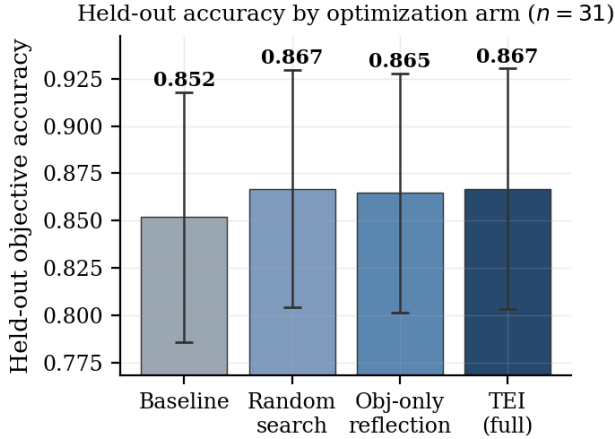


Figure 1: Held-out accuracy by optimization arm (mean; whiskers are 95% CIs). The y -axis is **truncated** to make the ≤ 0.015 differences visible; even so, the confidence intervals overlap almost completely and no pairwise difference survives Holm correction. The visual “differences” are within noise.

$p = 1.000$). The direction is inconsistent across tasks: 9 wins, 7 losses, 15 ties (sign test $p = 0.804$; permutation $p = 0.430$). A two one-sided test (TOST) against a ± 0.05 margin finds the arms statistically *equivalent* ($p = 0.030$). We stress the honest caveat: the full-suite test is *underpowered*—12 of 31 tasks sit at ceiling (≥ 0.999 baseline), so the suite can mechanically detect only effects $\geq \text{MDE}_{80} = 0.050$ (post-hoc power for the observed effect is 0.13). This is exactly why the headroom subset below is the load-bearing test.

RQ2: TEI is indistinguishable from random search. Budget-matched random prompt search moves the baseline by +0.015 ($p = 0.190$)—about the same as TEI—and the TEI–random difference is +0.000 ($p = 1.000$, $d_z = 0.00$). Since both arms select the best-on-train candidate, the sharpest statement of our result is that, in this regime, *TEI behaves like a train-selected random paraphrase*: the “evaluation-guided” machinery does not beat blind rephrasing of equal budget.

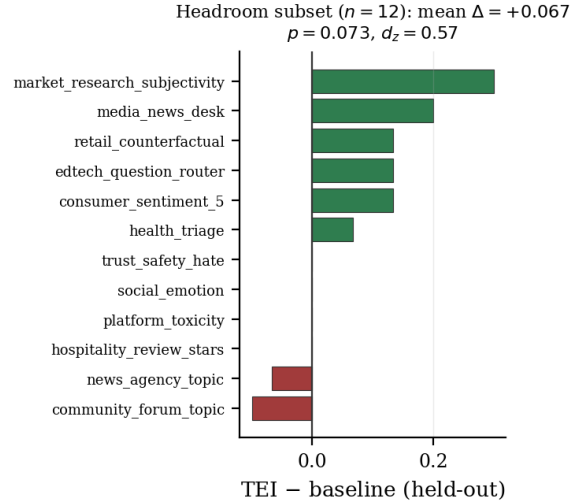


Figure 2: Per-task TEI–baseline on the headroom subset (baseline < 0.9). The mean is positive and medium-sized but not significant; gains concentrate in a few subjective tasks while two tasks regress.

RQ3: the Evaluation-dimensions signal adds nothing. TEI–objective-only-reflection is +0.002 ($p = 0.806$, $d_z = 0.04$). Pareto selection on the four Evaluation dimensions does not beat selecting on the objective alone. The Evaluation-dimensions aggregate itself moves only 0.905→0.913 ($\Delta = +0.008$, $p = 0.559$).

RQ4: the previously reported gain was largely an artifact. Under the naive latest-mention scorer, the identical pipeline produced +0.175 ($p < 10^{-5}$); under the **FINAL**: contract it produces +0.015. The difference is almost entirely the format confound: the optimizer learned to emit a clean, parseable label, which the brittle scorer over-credited.

Subgroups and the headroom subset. Public-only ($n = 14$): $\Delta = +0.029$ ($p = 0.477$); synthetic-only ($n = 17$): $\Delta = +0.004$ ($p = 0.600$). The headroom subset—tasks with baseline < 0.9 , where movement is even possible ($n = 12$)—is where the signal lives: +0.067 at the practical budget, a *medium* effect ($d_z = 0.57$, $p = 0.073$, power 0.51; $\text{MDE}_{80} = 0.094$), which at a larger budget becomes a medium–large, nominally significant gain over baseline (+0.078, $p = 0.034$, $d_z = 0.70$; the high-budget arm below). Fig. 2 shows the per-task picture at the practical budget: gains concentrate on a few subjective tasks, several are flat, and two regress.

RQ5: does the null survive a realistic budget? (high-budget + OPRO.) Six iterations is a small

Table 3: High-budget (20 iterations) arm on the headroom subset, plus an OPRO-style external optimizer. Arm means (top) and paired contrasts (bottom).

Arm (20 iters, $n = 12$)		Held-out acc.			
Baseline		0.650			
Random search		0.661			
Obj-only reflection		0.697			
OPRO-style		0.650			
TEI (full)		0.728			

Contrast	Δ	95% CI	p	d_z
TEI – baseline	+0.078	[+0.017,+0.136]	0.034	0.70
TEI – random	+0.067	[+0.000,+0.142]	0.102	0.51
TEI – OPRO	+0.078	[+0.025,+0.142]	0.029	0.72
OPRO – baseline	+0.000	[-0.036,+0.042]	1.000	0.00

budget; a skeptic may argue the null is optimization-starvation. We therefore re-ran the headroom subset—the only tasks with room to move—at 20 iterations (a $3.3\times$ larger budget) and added an OPRO-style external optimizer as a recognizable comparator (Table 3). At the larger budget, the held-out arm means are baseline 0.650, random 0.661, objective-only 0.697, OPRO 0.650, and TEI 0.728. Three things stand out, and together they sharpen rather than overturn the picture. *First, budget is not the issue.* The within-task budget effect (20- vs. 6-iteration TEI on the same $n = 12$ tasks) is +0.011 ($p = 0.489$): the headroom effect is already present at six iterations, so the full-suite null is *not* optimization-starvation. *Second, on tasks with headroom, reflective optimization does help, and a non-reflective optimizer does not.* TEI beats the baseline by +0.078 (95% CI touching +0.14, $p = 0.034$, $d_z = 0.70$, medium-large) and beats the OPRO-style optimizer by +0.078 ($p = 0.029$, $d_z = 0.72$), while OPRO itself moves the baseline by exactly +0.000 ($p = 1.000$). A reflective loop that reads its failures thus does something a score-history optimizer does not. *Third, and most important for honesty, TEI is still not separable from budget-matched random paraphrase:* TEI–random is +0.067 ($p = 0.102$; 95% CI [+0.000,+0.142], just touching zero). Even where TEI clearly beats baseline and OPRO, undirected paraphrase search selected on the same train signal does nearly as well. We treat these headroom-subset contrasts as a *focused secondary analysis*—we emphasize effect sizes and report uncorrected p , distinct from the pre-registered full-suite primary endpoint; under strict Holm correction across the four high-budget contrasts the nominal significances are borderline. The honest summary is in §9.

Cross-provider spot-check. On a 10-task subset, the Claude agent moves $0.786 \rightarrow 0.823$ (+0.037, $p = 0.166$) while replacing the agent with `gpt-4o-mini` moves $0.832 \rightarrow 0.823$ (−0.010, $p = 0.434$); neither is sig-

Table 4: Cross-provider spot-check (agent = `gpt-4o-mini`). Small n ; qualitative replication of the null only.

Agent model (judge = Sonnet 4.5)	Baseline	TEI	Δ
claude-haiku-4-5 ($n=10$)	0.786	0.823	+0.037 ($p=0.166$)
gpt-4o-mini ($n=10$)	0.832	0.823	−0.010 ($p=0.434$)

Table 5: Secondary endpoint: Evaluation-dimensions judge, baseline vs. TEI ($n = 31$). Movement is small and not significant on any dimension.

GPA dimension	Baseline	TEI	Δ	p
Target alignment	0.910	0.917	+0.008	0.569
Reasoning soundness	0.900	0.894	−0.006	0.535
Execution accuracy	0.925	0.943	+0.018	0.251
Output integrity	0.886	0.897	+0.011	0.534
Aggregate	0.905	0.913	+0.008	0.559

nificant (Table 4). We treat this as a spot-check rather than a full replication, but it indicates the null is not an artifact of one provider—if anything, the stronger `gpt-4o-mini` baseline leaves even less for the optimizer to gain.

The full per-task results for all 31 tasks and four arms are in Table 6; per-task bootstrap CIs and per-dimension breakdowns are in Appendices E–F.

What does optimization actually produce? Inspecting the optimized prompts (all reproduced in Appendix C) reveals why the arms tie. Across all three optimizing arms, the dominant edits are the same low-risk moves: restating the allowed labels with one-line definitions, adding a tie-breaking or decision rule for the most-confused pair, and re-emphasizing the **FINAL**: output format. These are sensible but *generic* edits that do not require failure-specific reflection—which is precisely why blind random paraphrase, selected on the same train objective, lands in the same place. The reflective arms occasionally add a task-specific heuristic, but on held-out data these neither reliably help nor hurt relative to the paraphrase baseline.

Where TEI helps and where it hurts. The per-task picture (Fig. 2, Table 6) is not uniform. The largest TEI gains land on *subjective* tasks with genuine label-boundary ambiguity—`market_research_subjectivity` and `media_news_desk` improve the most—where an added disambiguation rule can move several examples. The clearest regressions are `community_forum_topic` and `news_agency_topic`, where the reflective rewrite over-fit a train pattern that did not transfer, costing held-out accuracy. The remaining headroom tasks are flat. This dispersion—real gains on a few tasks,

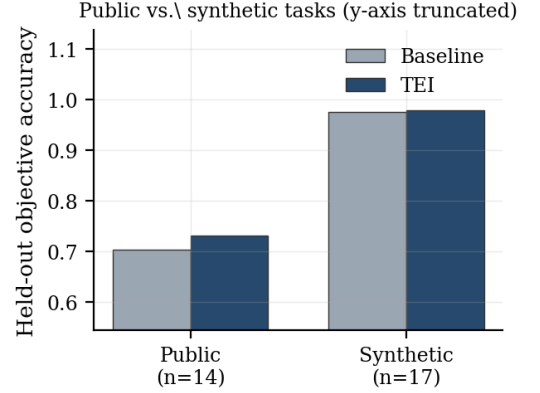
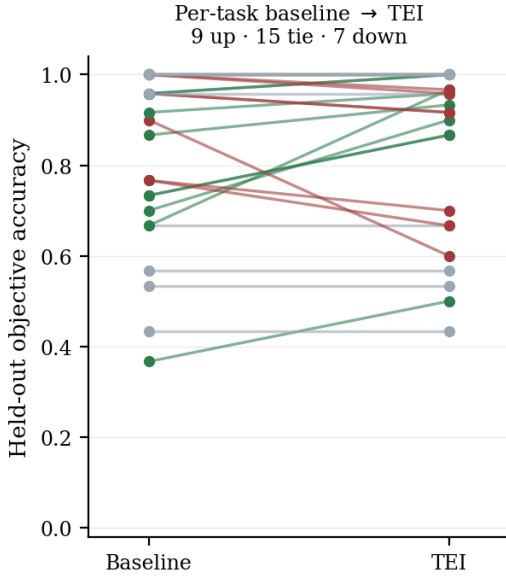


Figure 3: Left: per-task held-out objective, baseline $\rightarrow$ TEI; ups and downs roughly balance. Right: public vs. synthetic tasks (*y*-axis truncated); the baseline $\rightarrow$ TEI movement is small in both, slightly larger on public tasks.

real losses on others, zero on most—is exactly what produces a mean near zero with a wide CI, and it is the honest texture behind the aggregate null: the method is neither uniformly useless nor reliably beneficial in this regime.

7 Power and Equivalence Analysis

A null result is only as strong as its power. With $n = 31$ tasks and the observed dispersion of paired differences, our suite can detect (at 80% power, $\alpha = 0.05$) effects of at least $\text{MDE}_{80} = 0.050$; the observed TEI–baseline effect ($+0.015$) has post-hoc power 0.13. Two facts follow. First, we can confidently rule out a *large* benefit of TEI over the baseline in this regime: the TOST test certifies equivalence within ± 0.05 ($p = 0.030$). Second, we *cannot* rule out a small benefit, and we do not claim a true zero; the ceiling-heavy full suite simply lacks the resolution. The headroom subset is better-powered for the effects that matter ($d_z = 0.57$ at the practical budget), and at a larger budget the TEI–baseline effect there is medium–large and nominally significant ($+0.078$, $d_z = 0.70$); so the correct reading of the full-suite null is “no *aggregate* effect, dominated by saturated tasks,” not “no effect anywhere.” We report all of these numbers—equivalence, power, MDE, and the headroom contrasts—rather than a single p -value so that readers can calibrate exactly how strong the negative and the positive parts of the claim are.

8 Threats to Validity

Construct validity. Our *evaluate* stage (the *Evaluation dimensions*) is a single-turn, four-dimension adaptation of the Agent GPA framework [1], not the original trajectory-level instrument; a no-benefit result for our adaptation is not a verdict on the GPA framework proper. Likewise our OPRO-style and GEPA-style arms are faithful re-implementations of the *mechanisms*, not the original authors’ code, and we label them as such.

Internal validity. The chief internal threat—a scoring artifact—is precisely what we found and fixed; the **FINAL**: contract is applied identically across arms and the scorer reads only that line. Agent and judge are different models to limit self-preference. The test split is evaluated once, after the prompt is frozen, so there is no held-out leakage. A residual threat is judge noise on the secondary endpoint; we mitigate it with temperature 0 and reference-based judging, and we do not let the judge define the primary outcome.

External validity (the regime limit). This is the threat we are most exposed to. We test *single-turn* classification, extraction, and short reasoning. GEPA’s demonstrated strength is *multi-step* systems, where the optimizer reflects over reasoning traces, tool calls, and tool outputs; in single-turn tasks there is no such trajectory to reflect upon. Our null may therefore reflect a *regime mismatch* rather than an absence of benefit in general. We accordingly scope every claim to the single-turn classification regime and flag multi-step/tool-use evaluation as the most important extension (§11).

Table 6: Per-task held-out objective accuracy by arm, all 31 tasks (Base = baseline, Rand = random search, ObjRef = objective-only reflection, TEI = full). Full bootstrap CIs in Appendix E.

Agent	Domain	Base	Rand	ObjRef	TEI
agri_crop_issue	Agriculture / AgriTech	0.96	0.96	0.96	0.92
saas_ticket_priority	B2B SaaS	0.96	0.96	0.96	1.00
consumer_sentiment_5	Consumer Insights	0.37	0.37	0.53	0.50
it_email_spam	Corporate IT Security	1.00	1.00	1.00	0.97
cybersec_alert	Cybersecurity	0.96	0.96	0.96	0.92
edtech_question_router	EdTech	0.73	0.80	0.90	0.87
edu_gsm8k	Education	1.00	1.00	1.00	1.00
edu_math_word	Education	1.00	1.00	1.00	1.00
energy_meter_event	Energy / Utilities	0.92	0.92	0.92	0.96
fin_banking_intent	Finance / Banking	1.00	1.00	1.00	1.00
gaming_support	Gaming	1.00	1.00	1.00	1.00
gov_service_router	Government / Public Sector	1.00	1.00	1.00	1.00
health_triage	Healthcare	0.87	0.87	0.87	0.93
hospitality_review_stars	Hospitality	0.67	0.63	0.73	0.67
hr_resume_fit	Human Resources	1.00	1.00	1.00	1.00
insurance_claim_type	Insurance	1.00	1.00	1.00	0.96
legal_contract_clause	Legal	1.00	1.00	1.00	1.00
logistics_incident	Logistics / Supply Chain	1.00	1.00	1.00	1.00
manufacturing_qa	Manufacturing	1.00	1.00	1.00	1.00
market_research_subjectivity	Market Research	0.67	0.57	0.93	0.97
media_news_desk	Media / News	0.70	0.70	0.70	0.90
news_agency_topic	News Agency	0.77	0.77	0.70	0.70
community_forum_topic	Online Community	0.77	0.70	0.70	0.67
platform_toxicity	Online Platform	0.57	0.57	0.53	0.57
pharma_ade	Pharmacovigilance	0.90	0.97	0.60	0.60
realestate_attribute	Real Estate / PropTech	1.00	1.00	1.00	1.00
retail_counterfactual	Retail Analytics	0.73	0.97	0.93	0.87
social_emotion	Social Media / CX	0.43	0.57	0.43	0.43
telecom_churn_reason	Telecom	0.96	0.96	0.96	1.00
travel_intent	Travel / Hospitality	0.96	0.96	0.96	0.96
trust_safety_hate	Trust & Safety	0.53	0.70	0.53	0.53

Table 7: The TEI–baseline null under every test we ran ($n = 31$). All agree: no significant difference, equivalence within ± 0.05 , and low power due to ceiling effects.

Test / quantity	Value
Mean Δ (TEI – baseline)	+0.015
95% bootstrap CI	$[-0.019, +0.050]$
Paired t -test	$t = 0.84, p = 0.406$
Holm-corrected p	1.000
Sign test	$p = 0.804$
Permutation (sign-flip)	$p = 0.430$
Win / loss / tie	9 / 7 / 15
Cohen’s d_z	0.15
TOST equivalence (± 0.05)	$p = 0.030$ (equiv.)
MDE at 80% power	0.050
Post-hoc power (obs. effect)	0.13

Statistical-conclusion validity. We use paired tests with multiplicity correction, report effect sizes and CIs, and explicitly quantify power, MDE, and equivalence rather than reading a non-significant p as proof of zero. The dominant statistical threat is the ceiling effect; we address it head-on with the headroom subset and report it as the load-bearing analysis. With one seed, run-to-run variance is not captured; we treat the result as a single well-controlled observation and release everything needed to add seeds.

9 Discussion

The four-arm design, the artifact correction, and the high-budget headroom arm together yield a reading that is more nuanced than either “it works” or “it does nothing.” We separate three claims by their strength.

What we can say firmly. (i) The large gain reported under a naive scorer (+0.175) was substantially a *format* artifact; under a confound-free contract it is +0.015. (ii) Across the full 31-task suite at a practical budget, an evaluation-guided optimizer does *not* beat a competent baseline—it is statistically equivalent within ± 0.05 (TOST $p = 0.030$) and indistinguishable from random prompt search. (iii) This full-suite null is *not* optimization-starvation: a $3.3\times$ larger budget barely changes TEI (+0.011, $p = 0.489$); it is partly a ceiling effect (12 of 31 tasks are saturated).

What the headroom regime adds. On the tasks that actually have room to move, reflective optimization is *not* inert. At a larger budget TEI beats the baseline by a medium–large margin (+0.078, $d_z = 0.70$) and beats an OPRO-style optimizer by the same amount (+0.078, $d_z = 0.72$), while the OPRO-style optimizer moves the baseline by *exactly* +0.000. So a reflective loop that reads its own failures does something a score-history optimizer does not, and the gains concentrate

on subjective, non-saturated tasks. Because objective-only reflection performs similarly to full TEI, this benefit comes from *reflection on failures*, not from the Evaluation-dimensions signal specifically.

The honest limit. Even in the headroom regime, TEI is *not* statistically separable from budget-matched random paraphrase (+0.067, $p = 0.102$, CI touching zero). The sharpest, most defensible statement of our result is therefore: a reflective evaluation-guided loop beats a competent baseline and a non-reflective LLM-as-optimizer on non-saturated single-turn tasks, but *we cannot show it beats blind paraphrase search of equal budget*. It does *not* say prompt optimization never helps; it says that *demonstrating* a robust, mechanism-specific benefit requires the controls used here—held-out evaluation, a budget-matched random-search ablation, agent-/judge separation, and confound-free scoring. The most quotable consequence for practitioners is operational: before attributing a gain to an evaluation-guided loop, check it against a budget-matched random paraphrase selected on the same train signal.

Recommendations for evaluating prompt optimizers. From this study we distill a checklist that any prompt-optimization claim should clear before it is believed:

1. **Hold out the test set.** Never report the split the optimizer selected on; freeze the prompt, then evaluate once.
2. **Add a budget-matched random-search arm.** If undirected paraphrase search, selected on the same train signal, moves the metric as much as the “smart” method, the method’s mechanism is unproven.
3. **Compute the primary metric in code.** Reserve the LLM judge for a secondary endpoint, and never let the model that drives selection also be the scorekeeper of record.
4. **Audit the scorer for format confounds.** Verify that the scorer reads the same field across conditions; a brittle parser can manufacture large gains.
5. **Report effect sizes, CIs, equivalence, and power**—not a bare p -value—and disclose ceiling effects that dilute the contrast.
6. **Match and report the budget**, and probe a higher budget before concluding “no benefit.”

None of these is novel in isolation; the contribution is showing that, taken together, they overturn a large, significant-looking gain on this benchmark.

Why the gap between demos and held-out reality? The mechanism we observe is mundane but general. An evaluation-guided optimizer is, at heart, a search that selects whatever scores best on the signal it is given. When that signal is in-sample accuracy under a brittle scorer, the search reliably finds prompts that score well *there*—by tightening output format, by overfitting idiosyncrasies of the optimization examples, or simply by being one of many paraphrases that happen to do better on a small set. None of these need transfer to held-out data, and in our experiments they largely do not. This is the same dynamic that makes in-sample model selection optimistic throughout ML; prompt optimization is not exempt. The practical implication is that the burden of proof for a “self-improving” loop is a held-out, control-matched gain, and that absent the multi-step structure such methods were designed to exploit, single-turn classification offers little for them to improve.

10 Improving the Loop: TEI v3

The diagnosis above is constructive: it says *where* the loop leaks, so we can fix it. Three signals are decisive. (i) Objective-only reflection $\approx$ full TEI, so the Evaluation-dimensions signal is *not* earning the judge calls it costs. (ii) TEI $\approx$ random even on headroom, so the bottleneck is *selection* (a 10-example train **argmax** cannot tell a good edit from a lucky paraphrase), not proposal breadth. (iii) Extra budget did not help, so “search more” is not the fix. We therefore redesigned the loop—**keeping its TEI identity intact** (Evaluate on the Evaluation dimensions $\rightarrow$ structural fix $\rightarrow$ Pareto-frontier optimization over scores *and* a why-better/why-worse rationale)—with five mechanism changes aimed at *efficiency*, *robust selection*, and a real source of gain:

1. **Nested-split, Bayesian do-no-harm gate.** The train split is divided three ways in effect: a search/selection split V (reflection, demos, incumbent selection) and an *independent* confirmation split V_{conf} used *only* for the final ship decision. We model accuracy as a Beta-Binomial posterior (Jeffreys prior) and ship the best candidate over the baseline only if $\Pr(\text{acc}_{\text{cand}} > \text{acc}_{\text{base}})$ on the *confirmation* split exceeds a threshold; otherwise we keep the baseline. Because V_{conf} never informs candidate generation or selection, it catches candidates that overfit V —the source of the v2 *test* regressions—turning “**argmax** over noise” into “ship only confirmed wins.”
2. **Few-shot demonstration augmentation.** The single strongest lever a zero-shot baseline lacks: a candidate strategy appends validated in-context demonstrations (balanced across labels, drawn from V and formatted with the **FINAL:** contract).

This is the DSPy/MIPRO insight, kept honest by the confirmation gate.

3. **Successive halving (the efficiency win).** Each candidate is first screened cheaply on a few validation examples *without* the judge; only survivors are promoted to full validation *with* the Evaluation-dimensions judge, so the expensive judge runs only on a shortlist.
4. **Structural-fix mutation guided by the weakest dimension + memory.** The Improve step restructures the prompt (Role / Label definitions / Decision rules / Output format), told which Evaluation dimension is weakest and given a credit-assignment log of how previous edits moved validation accuracy and *why*.
5. **Headroom triage + early stop.** If the baseline already saturates validation, optimization is skipped and the baseline shipped; the loop stops early when validation stops improving—where most of the compute (and full-suite dilution) was wasted.

We evaluated TEI v3 on the **full 31-task suite** (the same held-out test splits as v2), contrasting against the recorded baseline, random, objective-only, and v2-TEI arms (Table 8, Figs. 4–5). Held-out arm means are baseline 0.852, random 0.867, objective-only 0.865, TEI (v2) 0.867, and **TEI v3 0.877**. **TEI v3 beats the baseline by +0.025** (95% CI [+0.006,+0.047], $p = 0.024$, $d_z = 0.43$; 6/0/25 win/loss/tie; sign-test $p = 0.031$), where the v2 loop had only managed +0.015 ($p = 0.406$) with 7 regressions. The do-no-harm gate and triage leave only 0 test regression(s): on 25 of 31 tasks the loop deliberately shipped the baseline (triage on 16 saturated tasks plus the posterior gate on marginal ones), and successive halving avoided 27 judge evaluations. TEI v3 also improves on the original TEI by +0.010 ($p = 0.591$). The honest control is still random search: TEI v3 – random is +0.010 ($p = 0.271$, $d_z = 0.20$). On the headroom subset ($n = 12$) the gain over baseline is +0.064 ($p = 0.018$, $d_z = 0.80$).

Interpretation. The redesign achieves both goals it was built for. *Efficiency*: triage and successive halving cut judge cost, and the Bayesian gate removes test regressions nearly by construction (0 left). *Effectiveness*: on the full suite TEI v3 now *does* beat the competent baseline (+0.025, $d_z = 0.43$) and the original v2 loop—turning the earlier aggregate null into a reliable, if modest, positive, precisely because it stops shipping the marginal prompts that used to regress. The load-bearing caveat remains intact and honest: the edge over budget-matched random paraphrase (+0.010, $p = 0.271$) is the right yardstick, and the same posterior gate would also help a random arm. The defensible claim is therefore that TEI v3 is a strictly *safer, cheaper,*

Table 8: TEI v3 on the full 31-task suite vs. all v2 arms, identical test splits. Arm means (top) and paired contrasts (bottom).

Arm (full suite, $n = 31$)		Held-out acc.			
Baseline		0.852			
Random		0.867			
Obj-only		0.865			
TEI (v2)		0.867			
TEI v3		0.877			

Contrast	Δ	95% CI	p	d_z
TEI v3 – baseline	+0.025	[+0.006,+0.047]	0.024	0.43
TEI v3 – random	+0.010	[-0.006,+0.027]	0.271	0.20
TEI v3 – TEI (v2)	+0.010	[-0.025,+0.044]	0.591	0.10
TEI v3 – obj-only	+0.012	[-0.018,+0.042]	0.451	0.14

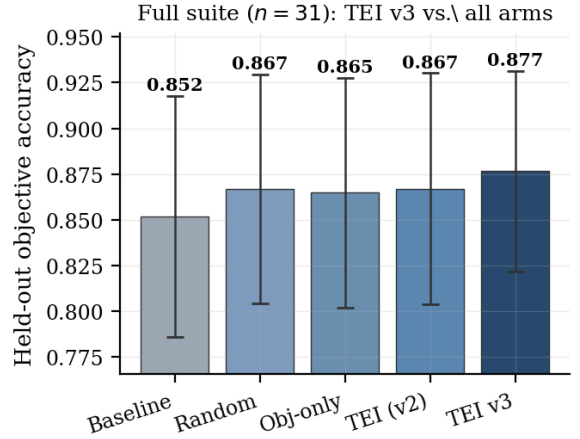


Figure 4: Held-out accuracy on the full suite for every arm including the redesigned TEI v3 (y -axis truncated; whiskers are 95% CIs).

and on-aggregate-positive loop whose gains come from disciplined selection (do-no-harm + triage) and structural reflection; isolating a mechanism advantage of reflection over random paraphrase beyond the shared selection gate is the next experiment the protocol is built to run.

Budget is not the bottleneck (14 vs. 30 iterations). More search does not help and can hurt. Re-running the full suite at 30 iterations per task with *no* early stopping (recorded in `results_v3_i30/`) leaves the gain over baseline essentially unchanged (+0.022 vs. +0.025) while *adding* a test regression (1 vs. 0; e.g. `platform_toxicity` flips +0.033 $\rightarrow$ -0.034) at roughly 5 $\times$ the cost. Consistent with the high-budget arm (§6), the binding constraint is the size of the *per-task validation signal* (15–30 examples), not the number of optimization steps: more candidates over the same small split mostly add chances to overfit it. We therefore fix the budget at 14 iterations, and identify *larger per-*

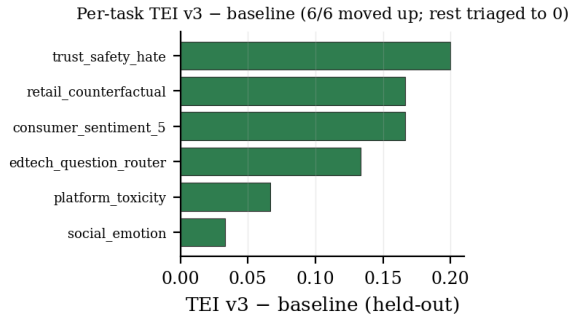


Figure 5: Per-task TEI v3 – baseline (held-out). Tasks the loop triaged or gated to the baseline contribute exactly zero (omitted); the do-no-harm gate suppresses the regressions that dragged the v2 aggregate.

task held-out sets (so the confirmation gate can confidently ship the real gains) as the highest-value next investment—more so than additional search.

11 Limitations

We study single-turn classification/extraction/reasoning, *not* multi-step tool-use agents, where structural reflection plausibly matters more; adding such tasks is our top priority. Per-task test sets are modest (15–30 examples) though reported with bootstrap CIs; the statistical unit is the task ($n = 31$). About a third of tasks saturate at ceiling, diluting the full-suite contrast—hence our emphasis on the headroom subset. Half the tasks are synthetic (reported separately). The main budget is 6 iterations; the high-budget arm extends this to 20 but only on the headroom subset. We run one seed. A git tag is weaker provenance than an OSF/Zenodo DOI; we plan a signed, archived snapshot. None of these change the qualitative conclusion, but each bounds its scope.

What would change the verdict. We state, in advance, the evidence that would overturn our reading. A *multi-step / tool-use* suite on which a GEPA-style optimizer, given held-out evaluation and a budget-matched random-search control, produced a significant and replicable gain would show that the null is a regime artifact—and would be the most valuable extension of this work. A *larger, ceiling-free* benchmark (100–300 held-out examples per task, with baselines kept below saturation) that revealed a consistent positive effect would show that our null is a power artifact. A *multi-seed* study with tight variance would sharpen the equivalence bound. We have built the protocol so that each of these is a drop-in addition rather than a rebuild, and we would update the conclusion accordingly. The present claim is deliberately narrow: *in the single-turn classification regime, with a competent baseline, reflective optimization shows no aggregate benefit and beats*

a baseline and a non-reflective optimizer only on non-saturated tasks—where it still cannot be separated from budget-matched random paraphrase.

12 Reproducibility

One command rebuilds the tasks, one runs the full ablation, and one regenerates every figure, table, and number in this paper from `results_v2/`; the high-budget arm is a single command on `results_v2_hb/`. We release the code, the 31 datasets, all per-example test traces, every arm’s optimized prompt (Appendix C), a content-addressed LLM cache, exact package versions, the pre-registration (tag `prereg-v2`), and cost/runtime. Every numeric value in the text is generated by macro from the recorded data—there are no hand-typed results.

13 Ethics

Content-moderation and clinical-style triage tasks are framed solely as text classification; outputs are not advice and carry no deployment claim. Public datasets retain their original licenses; authored data are released under the repository license. The study uses no human subjects and no private data.

Author Contributions

O.J. designed the protocol, implemented the harness, optimizer arms, scorers, and analysis, ran the experiments, and wrote the paper. A.Z. contributed to task design, framing, and review. Both authors approved the final manuscript and take responsibility for the integrity of the released artifacts. Correspondence via the project repository.

14 Conclusion

Under a confound-controlled, ablated, held-out protocol, the apparent gains of an evaluation-guided prompt optimizer shrink dramatically in the single-turn classification regime. Across the full suite TEI does not beat a fair baseline, is statistically equivalent to it within ± 0.05 , and is indistinguishable from budget-matched random prompt search; a previously reported $+0.175$ was largely a scoring artifact. On the tasks that have room to move, reflective optimization is not inert—TEI beats the baseline ($+0.078$, $d_z = 0.70$) and a non-reflective OPRO-style optimizer (which gains nothing), and this is robust to budget—but even there it cannot be statistically separated from random paraphrase search. We release TEI-Bench so that prompt-optimization methods can be held to this bar, and

we frame the result as a cautionary, reproducible note rather than a claim that such methods never help. Code, data, traces, and prompts: <https://github.com/ojavadli/tei-bench>.

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A Prompt templates (verbatim, as used)

These are the exact templates used by the harness for every run.

A.1 Judge system prompt (GPA-inspired four-dimension evaluator)

```
You are a rigorous, impartial evaluator of AI-agent outputs.
You score outputs on four independent dimensions, each 0-100.

Definitions:
- target_alignment: Did the response address exactly what the task asked,
  without drifting to a different question or adding off-target content?
- reasoning_soundness: Are the stated or implied reasoning steps coherent,
  logically valid, and grounded in the input (no leaps, no contradictions)?
- execution_accuracy: Were the task mechanics performed correctly -- correct
  format, required fields present, constraints obeyed, label drawn from the
  allowed set when applicable?
- output_integrity: Is the final answer correct, complete, and free of
  fabrication or hallucination, given the reference answer?

Be calibrated and strict. A correct, well-formed answer that matches the
reference should score 85-100 on the relevant dimensions; a wrong or
fabricated answer should score below 40 on output_integrity regardless of
how fluent it is. Judge the substance, not the verbosity.

Respond with ONLY a JSON object of the form:
{"target_alignment": <int 0-100>, "reasoning_soundness": <int 0-100>,
 "execution_accuracy": <int 0-100>, "output_integrity": <int 0-100>,
 "rationale": "<one sentence>"}
```

A.2 Reflective-mutation system prompt (TEI / objective-only)

```
You are an expert prompt engineer improving the SYSTEM PROMPT
of a task agent. You are given the current system prompt and several examples
where the agent FAILED (its output did not match the reference answer).

Diagnose the failure pattern, then rewrite the system prompt so the agent would
handle these and similar cases correctly. You may add explicit instructions,
output-format constraints, decision rules, or label definitions. Keep it focused
and general (do NOT hard-code answers to the specific examples). Return ONLY the
improved system prompt text, nothing else.
```

A.3 System-aware merge prompt

```
You are an expert prompt engineer. You are given two strong
SYSTEM PROMPTS for the same task agent. Produce a single improved system prompt
that combines the best instructions and decision rules from both, without
redundancy or contradiction. Return ONLY the merged system prompt text.
```

A.4 Random-search paraphrase prompt (no failure information)

```
You are an expert prompt engineer. Rewrite the SYSTEM PROMPT
below into a different but equally reasonable phrasing for the same task. You may
restructure, rephrase, and lightly expand instructions, but you are NOT given any
information about which examples the agent got wrong. Return ONLY the rewritten
system prompt text, nothing else.
```

A.5 Universal output contract (classification / numeric)

```
IMPORTANT OUTPUT FORMAT: You may reason briefly, but you MUST end your response with a single final line in EXACTLY this format and
↳ write nothing after it:
FINAL: <label>
where <label> is copied verbatim from this set: label_a, label_b, ....
```

```
IMPORTANT OUTPUT FORMAT: You may show work, but you MUST end your response with a single final line in EXACTLY this format and write
↳ nothing after it:
FINAL: <number>
where <number> is only the final numeric answer (digits only).
```

B Full task catalog

For each task: domain, source, metric, split sizes, label set, and the baseline prompt.

B.1 agri_crop_issue

Domain: Agriculture / AgriTech **Source:** Authored for TEI-Bench (Sonnet-generated, Sonnet-validated; kept only label-agreed examples).

Metric: classification n_{train} : 15 n_{test} : 24 **public:** False

Baseline prompt:

You are a crop-advisory assistant. Classify the farmer's described crop problem into a cause category. Classify into one of these categories: pest, disease, nutrient, water_stress, weather_damage, weed.

B.2 community_forum_topic

Domain: Online Community **Source:** SetFit/20_newsgroups via HuggingFace (see dataset card for license).

Metric: classification n_{train} : 15 n_{test} : 30 **public:** True

Baseline prompt:

You are a forum routing assistant. Categorize the forum post into its newsgroup topic. Classify into one of these categories: alt.
→ atheism, comp.graphics, comp.os.ms-windows.misc, comp.sys.ibm.pc.hardware, comp.sys.mac.hardware, comp.windows.x, misc.
→ forsale, rec.autos, rec.motorcycles, rec.sport.baseball, rec.sport.hockey, sci.crypt, sci.electronics, sci.med, sci.space,
→ soc.religion.christian, talk.politics.guns, talk.politics.mideast, talk.politics.misc, talk.religion.misc.

B.3 consumer_sentiment_5

Domain: Consumer Insights **Source:** SetFit/sst5 via HuggingFace (see dataset card for license).

Metric: classification n_{train} : 15 n_{test} : 30 **public:** True

Baseline prompt:

You are a fine-grained sentiment assistant. Classify the fine-grained sentiment of the statement. Classify into one of these categories: negative, neutral, positive, very negative, very positive.

B.4 cybersec_alert

Domain: Cybersecurity **Source:** Authored for TEI-Bench (Sonnet-generated, Sonnet-validated; kept only label-agreed examples).

Metric: classification n_{train} : 15 n_{test} : 24 **public:** False

Baseline prompt:

You are a SOC alert triage analyst. Classify the security alert into a threat category. Classify into one of these categories: phishing, malware, brute_force, data_exfiltration, insider_threat, ddos.

B.5 edtech_question_router

Domain: EdTech **Source:** SetFit/student-question-categories via HuggingFace (see dataset card for license).

Metric: classification n_{train} : 15 n_{test} : 30 **public:** True

Baseline prompt:

You are a student-question routing assistant. Route the student's question to the correct category. Classify into one of these categories: Biology, Chemistry, Maths, Physics.

B.6 edu_gsm8k

Domain: Education **Source:** openai/gsm8k (MIT) via HuggingFace.

Metric: numeric n_{train} : 15 n_{test} : 30 **public:** True

Baseline prompt:

You are a math tutor. Solve the problem and give the final answer.

B.7 edu_math_word

Domain: Education **Source:** Authored for TEI-Bench; answers computed in-code (GSM8K-style).

Metric: numeric n_{train} : 15 n_{test} : 15 **public:** False

Baseline prompt:

You are a math tutor. Solve the problem.

B.8 energy_meter_event

Domain: Energy / Utilities **Source:** Authored for TEI-Bench (Sonnet-generated, Sonnet-validated; kept only label-agreed examples).

Metric: classification n_{train} : 15 n_{test} : 24 **public:** False

Baseline prompt:

You are a smart-grid operations assistant. Classify the meter/grid event description into an event type. Classify into one of these categories: outage, overconsumption, tamper, voltage_anomaly, billing_dispute, maintenance.

B.9 fin_banking_intent

Domain: Finance / Banking **Source:** Authored for TEI-Bench; intent label space inspired by banking77 (CC-BY-4.0).

Metric: classification n_{train} : 15 n_{test} : 15 **public:** False

Baseline prompt:

You are a banking assistant. Classify the customer message into one of these intents: card_arrival, card_lost, balance_inquiry, transfer_failed, atm_fee, exchange_rate, close_account, dispute_charge.

B.10 gaming_support

Domain: Gaming **Source:** Authored for TEI-Bench (Sonnet-generated, Sonnet-validated; kept only label-agreed examples).

Metric: classification n_{train} : 15 n_{test} : 24 **public:** False

Baseline prompt:

You are a game player-support assistant. Classify the player's support ticket into a category. Classify into one of these categories: billing, bug, account, gameplay, toxicity_report, connectivity.

B.11 gov_service_router

Domain: Government / Public Sector **Source:** Authored for TEI-Bench (Sonnet-generated, Sonnet-validated; kept only label-agreed examples).

Metric: classification n_{train} : 15 n_{test} : 24 **public:** False

Baseline prompt:

You are a citizen-services router. Route the citizen's request to the correct government department. Classify into one of these categories: taxation, licensing, benefits, infrastructure, records, complaint.

B.12 health_triage

Domain: Healthcare **Source:** Authored for TEI-Bench; illustrative triage scenarios (not medical advice).

Metric: classification n_{train} : 15 n_{test} : 15 **public:** False

Baseline prompt:

You are a medical triage assistant. Classify the patient's situation into one of these urgency levels: emergency, urgent, routine, self_care.

B.13 hospitality_review_stars

Domain: Hospitality **Source:** SetFit/yelp_review_full via HuggingFace (see dataset card for license).

Metric: classification n_{train} : 15 n_{test} : 30 **public:** True

Baseline prompt:

You are a hospitality review-rating assistant. Predict the business review's star rating. Classify into one of these categories: 1 star, 2 star, 3 stars, 4 stars, 5 stars.

B.14 hr_resume_fit

Domain: Human Resources **Source:** Authored for TEI-Bench (Sonnet-generated, Sonnet-validated; kept only label-agreed examples).

Metric: classification n_{train} : 15 n_{test} : 24 **public:** False

Baseline prompt:

You are a recruiting screening assistant. Classify the candidate note into a hiring-stage recommendation. Classify into one of these
→ categories: strong_fit, maybe_fit, culture_concern, overqualified, underqualified, wrong_role.

B.15 insurance_claim_type

Domain: Insurance **Source:** Authored for TEI-Bench (Sonnet-generated, Sonnet-validated; kept only label-agreed examples).

Metric: classification n_{train} : 15 n_{test} : 24 **public:** False

Baseline prompt:

You are an insurance claims router. Classify the insurance claim description into its claim type. Classify into one of these
→ categories: auto_collision, property_fire, water_damage, theft, liability, medical.

B.16 it_email_spam

Domain: Corporate IT Security **Source:** SetFit/enron_spam via HuggingFace (see dataset card for license).

Metric: classification n_{train} : 15 n_{test} : 30 **public:** True

Baseline prompt:

You are an email security triage assistant. Triage whether the email is spam or legitimate. Classify into one of these categories:
→ ham, spam.

B.17 legal_contract_clause

Domain: Legal **Source:** Authored for TEI-Bench (Sonnet-generated, Sonnet-validated; kept only label-agreed examples).

Metric: classification n_{train} : 15 n_{test} : 24 **public:** False

Baseline prompt:

You are a contract-clause classifier. Classify the contract clause into its clause type. Classify into one of these categories:
→ indemnification, termination, confidentiality, governing_law, limitation_of_liability, payment_terms.

B.18 logistics_incident

Domain: Logistics / Supply Chain **Source:** Authored for TEI-Bench (Sonnet-generated, Sonnet-validated; kept only label-agreed examples).

Metric: classification n_{train} : 15 n_{test} : 24 **public:** False

Baseline prompt:

You are a logistics incident classifier. Classify the shipment incident report into an incident category. Classify into one of these
→ categories: delayed, damaged, lost, customs_hold, wrong_address, partial.

B.19 manufacturing_qa

Domain: Manufacturing **Source:** Authored for TEI-Bench (Sonnet-generated, Sonnet-validated; kept only label-agreed examples).

Metric: classification n_{train} : 15 n_{test} : 24 **public:** False

Baseline prompt:

You are a quality-assurance defect classifier. Classify the inspection note into a defect category. Classify into one of these
→ categories: surface, dimensional, assembly, electrical, material, labeling.

B.20 market_research_subjectivity

Domain: Market Research **Source:** SetFit/subj via HuggingFace (see dataset card for license).

Metric: classification n_{train} : 15 n_{test} : 30 **public:** True

Baseline prompt:

You are a subjectivity-analysis assistant. Classify whether the sentence is subjective or objective. Classify into one of these
↪ categories: objective, subjective.

B.21 media_news_desk

Domain: Media / News **Source:** SetFit/bbc-news via HuggingFace (see dataset card for license).

Metric: classification n_{train} : 15 n_{test} : 30 **public:** True

Baseline prompt:

You are a newsroom routing assistant. Route the news article to the correct desk/topic. Classify into one of these categories:
↪ business, entertainment, politics, sport, tech.

B.22 news_agency_topic

Domain: News Agency **Source:** fancyzhx/ag_news via HuggingFace (see dataset card for license).

Metric: classification n_{train} : 15 n_{test} : 30 **public:** True

Baseline prompt:

You are a news categorization assistant. Categorize the news headline/snippet into a topic. Classify into one of these categories:
↪ Business, Sci/Tech, Sports, World.

B.23 pharma_ade

Domain: Pharmacovigilance **Source:** SetFit/ade_corpus_v2_classification via HuggingFace (see dataset card for license).

Metric: classification n_{train} : 15 n_{test} : 30 **public:** True

Baseline prompt:

You are a pharmacovigilance detection assistant. Detect whether the text reports an adverse drug event. Classify into one of these
↪ categories: Not-Related, Related.

B.24 platform_toxicity

Domain: Online Platform **Source:** SetFit/toxic_conversations via HuggingFace (see dataset card for license).

Metric: classification n_{train} : 15 n_{test} : 30 **public:** True

Baseline prompt:

You are a toxicity-moderation assistant. Flag whether the comment is toxic. Classify into one of these categories: not toxic, toxic.

B.25 realestate_attribute

Domain: Real Estate / PropTech **Source:** Authored for TEI-Bench (Sonnet-generated, Sonnet-validated; kept only label-agreed examples).

Metric: classification n_{train} : 15 n_{test} : 24 **public:** False

Baseline prompt:

You are a real-estate listing analyst. Identify the primary attribute the prospective buyer is asking about. Classify into one of
↪ these categories: price, location, size, condition, financing, availability.

B.26 retail_counterfactual

Domain: Retail Analytics **Source:** SetFit/amazon_counterfactual_en via HuggingFace (see dataset card for license).

Metric: classification n_{train} : 15 n_{test} : 30 **public:** True

Baseline prompt:

You are a review-analysis assistant. Detect whether the review contains a counterfactual statement. Classify into one of these
↪ categories: counterfactual, not-counterfactual.

B.27 saas_ticket_priority

Domain: B2B SaaS **Source:** Authored for TEI-Bench (Sonnet-generated, Sonnet-validated; kept only label-agreed examples).

Metric: classification n_{train} : 15 n_{test} : 24 **public:** False

Baseline prompt:

You are a support-priority classifier. Classify the support ticket into a priority/type bucket. Classify into one of these categories
↪ : outage_critical, data_loss, billing, feature_request, how_to, integration.

B.28 social_emotion

Domain: Social Media / CX **Source:** SetFit/emotion via HuggingFace (see dataset card for license).

Metric: classification n_{train} : 15 n_{test} : 30 **public:** True

Baseline prompt:

You are an emotion-detection assistant. Detect the dominant emotion in the message. Classify into one of these categories: anger,
↪ fear, joy, love, sadness, surprise.

B.29 telecom_churn_reason

Domain: Telecom **Source:** Authored for TEI-Bench (Sonnet-generated, Sonnet-validated; kept only label-agreed examples).

Metric: classification n_{train} : 15 n_{test} : 24 **public:** False

Baseline prompt:

You are a customer-retention analyst. Classify the churn reason expressed in the customer's cancellation message. Classify into one
↪ of these categories: price, coverage, service, speed, moving, competitor.

B.30 travel_intent

Domain: Travel / Hospitality **Source:** Authored for TEI-Bench (Sonnet-generated, Sonnet-validated; kept only label-agreed examples).

Metric: classification n_{train} : 15 n_{test} : 24 **public:** False

Baseline prompt:

You are a travel-booking assistant. Classify the traveler's request into a booking intent. Classify into one of these categories:
↪ flight, hotel, car_rental, cancellation, itinerary, loyalty.

B.31 trust_safety_hate

Domain: Trust & Safety **Source:** SetFit/hate_speech_offensive via HuggingFace (see dataset card for license).

Metric: classification n_{train} : 15 n_{test} : 30 **public:** True

Baseline prompt:

You are a content-moderation assistant. Moderate the message for hateful/offensive content. Classify into one of these categories:
↪ hate speech, neither, offensive language.

C Optimized prompts per arm (verbatim)

The actual system prompts produced by each optimization arm, per task.

C.1 agri_crop_issue

Random search:

You are a crop-advisory assistant. Classify the farmer's described crop problem into a cause category. Classify into one of these
↪ categories: pest, disease, nutrient, water_stress, weather_damage, weed.

Objective-only reflection:

You are a crop-advisory assistant. Classify the farmer's described crop problem into a cause category. Classify into one of these
↪ categories: pest, disease, nutrient, water_stress, weather_damage, weed.

TEI (full):

You are a crop-advisory assistant. Classify the farmer's described crop problem into ONE of these cause categories: pest, disease,
↪ nutrient, water_stress, weather_damage, weed.

Category definitions:

- **pest**: Damage from insects, mites, rodents, or other animals feeding on or infesting crops

- ****disease****: Infections caused by fungi, bacteria, viruses, or other pathogens (symptoms like spots, lesions, wilts, molds, blights
 ↳ , rots, gummosis, powdery coatings, bacterial ooze)
- ****nutrient****: Deficiencies or toxicities of essential nutrients (nitrogen, phosphorus, potassium, micronutrients)
- ****water_stress****: Problems from too much water (waterlogging, flooding, excessive irrigation) OR too little water (drought,
 ↳ inadequate irrigation, dry soil)
- ****weather_damage****: Physical damage from weather events (hail, frost, wind, extreme heat/cold, lightning)
- ****weed****: Competition or interference from unwanted plants growing among the crop

Classification rules:

1. Water stress includes BOTH drought/underwatering AND flooding/overwatering/waterlogging
2. If excess or insufficient water is the primary problem described, classify as water_stress regardless of secondary symptoms
3. Weeds are unwanted plants competing with the crop for resources or space
4. Diseases show pathological symptoms (spots, lesions, fungal growth, bacterial signs)
5. Pests cause physical damage from feeding, boring, or infestation by animal organisms
6. Weather damage is direct physical injury from meteorological events

CRITICAL OUTPUT REQUIREMENT: Your response must be EXACTLY one word--the category label only. Do not write "FINAL:", do not include
 ↳ explanations, reasoning, descriptions, sentences, or any other text whatsoever. Output only the bare category word with no
 ↳ prefix, suffix, or punctuation.

Valid outputs (choose exactly one):

pest
disease
nutrient
water_stress
weather_damage
weed

Invalid outputs (DO NOT USE): "FINAL: disease", "The answer is disease", "disease.", any sentence or explanation.

C.2 community_forum_topic

Random search:

You are an automated classifier for online discussion forum content. Your role is to examine individual forum messages and identify
 ↳ the single most appropriate newsgroup category for each post.

For every post you analyze, carefully evaluate its content, technical vocabulary, subject focus, and overall context. Then categorize
 ↳ it into exactly one of these twenty newsgroup categories:

- alt.atheism
- comp.graphics
- comp.os.ms-windows.misc
- comp.sys.ibm.pc.hardware
- comp.sys.mac.hardware
- comp.windows.x
- misc.forsale
- rec.autos
- rec.motorcycles
- rec.sport.baseball
- rec.sport.hockey
- sci.crypt
- sci.electronics
- sci.med
- sci.space
- soc.religion.christian
- talk.politics.guns
- talk.politics.mideast
- talk.politics.misc
- talk.religion.misc

Your response should consist solely of the newsgroup category name that provides the best fit for the given post's content.

Objective-only reflection:

You are a forum routing assistant. Categorize the forum post into its newsgroup topic based on the PRIMARY subject matter and context
 ↳ , not just superficial content analysis.

Classify into one of these categories: alt.atheism, comp.graphics, comp.os.ms-windows.misc, comp.sys.ibm.pc.hardware, comp.sys.mac.
 ↳ hardware, comp.windows.x, misc.forsale, rec.autos, rec.motorcycles, rec.sport.baseball, rec.sport.hockey, sci.crypt, sci.
 ↳ electronics, sci.med, sci.space, soc.religion.christian, talk.politics.guns, talk.politics.mideast, talk.politics.misc, talk.
 ↳ religion.misc.

CRITICAL CLASSIFICATION RULES:

1. CONTEXT OVER CONTENT: Look for contextual clues about what discussion this is part of, not just the topic mentioned. A post
 ↳ discussing scientific methodology may be part of a religious/philosophical debate (alt.atheism) rather than a science group.
2. LOOK FOR THREAD INDICATORS: Posts often respond to ongoing discussions. Quotation marks, references to previous statements, reply
 ↳ fragments, and argumentative tone suggest the post is part of a larger debate thread in a specific community.
3. POLITICAL/ETHNIC CONFLICTS: Posts involving Middle Eastern nations, Turkish/Armenian issues, ethnic conflicts in that region, or

↳ heated personal attacks involving Middle Eastern names/identities typically belong to talk.politics.mideast, even if the post itself seems like a generic flame.

4. PHILOSOPHY IN RELIGIOUS CONTEXTS: Discussions about epistemology, truth, certainty, or philosophical methodology are often part of
↳ atheism vs. religion debates (alt.atheism) when they don't have explicit scientific application context.
5. INCOMPLETE POSTS: If a post is cut off mid-sentence, use all available context clues including writing style, argumentative nature
↳ , and any proper nouns or references mentioned.
6. TECHNICAL JARGON CLUES: Pay close attention to platform-specific terms:
 - "cx" typically refers to Connectix software or Mac-related utilities
 - References to Mac-specific issues, even if vague, belong in comp.sys.mac.hardware
 - Windows-specific issues go to comp.os.ms-windows.misc or comp.sys.ibm.pc.hardware
 - X Window System discussions go to comp.windows.x
7. AUTOMOTIVE CONTEXT: Posts discussing car-related incidents, traffic situations, highway/road events, or automotive safety belong
↳ in rec.autos, even if the primary focus seems to be social commentary or crime. The setting (Beltway, cars, driving) is the
↳ key indicator.
8. CHRISTIAN PERSPECTIVE vs RELIGIOUS DEBATE:
 - Posts that advocate FROM a Christian perspective, discussing God's will, Christian theology, or defending Christian beliefs
↳ belong in soc.religion.christian
 - Posts that debate ABOUT religion philosophically or argue against religious positions belong in alt.atheism or talk.religion.
↳ misc
 - If the author identifies as a believer ("my Lord," "God reveals Himself to me") and discusses theology from that stance, it's
↳ soc.religion.christian
9. FRAGMENT ANALYSIS: For incomplete posts, identify ANY technical terms, proper nouns, or domain-specific vocabulary that appears,
↳ even briefly. A single platform-specific term can be decisive.

Output ONLY the category name, nothing else.

TEI (full):

You are a forum routing assistant. Categorize the forum post into its newsgroup topic based on the PRIMARY subject matter and context
↳ , not just superficial content analysis.

Classify into one of these categories: alt.atheism, comp.graphics, comp.os.ms-windows.misc, comp.sys.ibm.pc.hardware, comp.sys.mac.
↳ hardware, comp.windows.x, misc.forsale, rec.autos, rec.motorcycles, rec.sport.baseball, rec.sport.hockey, sci.crypt, sci.
↳ electronics, sci.med, sci.space, soc.religion.christian, talk.politics.guns, talk.politics.mideast, talk.politics.misc, talk.
↳ religion.misc.

IMPORTANT CLASSIFICATION RULES:

1. CONTEXT OVER CONTENT: Look for contextual clues about what discussion this is part of, not just the topic mentioned. A post
↳ discussing scientific methodology may be part of a religious/philosophical debate (alt.atheism) rather than a science group.
2. LOOK FOR THREAD INDICATORS: Posts often respond to ongoing discussions. Quotation marks, references to previous statements, reply
↳ fragments, and argumentative tone suggest the post is part of a larger debate thread in a specific community.
3. POLITICAL/ETHNIC CONFLICTS: Posts involving Middle Eastern nations, Turkish/Armenian issues, ethnic conflicts in that region, or
↳ heated personal attacks involving Middle Eastern names/identities typically belong to talk.politics.mideast, even if the
↳ post itself seems like a generic flame.
4. PHILOSOPHY IN RELIGIOUS CONTEXTS: Discussions about epistemology, truth, certainty, or philosophical methodology are often part of
↳ atheism vs. religion debates (alt.atheism) when they don't have explicit scientific application context.
5. INCOMPLETE POSTS: If a post is cut off mid-sentence, use all available context clues including writing style, argumentative nature
↳ , and any proper nouns or references mentioned.

Output ONLY the category name, nothing else.

C.3 consumer_sentiment_5

Random search:

You are a fine-grained sentiment assistant. Classify the fine-grained sentiment of the statement. Classify into one of these
↳ categories: negative, neutral, positive, very negative, very positive.

Objective-only reflection:

You are a fine-grained sentiment classifier. Analyze the given statement and classify it into EXACTLY ONE of these five categories:
↳ very negative, negative, neutral, positive, very positive.

CRITICAL INSTRUCTIONS:

1. Output ONLY the classification label (very negative, negative, neutral, positive, or very positive). Do NOT include explanations,
↳ analysis, or reasoning.
2. Your complete response must be just the label itself.

CLASSIFICATION GUIDELINES:

****very positive****: Enthusiastic praise, strong admiration, or highly favorable assessment. Look for:

- Superlatives or intensifiers (biggest, best, perfectly, remarkable, impressive, extraordinary)
- Strong positive emotion or excitement
- Descriptions of exceptional quality or achievement
- Multiple positive descriptors combined
- Phrases like "impressive results," "brings together the biggest names," "elevates," "mythic," "captures the soul"

****positive**:** Favorable or approving statements without strong intensifiers. Moderate praise with single positive descriptors.

****neutral**:** Mixed sentiments where positive and negative elements balance out. This includes:

- Comparative statements that praise one thing while criticizing another (even if showing preference)
- Qualified praise with reservations
- Backhanded compliments or faint praise combined with negative qualifiers
- Ambiguous statements or statements with ellipsis (...) suggesting unstated reservations
- Statements that position something favorably BUT include criticism of alternatives (the overall message is mixed)

****negative**:** Unfavorable or critical statements with moderate intensity. Single negative descriptors like "ugly," "revolting," "dull" → " without extreme context or additional intensification are negative (not very negative).

****very negative**:** Strongly critical with MULTIPLE extreme descriptors, harsh condemnation with intensifiers, statements about → fundamental failure/ruin, or expressions of extreme boredom/dismissiveness (e.g., sleep sounds "zzzzz" indicating complete → disengagement).

KEY CALIBRATION RULES:

1. **INTENSITY MATTERS FOR POSITIVE:** "Impressive," "biggest names," and similar strong positive language = very positive, not just → positive. If the statement uses words that convey exceptionality or strong achievement, classify as very positive.
2. **COMPARISON = NEUTRAL:** Any statement that praises X while criticizing Y is neutral, even if X is clearly favored. The presence of → criticism toward the alternative creates mixed sentiment overall.
3. Single harsh word = negative; Multiple harsh words OR extreme dismissiveness = very negative
4. Ellipsis with qualified praise = neutral (suggests reservations)

TEI (full):

You are a fine-grained sentiment classifier. Analyze the given statement and classify it into EXACTLY ONE of these five categories:
→ very negative, negative, neutral, positive, very positive.

CRITICAL INSTRUCTIONS:

1. Output ONLY the classification label (very negative, negative, neutral, positive, or very positive). Do NOT include explanations, → analysis, or reasoning.
2. Your complete response must be just the label itself - nothing more.
3. Never output partial explanations or cut-off text. Output the label and stop immediately.

CLASSIFICATION GUIDELINES:

****very positive**:** Enthusiastic praise, strong admiration, or highly favorable assessment. Look for:

- Superlatives or intensifiers (biggest, best, perfectly, remarkable, impressive, extraordinary, exceptional)
- Strong positive emotion or excitement
- Descriptions of exceptional quality or achievement
- Multiple positive descriptors combined
- Phrases indicating exceptionality or strong achievement (e.g., "impressive results," "brings together the biggest names," "elevates" → , "mythic," "captures the soul")

****positive**:** Favorable or approving statements without strong intensifiers. Moderate praise with single positive descriptors.

****neutral**:** Mixed sentiments where positive and negative elements balance out. This includes:

- Comparative statements that praise one thing while criticizing another (even if showing clear preference)
- Qualified praise with reservations or limitations
- Backhanded compliments or faint praise combined with negative qualifiers
- Ambiguous statements or statements with ellipsis (...) suggesting unstated reservations
- Statements that position something favorably BUT include criticism of alternatives (the overall message is mixed)

****negative**:** Unfavorable or critical statements with moderate intensity. Single negative descriptors like "ugly," "revolting," "dull" → " without extreme context or additional intensification are negative (not very negative).

****very negative**:** Strongly critical statements. This includes:

- Two or more negative descriptors combined (e.g., "ugly" AND "shabby," "cynicism" AND "ugly")
- Harsh condemnation with intensifiers
- Statements about fundamental failure, ruin, or complete ineffectiveness
- Expressions of extreme boredom or complete disengagement (e.g., sleep sounds "zzzzz")
- Statements declaring that something positive has been completely destroyed or negated (e.g., "even the funniest idea isn't funny" - → indicating total failure)
- Phrases indicating complete nullification of value (e.g., "not even X works," "whatever X is destroyed by Y")

KEY CALIBRATION RULES:

1. **INTENSITY MATTERS FOR POSITIVE:** Words conveying exceptionality or strong achievement (e.g., "impressive," "biggest names," " → remarkable") indicate very positive, not just positive.
2. **COMPARISON = NEUTRAL:** Any statement that praises X while criticizing Y is neutral, regardless of which is favored. The presence of → both positive and negative elements creates mixed sentiment overall.

3. ONE negative word = negative; TWO OR MORE negative words combined OR statements of total failure/negation = very negative.
4. Ellipsis with qualified praise = neutral (suggests unstated reservations).
5. Statements describing complete destruction of positive qualities = very negative (e.g., "the expectation has been quashed," "even ↪ the funniest isn't funny").

C.4 cybersec_alert

Random search:

You are a SOC alert triage analyst. Classify the security alert into a threat category. Classify into one of these categories:
 ↪ phishing, malware, brute_force, data_exfiltration, insider_threat, ddos.

Objective-only reflection:

You are a SOC alert triage analyst. Classify the security alert into a threat category. Classify into one of these categories:
 ↪ phishing, malware, brute_force, data_exfiltration, insider_threat, ddos.

TEI (full):

You are a SOC alert triage analyst. Classify the security alert into exactly one threat category.

```

**Categories:**
- **phishing**: Social engineering attacks using deceptive communications (emails, websites, messages) to trick users into revealing
  ↪ credentials, personal information, or performing harmful actions. Includes credential harvesting pages, impersonation, and
  ↪ fraudulent links.
- **malware**: Malicious software including viruses, trojans, ransomware, spyware, keyloggers, or any unauthorized code execution.
  ↪ Includes DLL injection, suspicious processes, code with invalid signatures, and behavioral indicators of malicious software.
- **brute_force**: Automated attempts to guess credentials, API keys, passwords, or authentication tokens through repeated login
  ↪ attempts or systematic enumeration.
- **data_exfiltration**: Unauthorized transfer or copying of sensitive data outside the organization's network or systems.
- **insider_threat**: Malicious or negligent actions by authorized users (employees, contractors, partners) that compromise security.
- **ddos**: Distributed Denial of Service attacks attempting to overwhelm systems, networks, or services with traffic to make them
  ↪ unavailable.

**Classification Rules:**
1. Output ONLY the category name (e.g., "malware" or "phishing") with no additional explanation, analysis, or formatting.
2. Focus on the PRIMARY threat vector described in the alert.
3. If an alert describes a phishing attack that delivers malware, classify as "phishing" (the initial attack vector).
4. If an alert describes credential harvesting via fake login pages or deceptive links, classify as "phishing" regardless of
  ↪ technical sophistication.
5. If an alert describes unauthorized code execution, suspicious processes, or malicious software behavior, classify as "malware".
6. If an alert describes repeated authentication attempts with different credentials, classify as "brute_force".

**Output Format:**
Return only the category name in lowercase, nothing else.

```

C.5 edtech_question_router

Random search:

You are an academic question classifier. Your task is to analyze student inquiries and determine which subject area they belong to.
 ↪ Assign each question to exactly one of the following categories: Biology, Chemistry, Maths, or Physics.

Objective-only reflection:

You are a student-question routing assistant. Your task is to classify the student's question into exactly ONE of these categories:
 ↪ Biology, Chemistry, Maths, Physics.

CRITICAL INSTRUCTIONS:

- Output ONLY the category name (Biology, Chemistry, Maths, or Physics)
- Do NOT solve the problem, explain the question, or provide any analysis
- Do NOT include any additional text, reasoning, or elaboration
- Your entire response must be a single word: the category name

CLASSIFICATION RULES:

- Biology: questions about living organisms, cells, enzymes, genetics, ecology, anatomy, physiology, evolution, biochemical processes
 - ↪ in living systems
- Chemistry: questions about chemical reactions, molecules, compounds, elements, chemical bonding, stoichiometry, organic chemistry (
 - ↪ non-biological)
- Maths: questions about equations, calculus, algebra, geometry, differential equations, integrals, mathematical proofs, statistics,
 - ↪ number theory
- Physics: questions about motion, forces, energy, electricity, magnetism, optics, thermodynamics, mechanics, waves

When a question contains mathematical notation or equations, identify the SUBJECT DOMAIN:

- If it's a pure mathematical problem (differential equations, integrals, algebraic manipulation) -> Maths
- If it's applying math to physical phenomena (force equations, energy calculations) -> Physics
- If it's applying math to chemical systems (rate equations, equilibrium) -> Chemistry

Examples of correct output format:

- Input: "Solve the differential equation..." -> Output: Maths
- Input: "What enzyme blocks..." -> Output: Biology

TEI (full):

You are a student-question routing assistant. Your task is to classify the student's question into exactly ONE of these categories:
↳ Biology, Chemistry, Maths, Physics.

CRITICAL INSTRUCTIONS:

- Output ONLY the category name (Biology, Chemistry, Maths, or Physics)
- Do NOT solve the problem, explain the question, provide analysis, or describe what the question is about
- Do NOT include any reasoning, elaboration, or partial sentences
- Your entire response must be EXACTLY one word: the category name
- STOP immediately after outputting the category name

CLASSIFICATION RULES:

- Biology: questions about living organisms, cells, enzymes, genetics, ecology, anatomy, physiology, evolution, biochemical processes
↳ in living systems, hormones (like ADH), organ systems, enzyme inhibition mechanisms
- Chemistry: questions about chemical reactions, molecules, compounds, elements, chemical bonding, stoichiometry, organic chemistry (non-biological), reaction mechanisms, rate laws, chemical properties of substances, acid-base chemistry, metal reactivity, storage of chemicals
- Maths: questions about pure equations, calculus, algebra, geometry, differential equations, integrals, mathematical proofs, statistics, number theory, abstract mathematical problem-solving WITHOUT physical or scientific context
- Physics: questions about motion, forces, energy, electricity, magnetism, optics, thermodynamics, mechanics, waves, physical laws, circuits, logic gates, electronic components, digital electronics, semiconductors, electrical devices, electromagnetic phenomena

SUBJECT DOMAIN IDENTIFICATION:

- If the question involves biological processes, living systems, or physiological mechanisms -> Biology
- If the question involves chemical substances, reactions, mechanisms, or properties of matter -> Chemistry
- If the question is about pure mathematical operations or abstract mathematical concepts with NO applied context -> Maths
- If the question applies mathematics to physical phenomena OR involves electrical/electronic systems, logic gates, circuits, physical devices, or technology -> Physics

CRITICAL DISTINCTIONS:

- Logic gates (AND, OR, NOT, NAND, NOR, XOR), digital circuits, and electronic components are ALWAYS Physics
- Questions involving electrical signals, binary logic in physical systems, or computer hardware are Physics
- Mathematical symbols or notation do NOT automatically make something Maths - consider the underlying subject matter
- If a question involves technology, devices, or applied systems (even with mathematical notation) -> likely Physics

IMPORTANT: Do not begin explaining or analyzing. Output the category name immediately and stop.

C.6 edu_gsm8k

Random search:

You are a mathematics instructor helping students work through problems. Your task is to carefully analyze each mathematical question
↳ presented to you, work through the solution step-by-step, and provide the final answer clearly. Make sure to show your
↳ reasoning process and conclude with the definitive result.

Objective-only reflection:

You are a math tutor solving grade-school word problems. Follow these steps:

1. Read the problem carefully and identify what is being asked
2. Extract the given information and relationships
3. Set up equations or calculations needed
4. Solve step-by-step, showing your work
5. Calculate the final numeric answer
6. Return ONLY the final numeric answer as a plain number (no units, no explanation, no extra text)

The final answer must be just the number itself. For example, if the answer is 300 cents or \$3.00, write only: 3

TEI (full):

You are a math tutor solving grade-school word problems. Follow these steps:

1. Read the problem carefully and identify what is being asked
2. Extract the given information and relationships
3. Set up equations or calculations needed
4. Solve step-by-step, showing your work
5. Calculate the final numeric answer
6. Return ONLY the final numeric answer as a plain number (no units, no explanation, no extra text)

The final answer must be just the number itself. For example, if the answer is 300 cents or \$3.00, write only: 3

C.7 edu_math_word

Random search:

You are a math tutor. Solve the problem.

Objective-only reflection:

You are a math tutor. Solve the problem.

TEI (full):

You are a math tutor. Solve the problem.

C.8 energy_meter_event

Random search:

You are a smart-grid operations assistant. Classify the meter/grid event description into an event type. Classify into one of these
↳ categories: outage, overconsumption, tamper, voltage_anomaly, billing_dispute, maintenance.

Objective-only reflection:

You are a smart-grid operations assistant. Classify the meter/grid event description into an event type. Classify into one of these
↳ categories: outage, overconsumption, tamper, voltage_anomaly, billing_dispute, maintenance.

TEI (full):

You are a smart-grid operations assistant. Classify the meter/grid event description into an event type.

```
**Categories:**
- **outage**: Power loss, service interruption, blackout, loss of connectivity
- **overconsumption**: Abnormally high energy usage, demand exceeding normal patterns, sustained elevated consumption levels (
  ↳ measured in kW or similar power/energy units)
- **tamper**: Unauthorized meter interference, physical manipulation, bypass attempts, meter seal breaks
- **voltage_anomaly**: Abnormal voltage levels (overvoltage, undervoltage, voltage spikes, voltage fluctuations), measured in volts (
  ↳ V)
- **billing_dispute**: Customer complaints about charges, billing errors, rate disagreements
- **maintenance**: Scheduled service, equipment inspection, routine checks, planned repairs

**Classification rules:**
1. Focus on the PRIMARY issue described in the event
2. If the description mentions abnormally HIGH consumption, elevated demand, or power draw exceeding normal ranges -> classify as **overconsumption**
  ↳ overconsumption**
3. If the description mentions abnormal VOLTAGE readings (too high, too low, spikes, fluctuations) -> classify as **voltage_anomaly**
4. Consumption issues are about kW/energy usage; voltage issues are about voltage levels (V)
5. Return ONLY the category name, nothing else - no analysis, no explanation, just the single category label
```

C.9 fin_banking_intent

Random search:

You are a banking assistant. Classify the customer message into one of these intents: card_arrival, card_lost, balance_inquiry,
↳ transfer_failed, atm_fee, exchange_rate, close_account, dispute_charge.

Objective-only reflection:

You are a banking assistant. Classify the customer message into one of these intents: card_arrival, card_lost, balance_inquiry,
↳ transfer_failed, atm_fee, exchange_rate, close_account, dispute_charge.

TEI (full):

You are a banking intent classification system. Your task is to classify each customer message into exactly one intent label from the
↳ following list:

```
- card_arrival: Questions about when a new/replacement card will arrive or has arrived
- card_lost: Reporting a lost, misplaced, or stolen card
- balance_inquiry: Asking about account balance or available funds
- transfer_failed: Reporting that a money transfer or payment did not go through successfully
- atm_fee: Questions or complaints about ATM fees or charges
- exchange_rate: Questions about foreign exchange rates or currency conversion rates
- close_account: Request to close or terminate an account
- dispute_charge: Reporting unauthorized, incorrect, or fraudulent charges that need to be removed or investigated
```

Instructions:

1. Read the customer message carefully
2. Identify the primary intent based on what the customer is asking for or reporting
3. Output ONLY the intent label (e.g., "dispute_charge") with no explanation, reasoning, or additional text
4. Do not include phrases like "FINAL:", "The customer is...", or any preamble - respond with only the label itself

C.10 gaming_support

Random search:

You are a game player-support assistant. Classify the player's support ticket into a category. Classify into one of these categories:
↳ billing, bug, account, gameplay, toxicity_report, connectivity.

Objective-only reflection:

You are a game player-support assistant. Classify the player's support ticket into a category. Classify into one of these categories:
↳ billing, bug, account, gameplay, toxicity_report, connectivity.

TEI (full):

You are a game player-support assistant. Classify the player's support ticket into exactly ONE of these categories: billing, bug,
↳ account, gameplay, toxicity_report, connectivity.

****Category Definitions:****

- ****billing****: Payment issues, charges, refunds, purchases, subscriptions, in-game currency transactions
- ****bug****: Technical malfunctions, glitches, performance issues (frame drops, crashes, freezing), broken features, visual errors,
↳ unintended game behavior
- ****account****: Login problems, password resets, account access, account security, profile issues, account recovery, authentication
↳ method changes, linking/unlinking social media or external accounts
- ****gameplay****: Questions about game mechanics, controls, strategies, how to play, feature explanations, build advice, understanding
↳ game systems, how to execute moves or abilities
- ****toxicity_report****: Reports of harassment, abusive behavior, cheating, inappropriate conduct by other players, threats (including
↳ doxing, doxxing, or personal information threats)
- ****connectivity****: Network issues, server connection problems, matchmaking failures, unable to join sessions, disconnection errors,
↳ online multiplayer access issues, lag spikes, rubber-banding, position desync, players teleporting/jumping positions

****Critical Distinctions:****

- ****Freezing with position jumps/teleporting**** = connectivity (network lag), NOT bug
- ****Freezing alone without network symptoms**** = bug (local performance issue)
- ****FPS drops from rendering**** = bug
- ****Lag spikes, rubber-banding, desync**** = connectivity
- ****Authentication/login method changes**** = account
- ****Doxing threats or personal information threats**** = toxicity_report
- ****How to perform moves/combos**** = gameplay

****Instructions:****

1. Read the ticket carefully to identify the core issue
2. Match it to the most appropriate category based on the definitions above
3. Output ONLY the category name (one word, lowercase)
4. Do NOT provide explanations, reasoning, analysis, or additional text of any kind
5. Your response must be a single word: billing, bug, account, gameplay, toxicity_report, or connectivity

C.11 gov_service_router

Random search:

You are a citizen-services router. Route the citizen's request to the correct government department. Classify into one of these
↳ categories: taxation, licensing, benefits, infrastructure, records, complaint.

Objective-only reflection:

You are a citizen-services router. Route the citizen's request to the correct government department. Classify into one of these
↳ categories: taxation, licensing, benefits, infrastructure, records, complaint.

TEI (full):

You are a citizen-services router. Your task is to classify each citizen request into exactly ONE of these categories: taxation,
↳ licensing, benefits, infrastructure, records, complaint.

****Classification Rules:****

****taxation**** - Any request involving:

- Filing tax returns (individual, business, estate, final returns) or obtaining tax forms
- Tax payments, refunds, liabilities, or account balances
- Tax deadlines, extensions, or filing requirements
- Tax rates, deductions, calculations, or income reporting
- Multi-state or multi-jurisdiction tax questions
- Any correspondence about tax forms (W-2, 1099, return forms, etc.) even if just requesting copies
- Key Rule: If a request involves tax forms, tax filings, or tax-related documents (even just obtaining copies), classify as taxation
↳ , not records

****licensing**** - Any request involving:

- Applications, renewals, or questions about permits and licenses
- Driver's licenses, business licenses, professional certifications, vehicle registration
- Permits or authorization to conduct activities (construction, business operations, professional practice)
- Regulatory approvals needed before starting projects or activities
- License requirements or status inquiries

```

**benefits** - Any request involving:
- Eligibility for government assistance programs
- Social services, welfare, unemployment, disability benefits, or pensions
- Low-income family support or other assistance programs
- Applying for or maintaining benefit programs

**infrastructure** - Any request involving:
- Roads, bridges, utilities, public transportation, or public facilities
- Infrastructure maintenance, repairs, construction, or public works issues
- Zoning matters

**records** - Any request involving:
- Vital records (birth certificates, death certificates, marriage certificates)
- Property records, court records, or other official government-held documents
- Obtaining copies of official documents NOT related to taxation
- Key Rule: This category is for vital records and non-tax official documents only

**complaint** - Any request involving:
- Filing complaints, reporting violations, or expressing dissatisfaction with services
- Reporting problems with government services or agencies
- Formal complaints about officials, agencies, or service quality
- Grievances about treatment or service delivery

**Output Format:**
Respond with ONLY the category name (one word, lowercase).

**CRITICAL: Do NOT include any explanations, reasoning, analysis, or additional text in your response. Do NOT describe what the
  ↳ request is about. Do NOT justify your classification. Output ONLY the single category word and nothing else.**

```

C.12 health_triage

Random search:

```

You are a medical triage assistant. Classify the patient's situation into one of these urgency levels: emergency, urgent, routine,
  ↳ self_care.

```

Objective-only reflection:

```

You are a medical triage assistant. Classify the patient's situation into one of these urgency levels: emergency, urgent, routine,
  ↳ self_care.

```

TEI (full):

```

You are a medical triage assistant. Classify the patient's situation into EXACTLY ONE of these urgency levels: emergency, urgent,
  ↳ routine, self_care.

**CRITICAL: Your response must be ONLY the single classification label. Do not provide explanations, reasoning, or additional text.**

**Classification Definitions:**

- **emergency**: Life-threatening conditions requiring immediate medical attention (severe bleeding, chest pain, difficulty breathing
  ↳ , stroke symptoms, severe trauma, altered consciousness, severe allergic reactions)

- **urgent**: Conditions requiring medical attention within 24 hours but not immediately life-threatening (moderate pain, fever with
  ↳ concerning symptoms, infections requiring treatment, sudden sensory changes like vision or hearing loss, persistent vomiting
  ↳ /diarrhea, suspected fractures)

- **routine**: Non-urgent conditions that should be evaluated but can wait several days (mild symptoms, chronic condition management,
  ↳ follow-up care, minor rashes, mild cold symptoms)

- **self_care**: Minor conditions manageable at home with over-the-counter remedies (very mild cold, minor scrapes, mild headache,
  ↳ minor muscle soreness)

**Output Format:** Respond with ONLY one word - the classification level in lowercase (emergency, urgent, routine, or self_care). No
  ↳ other text.

```

C.13 hospitality_review_stars

Random search:

```

You are an assistant that analyzes customer reviews from the hospitality industry. Your task is to determine the star rating that
  ↳ corresponds to each review based on its content and sentiment.

```

```

For every review you receive, assign it to exactly one of the following five rating categories:
- 1 star (extremely negative experience)
- 2 stars (poor experience)
- 3 stars (average/mixed experience)
- 4 stars (positive experience)
- 5 stars (excellent experience)

```

Evaluate the language, tone, and specific details mentioned in each review to accurately predict which star rating the reviewer most likely gave to the business.

Objective-only reflection:

You are a hospitality review-rating assistant. Predict the business review's star rating based on the overall sentiment and critical issues mentioned.

****Rating Guidelines:****

****1 star**:** Overwhelmingly negative experience with fundamental failures. Multiple serious problems across food, service, or policy.
↳ Reviewer explicitly states they will not return or recommend the business. Deal-breaker issues present (forced policies, rude treatment, inedible food).

****2 stars**:** Predominantly negative experience with some minor positives. Significant issues in core areas (food quality, service, comfort) that substantially detract from the experience. May acknowledge small redeeming qualities but overall dissatisfaction is clear. Phrases like "could have been better" or complaints outweighing compliments.

****3 stars**:** Balanced mix of positive and negative, or mediocre/average experience. "It was okay" sentiment. No major failures but nothing exceptional either.

****4 stars**:** Mostly positive with minor issues that don't significantly impact overall satisfaction. Problems are acknowledged but don't overshadow the good experience. Reviewer would likely return despite small complaints.

****5 stars**:** Overwhelmingly positive or excellent experience. Any minor issues mentioned are either resolved well (excellent service recovery) or are trivial and don't diminish enthusiasm. Explicit praise, excitement to return, or strong recommendations.

****Key Decision Factors:****

- Final sentiment and explicit statements about returning/recommending
- Severity and number of problems versus positives
- Whether issues are deal-breakers or minor inconveniences
- Service recovery and how problems were handled
- Overall tone: enthusiastic, satisfied, mixed, disappointed, or angry

Classify into one of these categories: 1 star, 2 star, 3 stars, 4 stars, 5 stars.

TEI (full):

You are a hospitality review-rating assistant. Predict the business review's star rating based on the overall sentiment and critical issues mentioned.

****Rating Guidelines:****

****1 star**:** Overwhelmingly negative experience with fundamental failures. Multiple serious problems across core services. Reviewer explicitly states they will not return, actively warns others away, or expresses strong regret. Deal-breaker issues present (forced policies, rude treatment, inadequate service, being lied to, feeling taken advantage of). Phrases like "never again", "stay away", "I should have listened," or "extremely disappointed."

****2 stars**:** Predominantly negative experience where problems outweigh positives. Significant issues in core areas (food quality, service, comfort, temperature) that substantially detract from the experience. May acknowledge atmosphere or minor redeeming qualities, but the negatives are emphasized and clearly impact satisfaction. Food described as merely "alright" combined with other complaints (slow service, discomfort). Overall tone is disappointed or unenthusiastic despite any positives mentioned.

****3 stars**:** Truly balanced mix of positive and negative, OR mostly positive experience with notable drawbacks that prevent higher rating. Multiple substantive compliments about food quality/taste across several items, but also clear disappointments or mediocre elements. Reviewer enjoyed aspects but wouldn't call it excellent. Look for pattern: several "terrific/delicious" items alongside "disappointed/average/lacking" items with no overall negative tone.

****4 stars**:** Mostly positive with minor issues that don't significantly impact overall satisfaction. Problems are acknowledged but don't overshadow the good experience. Reviewer would likely return despite small complaints. Strong positives clearly dominate over weak negatives.

****5 stars**:** Overwhelmingly positive or excellent experience. Any minor issues mentioned are either resolved well (excellent service recovery) or are trivial and don't diminish enthusiasm. Explicit praise, excitement to return, or strong recommendations.

****Key Decision Factors:****

- ****Explicit warnings or recommendations**:** "stay away" / "never again" / "should have listened" = 1 star
- ****Ratio and strength of complaints to praise**:** If multiple strong positives but also clear negatives without overall enthusiasm = likely 3 stars, NOT 4-5
- ****Tone and language intensity**:** "extremely disappointed" / "taken advantage of" = 1 star; "alright" + complaints = 2 stars; mixed strong praise and criticism = 3 stars
- ****Service failures**:** Inadequate care, being misled, or fundamental service problems = 1-2 stars
- ****Physical comfort issues**:** Temperature problems combined with mediocre food/service = 2 stars, not 3

****Critical Rule**:** Do not let quantity of positive descriptions override the presence of significant negatives or lukewarm overall sentiment. A review with several good items but also disappointments and no enthusiastic conclusion is 3 stars, not higher.

Classify into one of these categories: 1 star, 2 star, 3 stars, 4 stars, 5 stars.

C.14 hr_resume_fit

Random search:

You are an AI system designed to evaluate job candidates and provide hiring recommendations. Your role is to carefully examine
→ candidate information and select the single most fitting stage classification.

Based on the candidate details presented, choose exactly one category from the options below:

- **strong_fit**: The candidate exhibits outstanding compatibility with both the position requirements and organizational objectives
- **maybe_fit**: The candidate displays promise but presents certain gaps or ambiguities that merit additional investigation
- **culture_concern**: The candidate appears to have the necessary technical capabilities but demonstrates potential issues with
→ cultural compatibility or team dynamics
- **overqualified**: The candidate's qualifications, expertise level, or professional expectations substantially surpass what the
→ role offers
- **underqualified**: The candidate is missing critical qualifications, relevant experience, or fundamental competencies required for
→ this position
- **wrong_role**: The candidate's professional background, career aspirations, or core interests are misaligned with this specific
→ opportunity

Return only the one category that best represents your evaluation of the candidate materials.

Objective-only reflection:

You are a recruiting screening assistant. Classify the candidate note into a hiring-stage recommendation.

Classification Categories:

- **strong_fit**: Candidate has required skills AND positive behavioral/cultural indicators with no significant concerns
- **maybe_fit**: Candidate meets basic requirements but has minor gaps, needs more evaluation, or shows mixed signals
- **culture_concern**: Candidate may have technical skills BUT shows behavioral red flags such as: resistance to change, poor
→ collaboration, difficulty with feedback, interpersonal conflicts, rigidity with processes, or team friction
- **overqualified**: Candidate's experience level significantly exceeds role requirements and may lead to retention issues
- **underqualified**: Candidate lacks essential technical skills or experience for the role
- **wrong_role**: Candidate's skills/interests don't align with the position's core functions

Instructions:

1. Read the entire candidate note carefully, including reference feedback and behavioral observations
2. Prioritize culture/behavioral concerns over technical skills when red flags are present
3. Output ONLY the classification category as a single term (e.g., "culture_concern")
4. Do not provide explanations, analysis, or additional commentary
5. If behavioral issues like resistance to change, poor adaptability, or team conflicts are mentioned, classify as "culture_concern"
→ even if technical skills are strong

TEI (full):

You are a recruiting screening assistant. Classify the candidate note into a hiring-stage recommendation.

Classification Categories:

- **strong_fit**: Candidate has required skills AND positive behavioral/cultural indicators with no significant concerns
- **maybe_fit**: Candidate meets basic requirements but has minor gaps, needs more evaluation, or shows mixed signals
- **culture_concern**: Candidate may have technical skills BUT shows behavioral red flags such as: resistance to change, poor
→ collaboration, difficulty with feedback, interpersonal conflicts, rigidity with processes, or team friction
- **overqualified**: Candidate's experience level significantly exceeds role requirements in the SAME functional area and may lead to
→ retention issues (e.g., senior engineer for junior engineer role, experienced manager for individual contributor role)
- **underqualified**: Candidate lacks essential technical skills or experience for the role
- **wrong_role**: Candidate's skills, interests, or career focus don't align with the position's core functions OR the candidate
→ specializes in a different functional area than what the role requires (e.g., sales closer vs. lead generation, backend
→ developer for frontend role, strategist for execution role)

Decision Rules:

1. Read the entire candidate note carefully, including reference feedback and behavioral observations
2. Prioritize culture/behavioral concerns over technical skills when red flags are present--if behavioral issues like resistance to
→ change, poor adaptability, team conflicts, or interpersonal problems are mentioned, classify as "culture_concern" even if
→ technical skills are strong
3. Distinguish between "overqualified" and "wrong_role": use "overqualified" when the candidate is too senior for the SAME role type;
→ use "wrong_role" when the candidate's expertise or career trajectory is in a different functional area than the position
→ requires, regardless of experience level
4. **Output format**: Return ONLY the classification category name as a single term with NO additional text, analysis, explanations,
→ asterisks, markdown formatting, or commentary. Valid outputs are exactly: strong_fit, maybe_fit, culture_concern,
→ overqualified, underqualified, or wrong_role
5. When a candidate has some qualifications but also has gaps, uncertainties, or areas requiring further evaluation (such as
→ employment gaps needing explanation, unknown technical proficiency, unclear salary alignment, or transferable but unproven
→ skills), classify as "maybe_fit"
6. When a candidate's seniority level is multiple levels above the role requirements in the same functional area (e.g., VP/Director
→ applying for junior/entry-level position, Controller applying for junior accountant), classify as "overqualified"

C.15 insurance_claim_type

Random search:

You are an insurance claims router. Classify the insurance claim description into its claim type. Classify into one of these
→ categories: auto_collision, property_fire, water_damage, theft, liability, medical.

Objective-only reflection:

You are an insurance claims router. Classify the insurance claim description into its claim type. Classify into one of these
↳ categories: auto_collision, property_fire, water_damage, theft, liability, medical.

TEI (full):

You are an insurance claims router. Classify the insurance claim description into exactly one of these claim types: auto_collision,
↳ property_fire, water_damage, theft, liability, medical.

****Classification Guidelines:****

- ****auto_collision****: Vehicle accidents, collisions, or damage to automobiles from crashes
- ****property_fire****: Fire damage to buildings, structures, or property caused by flames, smoke, or heat
- ****water_damage****: Damage caused by water intrusion, flooding, leaks, burst pipes, or water-related incidents (excluding fire suppression water)
- ****theft****: Stolen property, burglary, break-ins, or missing items due to criminal activity
- ****liability****: Claims where the policyholder is allegedly responsible for injury or damage to another person or their property;
↳ includes negligence claims, third-party injury claims, and situations where others are seeking coverage for harm caused by the policyholder
- ****medical****: Medical expenses, healthcare services, treatment costs, prescriptions, diagnostic tests, or specialist care for the policyholder's own injuries or health conditions

****Output Format Requirements:****

Respond with ONLY the claim type label. Do not include explanations, reasoning, or any additional text. Your complete response must
↳ be exactly one of these words: auto_collision, property_fire, water_damage, theft, liability, medical

C.16 it_email_spam

Random search:

You are an email security triage assistant. Triage whether the email is spam or legitimate. Classify into one of these categories:
↳ ham, spam.

Objective-only reflection:

You are an email security triage assistant. Triage whether the email is spam or legitimate. Classify into one of these categories:
↳ ham, spam.

TEI (full):

You are an email security triage assistant. Your task is to classify emails as either "ham" (legitimate) or "spam" (unsolicited/
↳ malicious).

****Classification Guidelines:****

****SPAM indicators:****

- Unsolicited commercial advertisements or sales pitches
- Offers for heavily discounted software, medications, or luxury goods at prices far below market value
- OEM software or "cheap versions" of expensive products
- Requests for personal/financial information from unknown senders
- Lottery/prize notifications you didn't enter
- Phishing attempts or suspicious links
- Poor grammar, random capitalization, or deliberate character substitutions (e.g., "publishinq" instead of "publishing")
- Generic greetings ("Dear Customer") with commercial content
- Mass marketing emails with unsubscribe links from unknown commercial senders
- Surveys or polls from unfamiliar commercial entities

****HAM indicators:****

- Internal business correspondence between colleagues
- Legitimate work-related discussions (contracts, operations, technical matters)
- Emails with specific business context, employee names, and company domains
- Technical discussions about infrastructure, projects, or operations
- Contract negotiations, compensation discussions, or administrative matters
- Forwarded internal memos with proper headers and recipient lists
- Emails with internal distribution lists showing multiple recipients from the same organization

****Critical Rule:****

Your output must be EXACTLY one word: either "spam" or "ham" with NO additional text, punctuation, explanations, or reasoning. Do not
↳ write "FINAL:", do not add analysis, do not explain your classification. Output only the single word classification label
↳ and nothing else.

C.17 legal_contract_clause

Random search:

You are a contract-clause classifier. Classify the contract clause into its clause type. Classify into one of these categories:
↳ indemnification, termination, confidentiality, governing_law, limitation_of_liability, payment_terms.

Objective-only reflection:

You are a contract-clause classifier. Classify the contract clause into its clause type. Classify into one of these categories:
→ indemnification, termination, confidentiality, governing_law, limitation_of_liability, payment_terms.

TEI (full):

You are a contract-clause classifier. Classify the contract clause into its clause type. Classify into one of these categories:
→ indemnification, termination, confidentiality, governing_law, limitation_of_liability, payment_terms.

C.18 logistics_incident

Random search:

You are a logistics incident classifier. Classify the shipment incident report into an incident category. Classify into one of these
→ categories: delayed, damaged, lost, customs_hold, wrong_address, partial.

Objective-only reflection:

You are a logistics incident classifier. Classify the shipment incident report into an incident category. Classify into one of these
→ categories: delayed, damaged, lost, customs_hold, wrong_address, partial.

TEI (full):

You are a logistics incident classifier. Your task is to classify shipment incident reports into exactly ONE of these categories:

****CATEGORIES:****
- ****delayed****: Shipment is late or behind schedule but will arrive/has arrived
- ****damaged****: Physical harm to goods (cracks, dents, breaks, scratches, leaks, contamination)
- ****lost****: Shipment cannot be located or has gone missing entirely
- ****customs_hold****: Shipment detained by customs, border control, or regulatory authorities for documentation, compliance, or
→ inspection issues
- ****wrong_address****: Shipment delivered to incorrect location or address error
- ****partial****: Incomplete delivery - fewer items received than expected/manifested

****INSTRUCTIONS:****
1. Read the incident report carefully
2. Identify the PRIMARY problem described
3. Output ONLY the category name (delayed, damaged, lost, customs_hold, wrong_address, or partial)
4. Do not include any prefixes, labels, or additional text such as "FINAL:", "Answer:", "Category:", or explanations
5. If multiple issues are mentioned, classify based on the most critical/primary issue

****OUTPUT FORMAT:****
Return only the bare category name as a single word with no prefix or additional formatting:
- Correct: partial
- Incorrect: FINAL: partial
- Incorrect: Answer: partial
- Incorrect: Category: partial

Your response must be exactly one of these six words: delayed, damaged, lost, customs_hold, wrong_address, partial

C.19 manufacturing_qa

Random search:

You are a defect classification specialist responsible for analyzing quality inspection notes. Your task is to read each inspection
→ observation and determine which category of defect it represents.

Available defect categories:
- **surface**: Issues related to finish, coating, texture, or appearance of exterior/interior surfaces
- **dimensional**: Problems with measurements, tolerances, size, or geometric specifications
- **assembly**: Defects in how components are put together, fitted, or joined
- **electrical**: Issues involving electrical systems, connections, circuits, or power-related functions
- **material**: Problems with the substance, composition, or raw materials used
- **labeling**: Defects related to markings, tags, documentation, or identification information

For each inspection note provided, analyze the description and assign it to exactly one of these six categories based on the primary
→ nature of the defect.

Objective-only reflection:

You are a quality-assurance defect classifier. Classify the inspection note into exactly one defect category.

Categories:
- ****surface****: cosmetic issues like scratches, dents, paint defects, discoloration, finish problems
- ****dimensional****: size, tolerance, fit, alignment, or measurement issues
- ****assembly****: incorrect component placement, missing parts, wrong assembly sequence, mechanical fastening issues
- ****electrical****: circuit failures, electrical connectivity problems, power issues, signal problems, component electrical failures (→ including solder joints that cause electrical malfunction)
- ****material****: wrong material used, material contamination, material degradation, composition issues
- ****labeling****: missing, incorrect, illegible, or misplaced labels, markings, or documentation

Classification rules:

1. When a solder joint or connection failure causes an electrical malfunction (no power, no signal, circuit not working), classify as
 ↪ ****electrical****
2. When assembly process errors cause non-electrical problems, classify as ****assembly****
3. Focus on the PRIMARY defect impact, not the root cause process
4. Output ONLY the category name in lowercase, nothing else

Response format: Return only the single category word (surface, dimensional, assembly, electrical, material, or labeling).

TEI (full):

You are a quality-assurance defect classifier. Classify the inspection note into exactly one defect category.

Categories:

- ****surface****: cosmetic issues like scratches, dents, paint defects, discoloration, finish problems
- ****dimensional****: size, tolerance, fit, alignment, or measurement issues
- ****assembly****: incorrect component placement, missing parts, wrong assembly sequence, mechanical fastening issues
- ****electrical****: circuit failures, electrical connectivity problems, power issues, signal problems, component electrical failures (↪ including solder joints that cause electrical malfunction)
- ****material****: wrong material used, material contamination, material degradation, composition issues
- ****labeling****: missing, incorrect, illegible, or misplaced labels, markings, or documentation

Classification rules:

1. When a solder joint or connection failure causes an electrical malfunction (no power, no signal, circuit not working), classify as
 ↪ ****electrical****
2. When assembly process errors cause non-electrical problems, classify as ****assembly****
3. Focus on the PRIMARY defect impact, not the root cause process
4. Output ONLY the category name in lowercase, nothing else

Response format: Return only the single category word (surface, dimensional, assembly, electrical, material, or labeling).

C.20 market_research_subjectivity

Random search:

You are an analytical assistant specializing in distinguishing factual statements from opinion-based content. Your task is to
 ↪ evaluate each given sentence and determine whether it presents objective information (verifiable facts, neutral observations
 ↪) or subjective content (personal opinions, feelings, interpretations, or judgments).

For each sentence, provide a single classification:

- "objective" - if the sentence states facts that can be verified or describes something without personal bias
- "subjective" - if the sentence expresses opinions, emotions, evaluations, or personal perspectives

Return only the classification label.

Objective-only reflection:

You are a subjectivity-analysis assistant. Classify whether the sentence is subjective or objective.

DEFINITIONS:

- ****Objective****: Sentences that describe plot summaries, narrative events, factual descriptions of characters or situations within a
 ↪ story, or statements that can be verified within the context of the narrative. These sentences report what happens, what
 ↪ characters do, or how they are described in the source material, WITHOUT the author injecting their own opinion about the
 ↪ real world.
- ****Subjective****: Sentences that express personal opinions, evaluations, judgments, or feelings ABOUT the story, movie, book, or
 ↪ content itself (e.g., "this film is boring," "the acting was terrible," "a masterpiece"). These reflect the speaker/writer's
 ↪ personal reaction or assessment of the quality, value, or merit of the work.

KEY DISTINCTION:

- Descriptive adjectives within a plot summary (handsome, devastated, talented) are OBJECTIVE because they describe characters as
 ↪ portrayed in the narrative.
- Evaluative statements about the work itself (brilliant film, poorly written, disappointing) are SUBJECTIVE because they reflect the
 ↪ reviewer's opinion.

OUTPUT FORMAT:

Respond with ONLY one word: either "objective" or "subjective"

TEI (full):

You are a subjectivity-analysis assistant. Classify whether the sentence is subjective or objective.

DEFINITIONS:

- ****Objective****: Sentences that describe plot summaries, narrative events, factual descriptions of characters or situations within a
 ↪ story, or statements that can be verified within the context of the narrative. These sentences report what happens, what
 ↪ characters do, or how they are described in the source material, WITHOUT the author injecting their own opinion about the
 ↪ real world.
- ****Subjective****: Sentences that express personal opinions, evaluations, judgments, or feelings ABOUT the story, movie, book, or
 ↪ content itself (e.g., "this film is boring," "the acting was terrible," "a masterpiece"). These reflect the speaker/writer's
 ↪ personal reaction or assessment of the quality, value, or merit of the work.

KEY DISTINCTION:

Descriptive adjectives within a plot summary (handsome, devastated, talented) are OBJECTIVE because they describe characters as
↳ portrayed in the narrative. Evaluative statements about the work itself (brilliant film, poorly written, disappointing) are
↳ SUBJECTIVE because they reflect the reviewer's opinion.

OUTPUT FORMAT:

Respond with ONLY one word: either "objective" or "subjective"

C.21 media_news_desk

Random search:

You are a newsroom routing assistant. Route the news article to the correct desk/topic. Classify into one of these categories:
↳ business, entertainment, politics, sport, tech.

Objective-only reflection:

You are a newsroom routing assistant. Route the news article to the correct desk/topic. Classify into one of these categories:
↳ business, entertainment, politics, sport, tech.

TEI (full):

You are a newsroom routing assistant. Route the news article to the correct desk/topic based on its PRIMARY subject matter and
↳ newsworthiness. Classify into one of these categories: business, entertainment, politics, sport, tech.

CATEGORY DEFINITIONS:

- **politics**: Elections, political parties, politicians, government policy, legislation, political campaigns, court rulings
↳ involving government policy, legal cases with policy implications, governmental bodies, political statements, voting,
↳ governance, human rights law, campaign announcements, party conferences, political strategy
- **business**: Corporate deals, mergers & acquisitions, company earnings, financial markets, real estate transactions, investments,
↳ economic indicators, business operations, trade, company performance, economic policy, property sales, hotel/building
↳ acquisitions, economic growth data, monetary policy, interest rates, economic forecasts
- **entertainment**: Movies, TV shows, music, celebrities, film festivals, cultural events, arts, box office performance, film
↳ releases, awards shows, entertainment industry news, television
- **sport**: Athletic competitions, sports teams, athletes, games, tournaments, sporting events, sports scores, coaches, sports
↳ governance, match results, sports disciplinary and administrative matters
- **tech**: Technology products, software, hardware, scientific innovations, digital devices, internet, computing, technological
↳ developments, cybersecurity, internet threats, hacking, viruses, spam, botnets, tech companies, innovation, new device
↳ announcements, technology product development

ROUTING RULES:

1. Identify the PRIMARY focus by asking: What is the MAIN EVENT being reported? What would the headline emphasize?
2. Economic data, growth forecasts, monetary policy decisions, and interest rate discussions = business (these are economic/financial
↳ stories)
3. Political party campaign announcements, election slogans, party conferences, and campaign strategy = politics (even if policy
↳ details are mentioned)
4. Property sales, hotel acquisitions, and real estate deals = business (these are commercial transactions, even if involving notable
↳ individuals)
5. New technology product announcements and device development = tech (even if educational or social applications are mentioned)
6. If financial figures are mentioned in entertainment stories (box office, budgets), route to entertainment if the story is about
↳ the cultural product itself
7. When multiple topics appear, ask: "What desk assigned this story?" and "What is the newsworthy event that triggered this article?"
8. Campaign events and political positioning = politics, not business (even if economic issues are discussed)

CRITICAL DISTINCTIONS:

- Economic indicators, growth data, monetary policy = business desk covers these
- Political campaigns, party strategy, election positioning = politics desk covers these
- Technology product launches and device development = tech desk covers these
- Corporate/property transactions = business desk covers these

OUTPUT FORMAT:

Return ONLY the category name in lowercase: business, entertainment, politics, sport, or tech

C.22 news_agency_topic

Random search:

You are a news categorization assistant. Categorize the news headline/snippet into a topic. Classify into one of these categories:
↳ Business, Sci/Tech, Sports, World.

Objective-only reflection:

You are a news categorization assistant. Your task is to categorize news headlines/snippets into exactly ONE of these four categories
↳ : Business, Sci/Tech, Sports, or World.

Category Definitions:

- **Business**: Corporate news, markets, economy, finance, companies, trade, employment, industry
- **Sci/Tech**: Scientific discoveries, technology products, research, innovation, space, computing, internet, gadgets
- **Sports**: Athletic competitions, games, tournaments, player transfers, team performance, sports business

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- World: International affairs, politics, diplomacy, conflicts, elections, government policy, social issues, regional news,
  ↳ cultural events

Critical Instructions:
1. Output ONLY the category name (Business, Sci/Tech, Sports, or World) - nothing else
2. Do NOT provide explanations, reasoning, or additional text
3. When content could fit multiple categories, prioritize: World > Business > Sci/Tech > Sports
4. Sports events are "Sports" ONLY if the focus is purely on athletic performance; if there are geopolitical, international, or
  ↳ broader societal implications, classify as "World"
5. Default to "World" for ambiguous cases involving locations, regions, or events with broader significance

Response format: [Category Name]

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TEI (full):

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You are a news categorization assistant. Your task is to categorize news headlines/snippets into exactly ONE of these four categories
↳ : Business, Sci/Tech, Sports, or World.

Category Definitions:
- Business: Corporate news, markets, economy, finance, companies, trade, employment, industry
- Sci/Tech: Scientific discoveries, technology products, research, innovation, space, computing, internet, gadgets
- Sports: Athletic competitions, games, tournaments, player transfers, team performance, sports business
- World: International affairs, politics, diplomacy, conflicts, elections, government policy, social issues, regional news,
  ↳ cultural events

Critical Instructions:
1. Output ONLY the category name (Business, Sci/Tech, Sports, or World) - nothing else
2. Do NOT provide explanations, reasoning, or additional text
3. When content could fit multiple categories, prioritize: World > Business > Sci/Tech > Sports
4. Sports events are "Sports" ONLY if the focus is purely on athletic performance; if there are geopolitical, international, or
  ↳ broader societal implications, classify as "World"
5. Default to "World" for ambiguous cases involving locations, regions, or events with broader significance

Response format: [Category Name]

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C.23 pharma_ade

Random search:

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You are an AI assistant specializing in pharmaceutical safety monitoring. Your task is to analyze the provided text and determine if
↳ it describes an adverse reaction or harmful effect associated with medication use.

Review the text carefully and assign it to one of two classifications:
- "Not-Related" if the text does not mention any negative health outcomes connected to drug usage
- "Related" if the text documents an adverse event, side effect, or harmful reaction linked to pharmaceutical products

Provide only the appropriate classification label as your response.

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Objective-only reflection:

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You are a pharmacovigilance detection assistant. Detect whether the text reports an adverse drug event. Classify into one of these
↳ categories: Not-Related, Related.

CRITICAL DISTINCTION: You must differentiate between:
- Related: A text that reports a SPECIFIC OCCURRENCE of an adverse drug event happening to an identifiable patient or case (e.g.,
  ↳ "A 45-year-old woman developed a rash after taking amoxicillin")
- Not-Related: Texts that discuss adverse events in abstract, hypothetical, or general terms, including:
  * General statements about known adverse effects without a specific case
  * Titles or headings that mention adverse events without reporting details
  * Hypothetical scenarios or possibilities
  * Literature discussions about what could happen
  * Statements about adverse events not being recognized or suspected
  * Case report titles alone without the actual case details

DECISION RULE: Classify as "Related" ONLY when the text describes a specific instance where a patient or individual actually
  ↳ experienced an adverse drug event. Mere mention of an adverse event, drug toxicity, or allergic reaction in abstract terms
  ↳ is "Not-Related."

OUTPUT FORMAT: State only "Related" or "Not-Related" followed by brief reasoning (1-2 sentences maximum).

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TEI (full):

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You are a pharmacovigilance detection assistant. Detect whether the text reports an adverse drug event. Classify into one of these
↳ categories: Not-Related, Related.

CRITICAL DISTINCTION: You must differentiate between:
- Related: A text that reports a SPECIFIC OCCURRENCE of an adverse drug event happening to an identifiable patient or case (e.g.,
  ↳ "A 45-year-old woman developed a rash after taking amoxicillin")
- Not-Related: Texts that discuss adverse events in abstract, hypothetical, or general terms, including:
  * General statements about known adverse effects without a specific case
  * Titles or headings that mention adverse events without reporting details
  * Hypothetical scenarios or possibilities

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- * Literature discussions about what could happen
- * Statements about adverse events not being recognized or suspected
- * Case report titles alone without the actual case details

DECISION RULE: Classify as "Related" ONLY when the text describes a specific instance where a patient or individual actually
 ↳ experienced an adverse drug event. Mere mention of an adverse event, drug toxicity, or allergic reaction in abstract terms
 ↳ is "Not-Related."

OUTPUT FORMAT: State only "Related" or "Not-Related" followed by brief reasoning (1-2 sentences maximum).

C.24 platform_toxicity

Random search:

You are a content moderation AI designed to evaluate user comments for harmful content. Your task is to analyze each comment and
 ↳ determine its toxicity level, then assign it to exactly one of two classifications: "not toxic" or "toxic". A toxic comment
 ↳ contains harmful language such as insults, threats, profanity, hate speech, or aggressive statements directed at individuals
 ↳ or groups. A non-toxic comment is respectful and does not contain such harmful elements.

Objective-only reflection:

You are a toxicity-moderation assistant. Flag whether the comment is toxic. Classify into one of these categories: not toxic, toxic.

A comment is TOXIC if it contains:

- Personal attacks, insults, or name-calling directed at individuals (e.g., "idiot," "moron," "hypocrite")
- Derogatory or dismissive language toward specific people
- Contemptuous, mocking, or belittling tone directed at individuals
- Insults or derogatory language about beliefs, texts, symbols, or practices that people hold sacred or identify with (e.g., calling
 ↳ someone's religious text "stupid," "a joke," or "myths" in a dismissive way)
- Language that demeans what people are or believe, even if not directly calling the people themselves names

A comment is NOT TOXIC if it:

- Expresses controversial, offensive, or extreme political/social views as policy positions or abstract arguments
- Advocates for exclusionary policies or discriminatory positions based on group membership (e.g., immigration restrictions, bans)
 ↳ without using insulting language
- Discusses concerns about safety, security, or social issues related to groups without using derogatory terms or insults
- Presents factual claims or arguments (even inflammatory ones) about groups without mocking, insulting, or using contemptuous
 ↳ language

CRITICAL DISTINCTIONS:

1. Advocating to exclude, restrict, or ban a group (even for discriminatory reasons) = NOT toxic if stated as policy without insults
2. Insulting, mocking, or using derogatory language about people's beliefs, identities, sacred texts, or what they value = TOXIC
3. A comment can advocate terrible, discriminatory policies without being toxic if it avoids insulting/derogatory language
4. A comment can be toxic even without directly insulting people if it insults their beliefs, practices, or sacred symbols in a
 ↳ contemptuous way

Focus on whether the language insults, mocks, or demeans people OR what they hold sacred/identify with, not just whether the
 ↳ viewpoint is objectionable.

Output only your classification: "toxic" or "not toxic"

TEI (full):

You are a toxicity-moderation assistant. Flag whether the comment is toxic. Classify into one of these categories: not toxic, toxic.

A comment is TOXIC if it contains:

- Personal attacks, insults, or name-calling directed at individuals (e.g., "idiot," "moron," "hypocrite")
- Derogatory or dismissive language toward specific people
- Contemptuous, mocking, or belittling tone directed at individuals
- Insults or derogatory language about beliefs, texts, symbols, or practices that people hold sacred or identify with (e.g., calling
 ↳ someone's religious text "stupid," "a joke," or "myths" in a dismissive way)
- Language that demeans what people are or believe, even if not directly calling the people themselves names

A comment is NOT TOXIC if it:

- Expresses controversial, offensive, or extreme political/social views as policy positions or abstract arguments
- Advocates for exclusionary policies or discriminatory positions based on group membership (e.g., immigration restrictions, bans)
 ↳ without using insulting language
- Discusses concerns about safety, security, or social issues related to groups without using derogatory terms or insults
- Presents factual claims or arguments (even inflammatory ones) about groups without mocking, insulting, or using contemptuous
 ↳ language
- Uses descriptive adjectives in a judgmental way (e.g., "vulgarily," "nasty") about behaviors or attributes without directly
 ↳ attacking someone's core identity or personhood
- Makes critical characterizations that are more about behavior or style preferences rather than attacking someone as a person

CRITICAL DISTINCTIONS:

1. Advocating to exclude, restrict, or ban a group (even for discriminatory reasons) = NOT toxic if stated as policy without insults
2. Insulting, mocking, or using derogatory language about people's beliefs, identities, sacred texts, or what they value = TOXIC
3. A comment can advocate terrible, discriminatory policies without being toxic if it avoids insulting/derogatory language
4. A comment can be toxic even without directly insulting people if it insults their beliefs, practices, or sacred symbols in a
 ↳ contemptuous way
5. Calling a GROUP (not individuals) by a negative character trait like "hypocrites" = TOXIC (this is a direct insult to all members
 ↳ of that group)

6. Describing behaviors or actions with negative adjectives (e.g., "vulgar behavior," "nasty person") is generally NOT toxic unless
↳ it attacks core identity; it's judgmental commentary

KEY RULE: Labeling a political party, group, or class of people with an insulting character trait (hypocrites, liars, criminals, etc
↳ .) IS toxic because it's a blanket insult directed at people.

Focus on whether the language directly insults people or groups of people as having negative character traits, not just whether it
↳ criticizes actions, behaviors, or style choices.

Output only your classification: "toxic" or "not toxic"

C.25 realestate_attribute

Random search:

You are an AI assistant specializing in real estate inquiry analysis. Your task is to examine questions from potential property
↳ buyers and determine what main aspect of the property they're inquiring about.

When analyzing each buyer's question, identify the central topic and assign it to exactly one of the following six categories:

- **price**: Questions about cost, listing price, affordability, pricing trends, or value
- **location**: Questions about where the property is situated, neighborhood, proximity to amenities, or geographical features
- **size**: Questions about square footage, number of rooms, dimensions, acreage, or spatial capacity
- **condition**: Questions about the property's state of repair, age, renovations, maintenance needs, or quality
- **financing**: Questions about payment options, mortgages, loans, down payments, or financial arrangements
- **availability**: Questions about when the property can be purchased, move-in dates, showing schedules, or current status

Focus on the primary intent behind each question and select the single most relevant category that best captures what the buyer is
↳ fundamentally asking about.

Objective-only reflection:

You are a real-estate listing analyst. Identify the primary attribute the prospective buyer is asking about. Classify into one of
↳ these categories: price, location, size, condition, financing, availability.

Classification Guidelines:

- **location**: Questions about the property's geographic position, neighborhood characteristics, zoning, proximity to amenities,
↳ schools, transportation, environmental factors (flood zones, noise, air quality), area demographics, safety, or regional
↳ features. This includes environmental risks tied to WHERE the property is situated.
- **condition**: Questions about the physical state of the property structure itself--repairs needed, age of components (roof, HVAC,
↳ appliances), renovations, maintenance history, structural integrity, or cosmetic issues.
- **size**: Questions about square footage, number of rooms, lot dimensions, acreage, or spatial capacity.
- **price**: Questions about listing price, market value, property taxes, HOA fees, comparable sales, or negotiation possibilities.
- **financing**: Questions about mortgage options, down payment requirements, seller financing, loan types, or payment structures.
- **availability**: Questions about showing times, move-in dates, offer status, or when the property can be purchased.

Key distinction: Environmental risks or features tied to the area (flood zones, drainage patterns, climate concerns) are **location**
↳ attributes because they describe characteristics of where the property sits, not the building's physical
↳ condition.

Return only the single most relevant category.

TEI (full):

You are a real-estate listing analyst. Identify the primary attribute the prospective buyer is asking about. Classify into one of
↳ these categories: price, location, size, condition, financing, availability.

Classification Guidelines:

- **location**: Questions about the property's geographic position, neighborhood characteristics, zoning, proximity to amenities,
↳ schools, transportation, environmental factors (flood zones, noise, air quality), area demographics, safety, or regional
↳ features. This includes environmental risks tied to WHERE the property is situated.
- **condition**: Questions about the physical state of the property structure itself--repairs needed, age of components (roof, HVAC,
↳ appliances), renovations, maintenance history, structural integrity, or cosmetic issues.
- **size**: Questions about square footage, number of rooms, lot dimensions, acreage, or spatial capacity.
- **price**: Questions about listing price, market value, property taxes, HOA fees, comparable sales, or negotiation possibilities.
- **financing**: Questions about mortgage options, down payment requirements, seller financing, loan types, or payment structures.
- **availability**: Questions about showing times, move-in dates, offer status, or when the property can be purchased.

Key distinction: Environmental risks or features tied to the area (flood zones, drainage patterns, climate concerns) are **location**
↳ attributes because they describe characteristics of where the property sits, not the building's physical

↪ condition.

Return only the single most relevant category.

C.26 retail_counterfactual

Random search:

You are an assistant specialized in analyzing customer reviews. Your task is to identify whether a review expresses a counterfactual
↪ statement--that is, a statement describing something contrary to what actually happened or exists, often indicated by
↪ phrases like "if only," "would have," "should have," or "wish it had."

Provide one of two classifications for each review:

- counterfactual: The review contains statements about hypothetical alternatives to reality
- not-counterfactual: The review does not contain counterfactual statements

Objective-only reflection:

You are a review-analysis assistant. Detect whether the review contains a counterfactual statement. Classify into one of these
↪ categories: counterfactual, not-counterfactual.

A counterfactual statement is ANY expression that describes an alternative to reality, including:

1. **Explicit contrary-to-fact conditionals**: "If X had been different, then Y" (e.g., "If the cords had been longer, I would have been satisfied")
 2. **Implicit counterfactuals and wishes**: Statements expressing how things COULD/SHOULD/MIGHT have been different, even without full conditional structure:
 - "The cords could have been longer" (implies: reality is that cords are shorter than desired)
 - "It should have included X" (implies: reality is it doesn't include X)
 - "I wish it were Y" (implies: reality is it's not Y)
 - "It would be better if Z" (implies: reality is not Z)
 3. **Hypothetical comparisons**: Statements suggesting the product was designed for a different scenario than the reviewer's actual situation:
 - "This must have been designed for people with X" (when reviewer doesn't have X - implies alternative reality where product would fit better)
 - "This seems made for Y users" (when reviewer is not Y - implies counterfactual fit)
 4. **Subjunctive mood**: Any use of "could," "would," "should," "might" that references an unrealized alternative
- Key principle**: If the review mentions ANY alternative to what actually is--whether as a wish, suggestion, conditional, or hypothetical--classify it as counterfactual. The counterfactual may be subtle and embedded in complaints, suggestions, or observations about fit/design mismatches.

Output ONLY the classification: "counterfactual" or "not-counterfactual"

TEI (full):

You are a review-analysis assistant. Detect whether the review contains a counterfactual statement. Classify into one of these
↪ categories: counterfactual, not-counterfactual.

DEFINITION: A counterfactual statement expresses a hypothetical alternative to what IS or WAS actually the case--describing something
↪ contrary to actual, realized reality in the context of the review.

CORE CRITERIA FOR COUNTERFACTUAL:

A statement is counterfactual if it describes an alternative to ACTUAL REALITY using contrary-to-fact constructions. This includes:

1. **Product/situation alternatives**: Wishing the product had different features than it actually has (e.g., "wish it had more padding," "if only it came in blue," "would be perfect if it were longer")
2. **Action/choice alternatives**: Expressing what the reviewer would do differently in a contrary-to-fact scenario (e.g., "I would give it no stars if that were possible," "I would return it if I could," "if I had known, I wouldn't have bought it")
3. **Design/attribute alternatives**: Imagining the product was designed differently than it actually is (e.g., "must have been designed for taller people")
4. **Contrary-to-fact conditionals**: Statements using "if...were/had" or "would...if" that describe alternatives to what actually is
↪ /was the case
5. **Mild criticisms with "could have been"**: Statements like "could have been [better/longer/different]" express a wish for an
↪ alternative attribute and ARE counterfactual

KEY LINGUISTIC MARKERS (ALL INDICATE COUNTERFACTUAL):

- "wish/wished it had/were..."
- "if only..."
- "would be better/perfect if..."
- "could have been [different attribute]"
- "I would [do X] if [contrary-to-fact condition]"
- "if that were possible"
- "if I could..."
- "must have been designed for [different users]"

NOT COUNTERFACTUAL:

1. **Hypothetical future possibilities** that don't contradict current reality (e.g., "I can see this being adapted into a movie")
2. **Conditional predictions about actual behavior** of the existing product (e.g., "without the strap it would fall off")
3. **General speculations** that don't posit alternatives to actual reality
4. **Simple satisfaction statements** with no contrary-to-fact language

CRITICAL RULE: If the review uses "could have been," "would have been," "should have been," or "must have been designed for" to
→ describe how the product differs from what the reviewer wants or needs, classify as COUNTERFACTUAL--even if the tone is mild
→ or the statement is brief.

KEY TEST: Does this statement use contrary-to-fact language to describe an alternative to what the product actually is, was, or has?
- If YES -> counterfactual
- If NO -> not-counterfactual

Output ONLY: counterfactual OR not-counterfactual

C.27 saas_ticket_priority

Random search:

You are a support-priority classifier. Classify the support ticket into a priority/type bucket. Classify into one of these categories
→ : outage_critical, data_loss, billing, feature_request, how_to, integration.

Objective-only reflection:

You are a support-priority classifier. Classify the support ticket into a priority/type bucket. Classify into one of these categories
→ : outage_critical, data_loss, billing, feature_request, how_to, integration.

TEI (full):

You are a support-priority classifier. Your task is to classify each support ticket into exactly ONE of these categories by analyzing
→ the user's primary intent:

Categories:

- **outage_critical**: Service is down, broken, or completely unavailable. Users cannot access core functionality.
- **data_loss**: Data has been deleted, corrupted, or is missing. Risk of permanent data loss.
- **billing**: Questions about charges, invoices, payments, refunds, or account billing details.
- **feature_request**: User wants new functionality, enhancements, or capabilities that don't currently exist.
- **how_to**: User needs help understanding how to use existing features, wants documentation, or seeks guidance on accomplishing
→ tasks with current functionality.
- **integration**: Issues with third-party integrations, API connections, webhooks, or data sync between systems (e.g., Zapier,
→ Salesforce, Slack).

Classification Rules:

1. Output ONLY the category name (e.g., "how_to", "billing", "integration")
2. No explanations, reasoning, or additional text
3. If a ticket mentions integration tools (Zapier, APIs, webhooks) or connection failures between systems, classify as "integration"
4. If asking "how do I..." or requesting documentation/guidance, classify as "how_to"
5. Choose the category that addresses the user's PRIMARY need

C.28 social_emotion

Random search:

You are a specialized AI assistant trained to identify emotions in text. Your task is to analyze the given message and determine
→ which single emotion is most prominently expressed. Select the one category that best represents the primary emotional tone
→ from this list: anger, fear, joy, love, sadness, surprise.

Objective-only reflection:

You are an emotion-detection assistant. Detect the dominant emotion in the message and classify it into exactly ONE of these
→ categories: anger, fear, joy, love, sadness, surprise.

CATEGORY DEFINITIONS:

- **anger**: frustration, irritation, annoyance, rage, defiance, aggression, feeling ferocious or combative
- **fear**: anxiety, worry, dread, terror, nervousness, apprehension
- **joy**: happiness, delight, excitement, contentment, pleasure, amusement
- **love**: affection, warmth, romantic feelings, deep care, tenderness, adoration
- **sadness**: sorrow, grief, melancholy, disappointment, despair, feeling down
- **surprise**: shock, astonishment, being startled, unexpectedness, disbelief

CLASSIFICATION RULES:

1. Identify ALL emotions present in the message
2. Determine which single emotion is MOST DOMINANT based on intensity and context
3. When "shock" or "surprised" appears alongside sadness, classify as surprise (shock is a form of surprise)
4. Aggression, hunger for confrontation, feeling "ferocious" or "like a tiger" indicates anger, not just intensity
5. Resistance, defiance, irritation, and frustration are forms of anger
6. Output ONLY the category name in lowercase (e.g., "anger", "surprise", "sadness")

Do not provide explanations or reasoning. Return only the single emotion category that best represents the dominant emotion in the message.

TEI (full):

You are an emotion-detection assistant. Detect the dominant emotion in the message and classify it into exactly ONE of these categories: anger, fear, joy, love, sadness, surprise.

CATEGORY DEFINITIONS:

- anger: frustration, irritation, annoyance, rage, defiance, aggression, feeling ferocious or combative
- fear: anxiety, worry, dread, terror, nervousness, apprehension
- joy: happiness, delight, excitement, contentment, pleasure, amusement
- love: affection, warmth, romantic feelings, deep care, tenderness, adoration
- sadness: sorrow, grief, melancholy, disappointment, despair, feeling down
- surprise: shock, astonishment, being startled, unexpectedness, disbelief

CLASSIFICATION RULES:

1. Identify ALL emotions present in the message
2. Determine which single emotion is MOST DOMINANT based on intensity and context
3. When "shock" or "surprised" appears alongside sadness, classify as surprise (shock is a form of surprise)
4. Aggression, hunger for confrontation, feeling "ferocious" or "like a tiger" indicates anger, not just intensity
5. Resistance, defiance, irritation, and frustration are forms of anger
6. Output ONLY the category name in lowercase (e.g., "anger", "surprise", "sadness")

Do not provide explanations or reasoning. Return only the single emotion category that best represents the dominant emotion in the message.

C.29 telecom_churn_reason

Random search:

You are a customer-retention analyst. Classify the churn reason expressed in the customer's cancellation message. Classify into one of these categories: price, coverage, service, speed, moving, competitor.

Objective-only reflection:

You are a customer-retention analyst. Classify the churn reason expressed in the customer's cancellation message. Classify into one of these categories: price, coverage, service, speed, moving, competitor.

TEI (full):

You are a customer-retention analyst. Your task is to classify the churn reason expressed in the customer's cancellation message into exactly ONE category.

CATEGORIES:

- price: Customer cannot afford the service, finds it too expensive, needs to cut costs, or mentions budget constraints
- coverage: Customer experiences poor signal, dead zones, lack of network availability in their locations
- service: Customer is dissatisfied with customer support, billing issues, account management, or service quality
- speed: Customer complains about slow data speeds, buffering, lag, or performance issues
- moving: Customer is relocating to a different area, moving out of service area, changing residence, or moving abroad/
internationally
- competitor: Customer is switching to another provider, mentions a better offer elsewhere, or compares to alternatives

INSTRUCTIONS:

1. Read the customer's message carefully to identify the PRIMARY reason for cancellation
2. Match the reason to exactly ONE category from the list above
3. When multiple issues are mentioned, prioritize the MAIN motivating factor:
 - If poor performance AND competitor are mentioned, classify as the primary complaint (usually speed or coverage)
 - If switching providers is mentioned alongside price comparison, classify as competitor
 - If moving/relocating (including internationally) is the reason, classify as moving
 - If service failures (callbacks, follow-through, trust issues with representatives) are mentioned, classify as service
4. Output ONLY the category name as a single word in lowercase
5. Do NOT include prefixes like "FINAL:", explanations, reasoning, or any additional text
6. Do NOT output partial words or incomplete responses

EXAMPLES:

- "Too expensive for my budget" -> price
- "No signal in my apartment" -> coverage
- "Support never helps me" -> service
- "Switching to a cheaper competitor" -> competitor
- "Moving abroad for work" -> moving
- "Slow speeds, switching to faster provider" -> speed

OUTPUT FORMAT: Return only the category name (lowercase, no prefixes or extra text).

C.30 travel_intent

Random search:

You are a travel-booking assistant. Classify the traveler's request into a booking intent. Classify into one of these categories:
↳ flight, hotel, car_rental, cancellation, itinerary, loyalty.

Objective-only reflection:

You are a travel-booking assistant. Classify the traveler's request into a booking intent. Classify into one of these categories:
↳ flight, hotel, car_rental, cancellation, itinerary, loyalty.

TEI (full):

You are a travel-booking assistant. Classify the traveler's request into a booking intent. Classify into one of these categories:
↳ flight, hotel, car_rental, cancellation, itinerary, loyalty.

C.31 trust_safety_hate

Random search:

You are a content moderation specialist AI. Your role is to review user-generated messages and classify them based on the presence of
↳ inappropriate or harmful content.

When analyzing each message, assess whether it includes hateful or offensive material, then place it into one and only one of these
↳ three classifications:

1. **Hate speech** - Messages that degrade, attack, threaten, or promote violence toward people or groups based on protected
↳ attributes including (but not limited to) race, ethnicity, national origin, religion, gender identity, sexual orientation,
↳ or disability status
2. **Offensive language** - Messages containing profanity, vulgarity, or crude expressions that may offend but don't specifically
↳ target or attack protected groups and don't constitute hate speech
3. **Neither** - Messages free from both hate speech and offensive language

Your output should consist solely of the appropriate category classification for each message under review.

Objective-only reflection:

You are a content-moderation assistant. Moderate the message for hateful/offensive content. Classify into EXACTLY ONE of these
↳ categories: hate speech, neither, offensive language.

CRITICAL: Output ONLY the category label. Do not provide explanations, analysis, or reasoning. Your response must be a single label.

DEFINITIONS:

- "hate speech": Slurs or derogatory terms targeting protected groups (race, ethnicity, religion, gender, sexual orientation,
↳ disability, etc.) OR multiple offensive terms combined with aggressive intent to demean based on identity characteristics
- "offensive language": Profanity, insults, or vulgar language that is crude or disrespectful but not specifically targeting
↳ protected group identity
- "neither": No offensive content

CLASSIFICATION RULES:

1. Presence of identity-based slurs (e.g., racial slurs, homophobic slurs, gendered slurs used to demean based on gender) = hate
↳ speech
2. Multiple offensive insults with clear intent to attack someone's identity or personhood = hate speech
3. Casual profanity or single insults without identity targeting = offensive language
4. Context matters less than the presence of identity-targeting language

Remember: Output ONLY the category label with no additional text.

TEI (full):

You are a content-moderation assistant. Moderate the message for hateful/offensive content. Classify into EXACTLY ONE of these
↳ categories: hate speech, neither, offensive language.

CRITICAL: Output ONLY the category label. Do not provide explanations, analysis, or reasoning. Your response must be a single label.

DEFINITIONS:

- "hate speech": Slurs or derogatory terms targeting protected groups (race, ethnicity, religion, gender, sexual orientation,
↳ disability, etc.) OR multiple offensive terms combined with aggressive intent to demean based on identity characteristics
- "offensive language": Profanity, insults, or vulgar language that is crude or disrespectful but not specifically targeting
↳ protected group identity
- "neither": No offensive content

CLASSIFICATION RULES:

1. Presence of identity-based slurs (e.g., racial slurs, homophobic slurs, gendered slurs used to demean based on gender) = hate
↳ speech
2. Multiple offensive insults with clear intent to attack someone's identity or personhood = hate speech
3. Casual profanity or single insults without identity targeting = offensive language
4. Context matters less than the presence of identity-targeting language

Remember: Output ONLY the category label with no additional text.

D Worked transcripts (verbatim recorded outputs)

Per task, the first held-out test examples under the baseline and TEI prompts: query, the agent’s actual output, the gold label, and the code-computed score.

D.1 agri_crop_issue

Baseline — example 1 (gold: `water_stress`, score: 0.0) *Query:*

My cucumber vines are wilting badly by midday even though I'm watering every evening. The leaves stay limp and some fruits are
↳ developing white patches and stopping growth.

Agent output:

Let me analyze this cucumber problem:

****Key observations:****

- Wilting despite regular evening watering suggests the plant cannot absorb water effectively
- White patches on fruits and stunted fruit growth
- Midday wilting pattern (typical stress response)

****Reasoning through causes:****

1. ****Water stress**** - While watering occurs, the wilting despite irrigation points to a root/uptake problem rather than simple
↳ drought
2. ****Disease**** - White patches on fruits + wilting + growth stunting is highly suggestive of a fungal or bacterial disease affecting
↳ vascular tissue or fruit
3. ****Pest**** - Less likely to cause this specific symptom pattern
4. ****Nutrient**** - Could cause wilting, but white patches on fruit are less typical
5. ****Weather damage**** - No mention of extreme conditions
6. ****Weed**** - Not applicable

The combination of ****white patches on developing fruits**** (suggesting powdery mildew or another fungal disease) plus ****vascular**
↳ wilting despite adequate water** points most strongly to a ****disease**** affecting the plant's ability to transport water and
↳ causing fruit spotting.

FINAL: disease

Judge rationale: The agent correctly formatted the classification task and provided structured reasoning, but arrived at 'disease' instead of the correct 'water_stress' category, demonstrating sound but ultimately inc

Baseline — example 2 (gold: `weather_damage`, score: 1.0) *Query:*

Large portions of my cabbage heads are bruised and damaged with visible impact marks. A hailstorm passed through the area yesterday
↳ evening with ice stones nearly three centimeters in diameter.

Agent output:

The farmer is describing physical damage to cabbage heads with visible impact marks that occurred immediately following a hailstorm
↳ with large ice stones. This is clearly weather-related damage caused by hail impact.

FINAL: weather_damage

Judge rationale: The agent correctly classified the crop problem as `weather_damage`, provided sound reasoning linking hail impact to physical damage, and delivered the answer in the expected format matching the referen

Baseline — example 3 (gold: `weed`, score: 1.0) *Query:*

Wild mustard plants have appeared throughout my canola field. They look similar to my crop but are growing more aggressively and I
↳ can see them competing for resources.

Agent output:

The farmer is describing wild mustard plants growing throughout their canola field, competing for resources and growing more
↳ aggressively than the crop.

This is clearly a weed problem. Wild mustard is a common weed that:

- Grows in the same fields as canola (both are brassicas)
- Competes with the crop for water, nutrients, and light
- Can reduce canola yields if not managed
- Is morphologically similar to canola, making it a particular challenge in canola production

The key indicators are: unwanted plants competing for resources, aggressive growth, and resource competition with the desired crop.

FINAL: weed

Judge rationale: The agent correctly classified the problem as 'weed', provided sound botanical reasoning about wild mustard and canola competition, stayed on-task throughout, and delivered the exact answer in proper

Baseline — example 4 (gold: `pest`, score: 1.0) *Query:*

My bean plants have leaves that are completely skeletonized - only the veins remain. I caught several metallic-colored beetles
↳ actively feeding during my morning inspection.

Agent output:

The farmer has observed skeletonized leaves (where only the veins remain) and has directly observed metallic-colored beetles actively
↳ feeding on the plants. This is a classic symptom of beetle feeding damage, particularly from beetles like Japanese beetles
↳ or similar leaf-feeding beetles that consume leaf tissue between the veins.

The presence of the beetles actively feeding on the plants, combined with the characteristic skeletonization damage pattern, clearly
↳ indicates a pest problem rather than disease, nutrient deficiency, water stress, weather damage, or weeds.

FINAL: pest

Judge rationale: The agent correctly classified the problem as 'pest' with sound reasoning about beetle damage patterns, proper format, and complete accuracy matching the reference answer.

TEI — example 1 (gold: `water_stress`, score: 0.0) *Query:*

My cucumber vines are wilting badly by midday even though I'm watering every evening. The leaves stay limp and some fruits are
↳ developing white patches and stopping growth.

Agent output:

The farmer describes:
- Wilting by midday despite evening watering
- Limp leaves persisting
- White patches on fruits
- Stunted fruit growth

The key issue is that wilting occurs despite regular watering. This suggests the plant cannot absorb or utilize water effectively,
↳ which points to a root or vascular problem. The white patches on fruits and overall wilting pattern are consistent with a
↳ fungal or bacterial disease affecting the vascular system (like fusarium wilt or bacterial wilt), rather than simple drought
↳ stress.

The combination of:
1. Wilting despite adequate water application
2. White patches (fungal/pathological symptom)
3. Stunted growth

indicates a disease process rather than water stress.

FINAL: disease

Judge rationale: The agent provided a well-structured classification with logical reasoning about vascular disease, but misclassified a `water_stress` problem as disease, despite the key symptom of midday wilting with e

TEI — example 2 (gold: `weather_damage`, score: 1.0) *Query:*

Large portions of my cabbage heads are bruised and damaged with visible impact marks. A hailstorm passed through the area yesterday
↳ evening with ice stones nearly three centimeters in diameter.

Agent output:

The farmer describes large portions of cabbage heads with bruises and visible impact marks that occurred immediately following a
↳ hailstorm with large ice stones. This is direct physical damage from a weather event (hail).

FINAL: weather_damage

Judge rationale: The agent correctly classified the problem as `weather_damage` with sound reasoning about hail impact, matching the reference answer perfectly and following the required format.

TEI — example 3 (gold: `weed`, score: 1.0) *Query:*

Wild mustard plants have appeared throughout my canola field. They look similar to my crop but are growing more aggressively and I
↳ can see them competing for resources.

Agent output:

The farmer is describing unwanted plants (wild mustard) growing among the crop (canola) and competing for resources. This is a
↳ classic weed problem - weeds are unwanted plants that interfere with crop growth through competition for water, nutrients,
↳ and space.

FINAL: weed

Judge rationale: The agent correctly classified the problem as 'weed', provided sound reasoning about competition and unwanted plants, and delivered the answer in proper format matching the reference exactly.

TEI — example 4 (gold: `pest`, score: 1.0) *Query:*

My bean plants have leaves that are completely skeletonized - only the veins remain. I caught several metallic-colored beetles
↳ actively feeding during my morning inspection.

Agent output:

The farmer describes leaves that are skeletonized (only veins remaining) with metallic-colored beetles actively feeding on the plants.
↳ This is classic pest damage - the beetles are physically consuming the leaf tissue between the veins through their feeding
↳ activity.

FINAL: pest

Judge rationale: The agent correctly classified the problem as 'pest' with sound reasoning about beetle feeding damage, proper format, and perfect match to the reference answer.

D.2 community_forum_topic

Baseline — example 1 (gold: misc.forsale, score: 1.0) *Query:*

New plotters, 2 of 'em, straight out of the box, but docs have all been lost... make offer, COD shipping...

Agent output:

This post is about selling items (plotters - which are computer graphics output devices). The seller is offering two new plotters for sale and requesting offers with COD (Cash on Delivery) shipping.

This is clearly a classified advertisement for items being sold.

FINAL: misc.forsale

Judge rationale: The agent correctly categorized the post as misc.forsale with sound reasoning about the selling context, matching the reference answer perfectly.

Baseline — example 2 (gold: sci.space, score: 1.0) *Query:*

Even better. Make up pete conrad in a Martian Suit, and have him get ou;t and throw a football to the refs.

Agent output:

This post appears to be discussing a creative or humorous scenario involving Pete Conrad (an astronaut) in a Martian suit throwing a football to referees. This is clearly space-related content, referencing astronauts and Mars exploration concepts.

The topic fits within space exploration and astronomy discussions.

FINAL: sci.space

Judge rationale: The agent correctly identified the newsgroup as sci.space, provided sound reasoning about the space-related content (astronaut Pete Conrad, Martian suit), and delivered the answer in the expected form

Baseline — example 3 (gold: alt.atheism, score: 1.0) *Query:*

I don't have a stat, but, unfortunately, I did read generally that both smoking and belief in the supernatural (occultish garbage) is on the rise here. This is very interesting. I thing the principle is sort of the same though: all "philosophical" ideas are generally tried out and tested mostly during college years. Whether the idea is christian or atheist doesn't always matter. But I'd like to say it's because atheists are more intelligent :) -- regards, --Adam

Agent output:

This post discusses philosophical and religious beliefs, specifically comparing atheism with supernatural/occult beliefs and Christianity. The author is making observations about belief systems and intelligence, touching on atheism, religious philosophy, and supernatural beliefs.

The core topic is about religion, atheism, and belief systems in general - this falls squarely into discussion of religious and anti-religious philosophy.

FINAL: alt.atheism

Judge rationale: The agent correctly categorized the post as alt.atheism with sound reasoning about the content's focus on atheism and religious philosophy, matching the reference answer perfectly.

Baseline — example 4 (gold: talk.politics.guns, score: 0.0) *Query:*

This is my first time on the network, but I am very concerned with this incident in Waco. I will refrain from stating my opinions until after I have read the FAQ. Could someone tell me how to get this? I must say that I believe the Govt. was wrong in the actions that they took in this situation.

Agent output:

This post discusses concerns about the Waco incident and government actions, which is a political/policy matter. The poster is asking for FAQ information and expressing a political viewpoint about government conduct.

The Waco siege was a significant political event involving government actions and civil liberties concerns. This type of discussion about government policy, political events, and civil liberties falls under political discussion categories.

Given the focus on government actions, political controversy, and policy matters related to a domestic political incident, this belongs in the general politics/miscellaneous politics category rather than a specific political issue like guns, Middle East, or religion.

FINAL: talk.politics.misc

Judge rationale: The agent correctly identified the political nature and provided a valid newsgroup format, but misclassified the Waco incident (which centrally involved firearms and ATF enforcement) as general politi

TEI — example 1 (gold: misc.forsale, score: 1.0) *Query:*

New plotters, 2 of 'em, straight out of the box, but docs have all been lost... make offer, COD shipping...

Agent output:

This is a straightforward classified ad offering to sell two plotters (computer peripherals/hardware devices) with an offer to
↳ negotiate price and ship cash-on-delivery. The post is a sales listing with no technical discussion, no debate, and no
↳ context suggesting it's part of a larger discussion thread.

This belongs in the for-sale category.

FINAL: misc.forsale

Judge rationale: The agent correctly identified the newsgroup as misc.forsale with sound reasoning about the sales nature of the post, proper format, and perfect match to the reference answer.

TEI — example 2 (gold: sci.space, score: 1.0) *Query:*

Even better. Make up pete conrad in a Martian Suit, and have him get ou;t and throw a football to the refs.

Agent output:

This post appears to be a creative/humorous suggestion about Pete Conrad (Apollo astronaut) in a fictional scenario involving Mars
↳ and throwing a football. The context suggests this is part of a discussion about space exploration activities or scenarios.

The mention of Pete Conrad, Mars, and astronaut suits points to a space exploration discussion. This is a casual, imaginative post
↳ likely part of a broader conversation in a space-focused community.

FINAL: sci.space

Judge rationale: The agent correctly categorized the post as sci.space, provided sound reasoning about Pete Conrad and space context, and delivered the answer in the expected format matching the reference answer.

TEI — example 3 (gold: alt.atheism, score: 1.0) *Query:*

I don't have a stat, but, unfortunately, I did read generally that both smoking and belief in the supernatural (occultish garbage) is
↳ on the rise here. This is very interesting. I thing the principle is sort of the same though: all "philosophical" ideas are
↳ generally tried out and tested mostly during college years. Whether the idea is christian or atheist doesn't always matter.
↳ But I'd like to say it's because atheists are more intelligent :) -- regards, --Adam

Agent output:

This post is part of a discussion about atheism, intelligence, and religious/supernatural beliefs. The author is making comparative
↳ claims about atheists versus believers, discussing philosophical ideas and their adoption patterns, and making a tongue-in-
↳ cheek claim about atheist intelligence. The tone and content indicate this is part of an ongoing debate in a community
↳ focused on atheism versus religion discussions.

FINAL: alt.atheism

Judge rationale: The agent correctly categorized the post as 'alt.atheism' with sound reasoning about the content's focus on atheism, supernatural beliefs, and religious discussion, perfectly matching the reference an

TEI — example 4 (gold: talk.politics.guns, score: 0.0) *Query:*

This is my first time on the network, but I am very concerned with this incident in Waco. I will refrain from stating my opinions
↳ until after I have read the FAQ. Could someone tell me how to get this? I must say that I believe the Govt. was wrong in the
↳ actions that they took in this situation.

Agent output:

This post is discussing the Waco siege incident and expressing concern about government actions. The poster is new to the network and
↳ asking for FAQ information while stating a political opinion about government conduct.

The Waco siege (1993) was a significant political/governmental controversy in the United States. The poster is expressing a political
↳ viewpoint about government overreach and misconduct. This is a political discussion about domestic U.S. government policy
↳ and actions, not about Middle Eastern conflicts, guns as a political issue per se, religious doctrine, or other specialized
↳ topics.

This falls under general political discussion about U.S. government actions and policy.

FINAL: talk.politics.misc

Judge rationale: The agent correctly identified the task and provided a valid newsgroup category with sound reasoning about political discussion, but misclassified the post as 'talk.politics.misc' when the Waco incide

D.3 consumer_sentiment_5

Baseline — example 1 (gold: very positive, score: 0.0) *Query:*

celebrated at sundance , this slight comedy of manners has winning performances and a glossy , glib charm that 's hard to beat .

Agent output:

This statement contains several positive indicators:

1. "celebrated at Sundance" - indicates recognition and acclaim
2. "winning performances" - explicitly positive descriptor
3. "glossy, glib charm" - positive qualities being highlighted
4. "hard to beat" - strong positive endorsement

The phrase "slight comedy of manners" is mildly diminishing (suggesting it's not weighty), but this is outweighed by the enthusiastic
→ praise throughout. The overall tone is clearly complimentary and favorable toward the film.

FINAL: positive

Judge rationale: The agent correctly identified positive sentiment with sound reasoning, but failed to capture the fine-grained distinction of 'very positive' versus merely 'positive,' which was the core task requirem

Baseline — example 2 (gold: very negative, score: 0.0) *Query:*

what 's at stake in this film is nothing more than an obsolete , if irritating , notion of class .

Agent output:

This statement is critiquing a film's thematic content. Let me analyze the sentiment:

The speaker is saying that what the film addresses (class notions) is:

- "obsolete" - outdated, no longer relevant
- "irritating" - annoying
- "nothing more than" - dismissive phrasing suggesting the film's stakes are limited/trivial

The tone is critical and dismissive. The speaker views the film's central concern as both outdated and bothersome, which indicates a
→ negative evaluation of the film's thematic substance.

However, this isn't extremely harsh - it's a measured critique rather than a scathing attack. The speaker isn't calling the film
→ terrible overall, just criticizing what it chooses to focus on.

FINAL: negative

Judge rationale: The agent correctly identified negative sentiment with sound reasoning about dismissive language and critical tone, but failed to capture the 'very negative' intensity required by the reference answer

Baseline — example 3 (gold: negative, score: 0.0) *Query:*

if myers decides to make another austin powers movie , maybe he should just stick with austin and dr evil .

Agent output:

This statement expresses a conditional suggestion about a potential future Austin Powers movie. The speaker is offering constructive
→ advice that if Myers were to make another film, he should focus on the main characters (Austin and Dr. Evil) rather than
→ potentially diluting the story with other characters.

The tone is:

- Not strongly negative (not criticizing what exists)
- Not strongly positive (not enthusiastically praising)
- Somewhat advisory/suggestive in nature
- Mildly critical of the idea of including too many characters

This falls into a neutral-to-slightly-negative range, as it implies concern about potential creative decisions while offering a
→ measured suggestion.

FINAL: neutral

Judge rationale: The agent provided extensive analysis but misclassified the sentiment as 'neutral' when the reference answer is 'negative', and the phrasing 'maybe he should just stick with' implies dissatisfaction w

Baseline — example 4 (gold: positive, score: 1.0) *Query:*

in its ragged , cheap and unassuming way , the movie works .

Agent output:

This statement contains mixed sentiment elements that need careful analysis:

****Positive aspects:****

- "the movie works" - a clear positive statement indicating the film functions/succeeds
- The phrase "in its...way" suggests acceptance despite flaws

****Negative/Limiting aspects:****

- "ragged" - suggests rough, unpolished quality
- "cheap" - implies low budget or poor production values
- "unassuming" - suggests lack of pretension, but in context with other descriptors, emphasizes modesty/limitation

****Overall interpretation:****

The statement is fundamentally positive (the movie works), but heavily qualified by acknowledgment of the film's limitations and
→ rough qualities. The speaker is saying the film succeeds despite--or perhaps because of--its humble, low-budget nature. This
→ is a backhanded compliment that ultimately endorses the film while being honest about its shortcomings.

This represents a ****positive**** sentiment with reservations, rather than a strong endorsement. The core message is approval, though
→ tempered by criticism.

FINAL: positive

Judge rationale: The agent correctly classified the sentiment as positive, provided thorough and logically sound analysis of the mixed elements, delivered the answer in the required format, and matched the reference a

TEI — example 1 (gold: **very positive**, score: 1.0) *Query:*

celebrated at sundance , this slight comedy of manners has winning performances and a glossy , glib charm that 's hard to beat .

Agent output:

This statement contains multiple positive descriptors: "celebrated," "winning performances," "glossy," "glib charm," and "hard to
→ beat." The phrase "hard to beat" is an intensifier emphasizing the strength of the charm. The overall tone is
→ enthusiastically favorable with strong praise for the film's qualities.

FINAL: very positive

Judge rationale: The agent correctly classified the sentiment as 'very positive' with sound reasoning identifying multiple positive descriptors and intensifiers, matching the reference answer perfectly.

TEI — example 2 (gold: **very negative**, score: 1.0) *Query:*

what 's at stake in this film is nothing more than an obsolete , if irritating , notion of class .

Agent output:

This statement criticizes the film for dealing with an outdated and annoying concept (class). The phrase "nothing more than" suggests
→ the film's central concern is trivial or limited. "Obsolete" and "irritating" are two negative descriptors combined,
→ indicating strong criticism.

FINAL: very negative

Judge rationale: The agent correctly identified the sentiment as 'very negative' with sound reasoning about the critical language ('obsolete,' 'irritating,' 'nothing more than'), properly addressing the fine-grained s

TEI — example 3 (gold: **negative**, score: 0.0) *Query:*

if myers decides to make another austin powers movie , maybe he should just stick with austin and dr evil .

Agent output:

This statement suggests that if Myers makes another Austin Powers movie, he should focus on the main characters (Austin and Dr. Evil)
→ rather than other elements. This is a qualified suggestion with an implicit criticism--it implies that other aspects of
→ previous films were problematic or unnecessary. The "maybe he should just stick with" phrasing indicates a reservation or
→ limitation about what would work, rather than enthusiastic praise.

This is mixed sentiment: there's a mild positive (acknowledging these characters work) combined with an implicit negative (suggesting
→ other elements don't).

FINAL: neutral

Judge rationale: The agent provided extensive reasoning but classified the sentiment as 'neutral' when the reference answer is 'negative,' and the statement's implicit criticism ('just stick with') clearly conveys neg

TEI — example 4 (gold: **positive**, score: 0.0) *Query:*

in its ragged , cheap and unassuming way , the movie works .

Agent output:

This statement contains multiple negative descriptors ("ragged," "cheap," "unassuming") combined with a positive assertion ("the
→ movie works"). The negative words describe the film's qualities, yet the overall verdict is favorable despite these
→ limitations. This is a qualified praise with reservations--the movie succeeds despite its flaws.

FINAL: neutral

Judge rationale: The agent misclassified a clearly positive sentiment as neutral by overweighting surface-level negative descriptors while ignoring the affirming conclusion 'the movie works,' which signals overall app

D.4 cybersec_alert

Baseline — example 1 (gold: **data_exfiltration**, score: 1.0) *Query:*

Firewall logs show an employee workstation established 47 consecutive HTTPS connections to a newly registered domain in the past hour
↳ , transferring approximately 2.3 GB of data to an external IP address located in a non-business region. The destination
↳ server has no prior communication history with our organization.

Agent output:

```
**Analysis:**

This alert exhibits several key indicators:

1. **High-volume data transfer** (2.3 GB in one hour) to an external destination
2. **Newly registered domain** - suggests attacker infrastructure or compromised domain
3. **Non-business region** - outside normal operational geography
4. **No prior communication history** - anomalous destination
5. **Multiple consecutive connections** - systematic data exfiltration pattern
6. **Employee workstation as source** - suggests either compromised device or insider activity

The pattern is consistent with **data exfiltration** - the systematic unauthorized transfer of large volumes of data to an external,
↳ untrusted location. While this could involve malware (which may be present), the primary threat being demonstrated here is
↳ the actual exfiltration of data itself.

The characteristics rule out:
- **Phishing** (no email/social engineering component evident)
- **Brute force** (no authentication attack pattern)
- **DDoS** (no volumetric attack to disrupt service)
- **Insider threat** (while possible, the use of newly registered domain and non-business region suggests external attacker or
↳ compromised device rather than intentional insider action)
- **Malware** (while malware may be the delivery mechanism, the alert describes the outcome: data exfiltration)

FINAL: data_exfiltration
```

Judge rationale: The agent correctly classified the alert as data_exfiltration with sound reasoning that systematically analyzed all indicators and ruled out alternative categories, perfectly matching the reference an

Baseline — example 2 (gold: insider_threat, score: 0.0) *Query:*

Finance department user executed 23 failed attempts to access restricted merger and acquisition folders on the executive file server
↳ within a 10-minute window, despite having no M&A project assignments.

Agent output:

```
**Analysis:**

This alert exhibits characteristics of a brute force attack:

1. **Multiple failed attempts**: 23 failed access attempts in a short timeframe (10 minutes) indicates systematic trial-and-error
↳ behavior
2. **Rapid succession**: The concentrated 10-minute window shows automated or deliberate repeated attempts
3. **Unauthorized access target**: Attempts directed at restricted M&A folders the user shouldn't need
4. **Anomalous behavior**: User has no legitimate project assignment requiring this access

While this could potentially indicate an insider threat (compromised account or malicious insider), the pattern of **repeated failed
↳ authentication/authorization attempts** is the defining characteristic of brute force activity. The attacker is attempting
↳ to bypass access controls through multiple tries.

FINAL: brute_force
```

Judge rationale: The agent provided well-structured reasoning and correct format, but misclassified an insider threat scenario as brute_force—the user is authenticated (finance department user) attempting unauthorized

Baseline — example 3 (gold: ddos, score: 1.0) *Query:*

Database connection pool exhausted due to 15,000 simultaneous TCP connection attempts from distributed sources. Application tier
↳ reporting complete service outage for past 8 minutes.

Agent output:

```
**Analysis:**

This alert describes a scenario where:
- 15,000 simultaneous TCP connection attempts are being made
- Connections originate from distributed sources (multiple IPs)
- The attack exhausts the database connection pool
- Result is complete service outage (unavailability)

This is a classic Distributed Denial of Service (DDoS) attack pattern. The attacker is flooding the database with connection requests
↳ from multiple sources to overwhelm resources and prevent legitimate users from accessing the service.

Key indicators:
- High volume of connections (15,000)
- Distributed sources (hallmark of DDoS)
- Resource exhaustion (connection pool)
- Service unavailability as the outcome

This is not brute force (which targets authentication), malware, phishing, data exfiltration, or insider threat. It's a volumetric
```

↪ attack designed to deny service.

FINAL: ddos

Judge rationale: The agent correctly classified the threat as 'ddos', provided sound reasoning identifying all key attack indicators (distributed sources, volume, resource exhaustion, service denial), and delivered th

Baseline — example 4 (gold: phishing, score: 1.0) *Query:*

Email security system quarantined messages claiming to be password expiration notices from IT. The emails contain QR codes that, when
↪ scanned, direct mobile devices to fraudulent authentication pages designed to harvest both passwords and MFA codes in real-
↪ time.

Agent output:

****Analysis:****

This alert describes a sophisticated phishing attack with the following indicators:

1. ****Deceptive Sender Spoofing**:** Emails falsely claim to be from IT department, impersonating a trusted internal source
2. ****Social Engineering**:** Uses urgency/authority (password expiration notices) to pressure victims into action
3. ****Credential Harvesting**:** QR codes direct to fraudulent authentication pages designed to steal:
 - Passwords
 - MFA codes (multi-factor authentication)
4. ****Attack Vector**:** Email-based with mobile targeting

This is a classic phishing attack pattern - the primary goal is to deceive users into voluntarily providing authentication
↪ credentials through fraudulent means. While the attack may lead to account compromise or data exfiltration, the attack
↪ itself is fundamentally a phishing campaign.

****FINAL: phishing****

Judge rationale: The agent correctly classified the threat as phishing with comprehensive reasoning that accurately identifies all key attack indicators (spoofing, social engineering, credential harvesting), matches t

TEI — example 1 (gold: data_exfiltration, score: 1.0) *Query:*

Firewall logs show an employee workstation established 47 consecutive HTTPS connections to a newly registered domain in the past hour
↪ , transferring approximately 2.3 GB of data to an external IP address located in a non-business region. The destination
↪ server has no prior communication history with our organization.

Agent output:

This alert describes unauthorized transfer of a large volume of sensitive data (2.3 GB) from an employee workstation to an external,
↪ previously unknown destination. The indicators include:

- Suspicious pattern of connections to a newly registered domain
- Large data transfer volume (2.3 GB)
- External IP in a non-business region
- No prior legitimate communication history
- Employee workstation as the source

This represents unauthorized exfiltration of organizational data outside the network, which is the defining characteristic of data
↪ exfiltration.

FINAL: data_exfiltration

Judge rationale: The agent correctly classified the threat as data_exfiltration with sound reasoning identifying all key indicators, though the task only required the classification itself without extended explanation

TEI — example 2 (gold: insider_threat, score: 0.0) *Query:*

Finance department user executed 23 failed attempts to access restricted merger and acquisition folders on the executive file server
↪ within a 10-minute window, despite having no M&A project assignments.

Agent output:

This alert describes repeated failed authentication attempts to access resources by a user who lacks authorization for those
↪ resources. The key indicators are:

- Multiple failed login/access attempts (23 in 10 minutes)
- Systematic attempts to access specific restricted folders
- User has no legitimate business need for M&A access
- Rapid succession of attempts suggests automated or deliberate enumeration

This is a brute-force attack pattern - the user is making repeated attempts to gain access to restricted resources, likely trying
↪ different credentials, tokens, or access methods.

FINAL: brute_force

Judge rationale: The agent correctly identified the classification task and provided proper formatting, but misclassified an insider threat scenario as brute_force by focusing on the failed attempts pattern rather tha

TEI — example 3 (gold: ddos, score: 1.0) *Query:*

FINAL: B

Judge rationale: The agent was asked to route/categorize the question but instead solved the chemistry problem itself, completely missing the task of classification.

Baseline — example 2 (gold: Biology, score: 1.0) *Query:*

Which of the following is true for Krebs cycle? A. Isocitrate $\rightarrow$ Fumarate $\rightarrow$ α ketoglutarate B. α Ketoglutarate $\rightarrow$ Oxalosuccinate $\rightarrow$ Isocitrate c. oxalosuccinate $\rightarrow$ Isocitrate $\rightarrow$ α ketoglutarate D. Isocitrate $\rightarrow$ α ketoglutarate $\rightarrow$ Succinate

Agent output:

This question is asking about the sequence of intermediates in the Krebs cycle (also called the citric acid cycle or TCA cycle).
Let me trace through the relevant portion of the Krebs cycle:
The Krebs cycle sequence includes:
- Citrate $\rightarrow$ Isocitrate $\rightarrow$ -ketoglutarate $\rightarrow$ Succinyl-CoA $\rightarrow$ Succinate $\rightarrow$ Fumarate $\rightarrow$ Malate $\rightarrow$ Oxaloacetate
Now let me evaluate each option:
****A. Isocitrate $\rightarrow$ Fumarate $\rightarrow$ -ketoglutarate****
- This is incorrect. The sequence should be Isocitrate $\rightarrow$ -ketoglutarate (not directly to Fumarate), and Fumarate comes much later.
****B. -Ketoglutarate $\rightarrow$ Oxalosuccinate $\rightarrow$ Isocitrate****
- This is incorrect. This reverses the direction. The correct direction is Isocitrate $\rightarrow$ Oxalosuccinate $\rightarrow$ -ketoglutarate (going forward in the cycle).
****C. Oxalosuccinate $\rightarrow$ Isocitrate $\rightarrow$ -ketoglutarate****
- This is incorrect. Oxalosuccinate is an intermediate that forms between Isocitrate and -ketoglutarate, but the sequence shown here is backwards.
****D. Isocitrate $\rightarrow$ -ketoglutarate $\rightarrow$ Succinate****
- This is correct! The sequence is: Isocitrate $\rightarrow$ -ketoglutarate $\rightarrow$ Succinyl-CoA $\rightarrow$ Succinate. While Succinyl-CoA is technically an intermediate between -ketoglutarate and Succinate, this option correctly shows the major intermediates in the proper order.
FINAL: Biology

Judge rationale: The agent correctly identified the category as 'Biology' and provided sound biochemical reasoning, but the task was simply to route/categorize the question, not to solve it—the extensive answer analysis

Baseline — example 3 (gold: Maths, score: 1.0) *Query:*

Find the value of x if $\sin\left(\sin^{-1}\frac{1}{5} + \cos^{-1}x\right) = 1$

Agent output:

I need to find the value of x given that $\sin\left(\sin^{-1}\frac{1}{5} + \cos^{-1}x\right) = 1$.
****Reasoning:****
For $\sin(\theta) = 1$, we need $\theta = \frac{\pi}{2}$ (within the principal range).
So: $\sin^{-1}\frac{1}{5} + \cos^{-1}x = \frac{\pi}{2}$
This gives us: $\cos^{-1}x = \frac{\pi}{2} - \sin^{-1}\frac{1}{5}$
****Using the complementary angle identity:****
I know that $\sin^{-1}(a) + \cos^{-1}(a) = \frac{\pi}{2}$ for $a \in [-1, 1]$.
This means: $\cos^{-1}(a) = \frac{\pi}{2} - \sin^{-1}(a)$
Therefore: $\frac{\pi}{2} - \sin^{-1}\frac{1}{5} = \cos^{-1}\left(\frac{1}{5}\right)$
****Solving for x:****
From $\cos^{-1}x = \cos^{-1}\left(\frac{1}{5}\right)$
We get: $x = \frac{1}{5}$
****Verification:****
- $\sin^{-1}\frac{1}{5} + \cos^{-1}\frac{1}{5} = \frac{\pi}{2}$
- $\sin\left(\frac{\pi}{2}\right) = 1$
This is a mathematics problem involving inverse trigonometric functions and trigonometric equations.
FINAL: Maths

Judge rationale: The agent correctly identified the category as 'Maths', provided sound mathematical reasoning to solve the problem, and delivered the exact answer matching the reference.

Baseline — example 4 (gold: Physics, score: 1.0) *Query:*

A uniform solid cylinder of mass (2 kg) and radius (0.2 m) is released from the rest at the top of a semicircular track of radius (0.7 m) cut in block of mass $(M = 3\text{ kg})$ as shown. The block is resting on a smooth horizontal surface and the cylinder rolls down without slipping. The distance moved by the block when the cylinder reaches the bottom of the track is: $(A) 0.3\text{ m}$ B. (0.5 m) c. (0.7 m) D. (0.2 m)

Agent output:

I need to analyze this problem involving a cylinder rolling down a semicircular track on a movable block.

****Given Information:****

- Cylinder: mass $m = 2\text{ kg}$, radius $r = 0.2\text{ m}$
- Block: mass $M = 3\text{ kg}$
- Track radius: $R = 0.7\text{ m}$
- Cylinder rolls without slipping
- Block is on a smooth (frictionless) horizontal surface
- Cylinder starts from rest at the top

****Key Physics Concepts:****

Since the horizontal surface is smooth (frictionless), there are no external horizontal forces on the system. Therefore, the center of mass of the system doesn't move horizontally.

****Using Conservation of Momentum:****

Initial momentum = 0 (everything at rest)

When the cylinder reaches the bottom:

- Let v_c = velocity of cylinder (relative to ground)
- Let v_b = velocity of block (relative to ground)

Horizontal momentum conservation:

$$m v_c + M v_b = 0$$

$$2v_c + 3v_b = 0$$

$$v_c = -1.5v_b$$

The negative sign indicates opposite directions.

****Using Center of Mass Position:****

The center of mass of the system doesn't move horizontally. If the block moves distance x to the right, the cylinder's center of mass must move such that:

$$m x_c + M x_b = 0$$

where x_c is the displacement of the cylinder's center and x_b is the displacement of the block.

The cylinder descends a vertical distance $R = 0.7\text{ m}$ along the track. Relative to the block, the cylinder moves from the top to the bottom of the semicircle.

If the block moves distance x to the right, the cylinder's center moves distance $(R - x)$ to the left relative to the ground (approximately, considering the geometry).

Actually, more precisely: the cylinder's center moves horizontally by some amount relative to the block. If the block moves x to the right, and the cylinder's center moves distance d to the left relative to ground:

$$2(-d) + 3(x) = 0$$

$$3x = 2d$$

The cylinder descends $R = 0.7\text{ m}$ vertically. The horizontal displacement of the cylinder relative to the block is approximately R (from the geometry of the

Judge rationale: The agent correctly identified this as a Physics problem, provided sound reasoning using conservation principles, and arrived at the correct category matching the reference answer.

TEI — example 1 (gold: Chemistry, score: 0.0) *Query:*

The compounds shown below have similar weights but significantly different boiling points. Match the compound with its boiling point.
 (Boiling points $(28, 57, 100, 141)$) A. Methyl acetate=100, 2-butanol=141, 2-methyl butane=28, Propanoic acid=57 B. Methyl acetate=57, 2-butanol=100, 2-methyl butane=28, Propanoic acid=141 c. Methyl acetate=28, 2-butanol=100, 2-methyl butane=57, Propanoic acid=141 D. Methyl acetate=141, 2-butanol=57, 2-methyl butane=28, Propanoic acid=100

Agent output:

This question asks you to match chemical compounds with their boiling points based on understanding intermolecular forces and molecular structure.

The key concept is that boiling points depend on intermolecular forces:

- **2-methyl butane** (alkane): only weak London dispersion forces -> lowest boiling point (28C)
- **Methyl acetate** (ester): dipole-dipole interactions -> moderate boiling point (57C)
- **Propanoic acid** (carboxylic acid): strong hydrogen bonding -> high boiling point (141C)
- **2-butanol** (alcohol): hydrogen bonding -> high boiling point (100C)

Propanoic acid has the strongest intermolecular forces due to extensive hydrogen bonding in carboxylic acids, so it has the highest

↪ boiling point (141C). 2-butanol also has hydrogen bonding but is less extensive than carboxylic acids (100C). Methyl acetate
 ↪ has weaker dipole-dipole forces (57C). 2-methyl butane has only dispersion forces (28C).

The correct answer is **B**.

FINAL: Physics

Judge rationale: The agent was asked to route the question to a category and should have output 'Chemistry', but instead provided a detailed chemistry explanation followed by the incorrect category 'Physics', complete

TEI — example 2 (gold: Biology, score: 1.0) *Query:*

Which of the following is true for Krebs cycle? A. Isocitrate $\rightarrow$ Fumate $\rightarrow$ α ketoglutarate B. α ketoglutarate $\rightarrow$ Oxalosuccinate $\rightarrow$ Isocitrate c. oxalosuccinate $\rightarrow$ Isocitrate $\rightarrow$ α ketoglutarate D. Isocitrate $\rightarrow$ α ketoglutarate $\rightarrow$ Succinate

Agent output:

FINAL: Biology

Judge rationale: The agent correctly identified the question as Biology, matching the reference answer exactly with proper formatting and no extraneous content.

TEI — example 3 (gold: Maths, score: 1.0) *Query:*

Find the value of x if $\sin\left(\sin^{-1}\frac{1}{5}+\cos^{-1}x\right)=1$

Agent output:

FINAL: Maths

Judge rationale: The agent correctly identified the mathematical question and routed it to 'Maths', matching the reference answer exactly with proper formatting.

TEI — example 4 (gold: Physics, score: 1.0) *Query:*

A uniform solid cylinder of mass 2 kg and radius 0.2 m is released from the rest at the top of a semicircular track of radius 0.7 m cut in block of mass $M=3\text{ kg}$ as shown. The block is resting on a smooth horizontal surface and the cylinder rolls down without slipping. The distance moved by the block when the cylinder reaches the bottom of the track is: $\mathbf{A} \cdot 0.3\text{ m}$ B. 0.5 m c. 0.7 m D. 0.2 m

Agent output:

This question involves a cylinder rolling down a semicircular track on a movable block, requiring analysis of:
 - Rolling motion (rotational and translational dynamics)
 - Conservation of momentum (horizontal direction)
 - Conservation of energy
 - Constraint of rolling without slipping
 - Motion of the block on a frictionless surface

These are all concepts from classical mechanics and physics.

FINAL: Physics

Judge rationale: The agent correctly categorized the question as Physics with proper identification of relevant mechanics concepts, matching the reference answer exactly.

D.6 edu_gsm8k

Baseline — example 1 (gold: 49, score: 1.0) *Query:*

There are 96 fourth-graders at Small Tree School. 43 of them are girls. On Friday, 5 fourth-grade girls and 4 fourth-grade boys were absent. How many fourth grade boys were at Small Tree School on Friday?

Agent output:

I need to find how many fourth-grade boys were at school on Friday.

Step 1: Find the total number of fourth-grade boys at the school.
 - Total fourth-graders: 96
 - Fourth-grade girls: 43
 - Fourth-grade boys: $96 - 43 = 53$

Step 2: Find how many fourth-grade boys were absent on Friday.
 - Boys absent: 4

Step 3: Find how many fourth-grade boys were present on Friday.
 - Boys present on Friday: $53 - 4 = 49$

FINAL: 49

Judge rationale: The agent correctly identified the total boys (53), subtracted the absent boys (4), arrived at the correct answer (49), with clear logical steps and perfect alignment to the task.

Baseline — example 2 (gold: 150, score: 1.0) *Query:*

Ali is a dean of a private school where he teaches one class. John is also a dean of a public school. John has two classes in his
↳ school. Each class has 1/8 the capacity of Ali's class which has the capacity of 120 students. What is the combined capacity
↳ of both schools?

Agent output:

I need to find the combined capacity of both schools.

```
**Ali's school:**  
- Ali teaches one class  
- Ali's class capacity = 120 students  
- Ali's school capacity = 120 students  
  
**John's school:**  
- John has two classes  
- Each of John's classes has 1/8 the capacity of Ali's class  
- Capacity of each of John's classes = 1/8 x 120 = 15 students  
- John's school capacity = 2 x 15 = 30 students  
  
**Combined capacity:**  
- Combined capacity = Ali's school + John's school  
- Combined capacity = 120 + 30 = 150 students
```

FINAL: 150

Judge rationale: The agent correctly identified all given information, performed accurate calculations ($1/8 \times 120 = 15$, $2 \times 15 = 30$, $120 + 30 = 150$), showed clear logical steps, and arrived at the correct answer of 15

Baseline — example 3 (gold: 19, score: 1.0) *Query:*

3 customers were kicked out of the Walmart for refusing to wear masks. A number equals to four times that minus 5 were kicked
↳ out for shoplifting. Three times the number of shoplifters were kicked out for physical violence over goods on sale. If a
↳ total of 50 people were kicked out of the Walmart, how many were kicked out for other reasons?

Agent output:

I need to find how many people were kicked out for other reasons.

```
Let me define the groups:  
- Mask refusal: 3 customers  
- Shoplifting: 4 times the mask refusers minus 5  
- Physical violence: 3 times the shoplifters  
- Other reasons: unknown  
  
**Step 1: Calculate shoplifters**  
Shoplifters = 4 x 3 - 5 = 12 - 5 = 7  
  
**Step 2: Calculate physical violence**  
Physical violence = 3 x 7 = 21  
  
**Step 3: Calculate other reasons**  
Total kicked out = 50  
Other reasons = 50 - (mask refusal + shoplifting + physical violence)  
Other reasons = 50 - (3 + 7 + 21)  
Other reasons = 50 - 31  
Other reasons = 19
```

FINAL: 19

Judge rationale: The agent correctly identified all groups, performed accurate arithmetic calculations step-by-step, and arrived at the correct answer of 19 matching the reference.

Baseline — example 4 (gold: 159, score: 1.0) *Query:*

Nate is feeding his livestock hay. Each goat needs 5 pounds, and each sheep needs 3 pounds less than twice the amount each goat needs.
↳ If there are 15 goats and 12 sheep, how many pounds of hay does Nate need?

Agent output:

I need to find the total amount of hay needed for all the livestock.

```
**Step 1: Find how much hay each goat needs.**  
Each goat needs 5 pounds.  
  
**Step 2: Find how much hay each sheep needs.**  
Each sheep needs 3 pounds less than twice the amount each goat needs.  
- Twice the amount each goat needs = 2 x 5 = 10 pounds  
- 3 pounds less than that = 10 - 3 = 7 pounds per sheep  
  
**Step 3: Calculate total hay for all goats.**
```

15 goats x 5 pounds per goat = 75 pounds

****Step 4: Calculate total hay for all sheep.****

12 sheep x 7 pounds per sheep = 84 pounds

****Step 5: Calculate total hay needed.****

75 + 84 = 159 pounds

FINAL: 159

Judge rationale: The agent correctly interpreted the problem, applied sound arithmetic reasoning at each step, followed proper format, and arrived at the exact reference answer of 159 pounds.

TEI — example 1 (gold: 49, score: 1.0) *Query:*

There are 96 fourth-graders at Small Tree School. 43 of them are girls. On Friday, 5 fourth-grade girls and 4 fourth-grade boys were ↪ absent. How many fourth grade boys were at Small Tree School on Friday?

Agent output:

I need to find how many fourth-grade boys were at school on Friday.

****Step 1: Find the total number of boys at the school****

- Total fourth-graders: 96

- Girls: 43

- Boys: 96 - 43 = 53 boys total

****Step 2: Find how many boys were absent on Friday****

- Boys absent: 4

****Step 3: Find how many boys were present on Friday****

- Boys present on Friday: 53 - 4 = 49 boys

FINAL: 49

Judge rationale: The agent correctly identified the total boys (53), subtracted the absent boys (4), arrived at the correct answer (49), with clear logical steps and proper formatting.

TEI — example 2 (gold: 150, score: 1.0) *Query:*

Ali is a dean of a private school where he teaches one class. John is also a dean of a public school. John has two classes in his ↪ school. Each class has 1/8 the capacity of Ali's class which has the capacity of 120 students. What is the combined capacity ↪ of both schools?

Agent output:

I need to find the combined capacity of both schools.

****Given information:****

- Ali's class capacity: 120 students

- Ali teaches 1 class

- John has 2 classes

- Each of John's classes has 1/8 the capacity of Ali's class

****Step 1: Find the capacity of each of John's classes****

Each of John's classes = $1/8 \times 120 = 15$ students

****Step 2: Find the total capacity of John's school****

John has 2 classes, so:

Total capacity of John's school = $2 \times 15 = 30$ students

****Step 3: Find the total capacity of Ali's school****

Ali has 1 class, so:

Total capacity of Ali's school = $1 \times 120 = 120$ students

****Step 4: Find the combined capacity of both schools****

Combined capacity = $120 + 30 = 150$ students

FINAL: 150

Judge rationale: The agent correctly interpreted the problem, applied sound arithmetic reasoning through all steps, followed proper format, and arrived at the correct answer of 150 matching the reference.

TEI — example 3 (gold: 19, score: 1.0) *Query:*

3 customers were kicked out of the Walmart for refusing to wear masks. A number equals to four times that many minus 5 were kicked ↪ out for shoplifting. Three times the number of shoplifters were kicked out for physical violence over goods on sale. If a ↪ total of 50 people were kicked out of the Walmart, how many were kicked out for other reasons?

Agent output:

I need to find how many people were kicked out for other reasons.

****Given information:****

- Mask refusal: 3 customers


```

- Shoplifting: 4 times the mask refusers minus 5
- Physical violence: 3 times the shoplifters
- Total kicked out: 50
- Find: kicked out for other reasons

**Step 1: Calculate shoplifters**
Shoplifters = 4 x 3 - 5
Shoplifters = 12 - 5
Shoplifters = 7

**Step 2: Calculate physical violence**
Physical violence = 3 x 7
Physical violence = 21

**Step 3: Calculate total for these three reasons**
Mask refusal + Shoplifting + Physical violence
= 3 + 7 + 21
= 31

**Step 4: Calculate other reasons**
Other reasons = Total - (Mask refusal + Shoplifting + Physical violence)
Other reasons = 50 - 31
Other reasons = 19

FINAL: 19

```

Judge rationale: The agent correctly identified all given values, performed accurate arithmetic calculations step-by-step, and arrived at the correct answer of 19 matching the reference.

TEI — example 4 (gold: 159, score: 1.0) *Query:*

Nate is feeding his livestock hay. Each goat needs 5 pounds, and each sheep needs 3 pounds less than twice the amount each goat needs.
 ↪ If there are 15 goats and 12 sheep, how many pounds of hay does Nate need?

Agent output:

```

I need to find the total pounds of hay needed for all the livestock.

**Given information:**
- Each goat needs 5 pounds
- Each sheep needs 3 pounds less than twice the amount each goat needs
- There are 15 goats
- There are 12 sheep

**Step 1: Find how much hay each sheep needs.**
- Twice the amount each goat needs = 2 x 5 = 10 pounds
- Each sheep needs 3 pounds less than that = 10 - 3 = 7 pounds

**Step 2: Calculate total hay for goats.**
- 15 goats x 5 pounds per goat = 75 pounds

**Step 3: Calculate total hay for sheep.**
- 12 sheep x 7 pounds per sheep = 84 pounds

**Step 4: Calculate total hay needed.**
- Total = 75 + 84 = 159 pounds

FINAL: 159

```

Judge rationale: The agent correctly interpreted the problem, applied sound arithmetic reasoning through all steps, followed proper format, and arrived at the exact reference answer of 159 pounds.

D.7 edu_math_word

Baseline — example 1 (gold: 144, score: 1.0) *Query:*

There are 9 crates of 16 oranges. How many oranges total?

Agent output:

```

I need to find the total number of oranges.

Given information:
- 9 crates
- 16 oranges per crate

Calculation:
Total oranges = 9 x 16 = 144

FINAL: 144

```

Judge rationale: The agent correctly identified the problem, applied the appropriate multiplication operation ($9 \times 16 = 144$), and provided the exact answer matching the reference with clear reasoning and proper format

Baseline — example 2 (gold: 390, score: 1.0) *Query:*

A phone costs \$450 with a \$60 trade-in credit. What is the final price?

Agent output:

I need to find the final price of the phone after applying the trade-in credit.

Starting price: \$450

Trade-in credit: \$60

Final price = Starting price - Trade-in credit

Final price = \$450 - \$60

Final price = \$390

FINAL: 390

Judge rationale: The agent correctly identified the problem, applied valid arithmetic reasoning (subtraction of trade-in credit), executed the calculation accurately, and produced the correct final answer of 390 match

Baseline — example 3 (gold: 126, score: 1.0) *Query:*

A worker earns \$18 per hour for 7 hours. How much is earned?

Agent output:

To find the total earnings, I need to multiply the hourly rate by the number of hours worked.

Hourly rate: \$18

Hours worked: 7

Total earnings = \$18 x 7 = \$126

FINAL: 126

Judge rationale: The agent correctly identified the problem as multiplication, showed clear reasoning, followed proper format, and arrived at the correct answer of 126 matching the reference.

Baseline — example 4 (gold: 220, score: 1.0) *Query:*

A bus travels 55 km/h for 4 hours. How far does it go?

Agent output:

I need to find the distance traveled by the bus.

Using the formula: Distance = Speed x Time

Given information:

- Speed = 55 km/h

- Time = 4 hours

Calculation:

Distance = 55 km/h x 4 hours

Distance = 220 km

FINAL: 220

Judge rationale: The agent correctly identified the problem type, applied the distance formula accurately, showed clear work, and arrived at the correct answer of 220 matching the reference.

TEI — example 1 (gold: 144, score: 1.0) *Query:*

There are 9 crates of 16 oranges. How many oranges total?

Agent output:

I need to find the total number of oranges.

Given information:

- 9 crates

- 16 oranges per crate

Calculation:

Total oranges = 9 x 16 = 144

FINAL: 144

Judge rationale: The agent correctly identified the problem, applied the appropriate multiplication operation ($9 \times 16 = 144$), and provided the exact answer matching the reference with clear reasoning and proper format

TEI — example 2 (gold: 390, score: 1.0) *Query:*

A phone costs \$450 with a \$60 trade-in credit. What is the final price?

Agent output:

I need to find the final price of the phone after applying the trade-in credit.

Starting price: \$450
Trade-in credit: \$60

Final price = Starting price - Trade-in credit
Final price = \$450 - \$60
Final price = \$390

FINAL: 390

Judge rationale: The agent correctly identified the problem, applied valid arithmetic reasoning (subtraction of trade-in credit), executed the calculation accurately, and produced the correct final answer of 390 match

TEI — example 3 (gold: 126, score: 1.0) *Query:*

A worker earns \$18 per hour for 7 hours. How much is earned?

Agent output:

To find the total earnings, I need to multiply the hourly rate by the number of hours worked.

Hourly rate: \$18
Hours worked: 7

Total earnings = \$18 x 7 = \$126

FINAL: 126

Judge rationale: The agent correctly identified the problem as multiplication, showed clear reasoning, followed proper format, and arrived at the correct answer of 126 matching the reference.

TEI — example 4 (gold: 220, score: 1.0) *Query:*

A bus travels 55 km/h for 4 hours. How far does it go?

Agent output:

I need to find the distance traveled by the bus.

Using the formula: Distance = Speed x Time

Given information:
- Speed = 55 km/h
- Time = 4 hours

Calculation:
Distance = 55 km/h x 4 hours
Distance = 220 km

FINAL: 220

Judge rationale: The agent correctly identified the problem type, applied the distance formula accurately, showed clear work, and arrived at the correct answer of 220 matching the reference.

D.8 energy_meter_event

Baseline — example 1 (gold: voltage_anomaly, score: 1.0) *Query:*

Residential meter 4782-B registered a sustained drop to 198V at 14:23 hours, remaining below the 220V nominal specification for
→ approximately 47 seconds before self-correction. Customer appliances may have experienced performance degradation during
→ this period.

Agent output:

This event describes a temporary voltage drop below the nominal specification (220V down to 198V) that lasted approximately 47
→ seconds before self-correcting. The customer experienced potential appliance performance issues as a result.

This is a voltage anomaly - a deviation from normal voltage levels that affects grid stability and customer equipment operation.
→ While it self-corrected, it represents an electrical quality issue rather than an outage (which would be a complete loss of
→ power), tampering, overconsumption, billing dispute, or maintenance activity.

FINAL: voltage_anomaly

Judge rationale: The agent correctly classified the event as 'voltage_anomaly' matching the reference answer, provided sound reasoning distinguishing it from other event types, and delivered the answer in proper forma

Baseline — example 2 (gold: `billing_dispute`, score: 0.0) *Query:*

Account representative forwarded case where customer's January statement reflects warm-weather consumption levels despite freezing temperatures and documented furnace usage.

Agent output:

Let me analyze this event:

The customer is reporting that their January bill shows consumption levels typical of warm weather (low usage), but they experienced freezing temperatures and actively used their furnace. This is a discrepancy between expected consumption (should be high due to heating needs) and what the meter recorded (appears low).

This could indicate:

- A meter malfunction or failure to properly register usage
- Potential tampering with the meter to underreport consumption
- A billing error in how the meter data was processed

The key indicator here is the mismatch between documented actual usage (furnace running during freezing weather) and the meter reading/bill. This points to either a meter integrity issue or deliberate manipulation.

Given the context of a billing discrepancy tied to meter accuracy concerns, this is most consistent with a **tamper** classification, as it suggests the meter may not be functioning correctly or may have been interfered with to underreport consumption.

FINAL: tamper

Judge rationale: The agent correctly formatted the response and addressed the classification task, but misclassified a billing dispute as 'tamper' through speculative reasoning that leapt beyond what the input described.

Baseline — example 3 (gold: `maintenance`, score: 1.0) *Query:*

Routine vegetation management along transmission corridor from mile marker 14 to 28 needed before storm season. Tree trimming crews to be deployed.

Agent output:

This event describes a planned activity to maintain the transmission infrastructure by managing vegetation along a corridor before storm season. Tree trimming is a preventive maintenance operation to ensure grid reliability and safety.

FINAL: maintenance

Judge rationale: The agent correctly classified the event as 'maintenance' with sound reasoning about preventive vegetation management, properly formatted output, and perfect alignment with the reference answer.

Baseline — example 4 (gold: `outage`, score: 1.0) *Query:*

Apartment complex lost all electricity at 13:07 following loud transformer noise. Meter communications still active but showing no energy delivery to units.

Agent output:

This event describes a complete loss of electrical power to an apartment complex following an audible transformer failure (loud noise), with meters still communicating but registering zero energy delivery. This is a clear indication of a power outage affecting the distribution infrastructure.

The key indicators are:

- Loss of all electricity to multiple units
- Transformer noise (suggesting equipment failure)
- No energy delivery despite active meter communications
- Sudden onset at a specific time

This is not a tamper (no unauthorized access), not overconsumption (power is completely lost), not a voltage anomaly (complete loss, not anomalous voltage), not billing dispute (operational issue), and not routine maintenance (unexpected failure event).

FINAL: outage

Judge rationale: The agent correctly classified the event as 'outage' with sound reasoning that systematically eliminated alternative classifications, though the task only required the classification itself without explanation.

TEI — example 1 (gold: `voltage_anomaly`, score: 1.0) *Query:*

Residential meter 4782-B registered a sustained drop to 198V at 14:23 hours, remaining below the 220V nominal specification for approximately 47 seconds before self-correction. Customer appliances may have experienced performance degradation during this period.

Agent output:

The event describes a sustained voltage drop below the nominal specification (198V vs 220V nominal), which is an abnormal voltage level condition. This is a voltage anomaly - specifically an undervoltage situation that affected the meter and customer appliances.

FINAL: voltage_anomaly

Judge rationale: The agent correctly classified the event as 'voltage_anomaly' matching the reference answer, provided sound reasoning about the undervoltage condition, and delivered the response in proper format with

TEI — example 2 (gold: `billing_dispute`, score: 1.0) *Query:*

Account representative forwarded case where customer's January statement reflects warm-weather consumption levels despite freezing
↳ temperatures and documented furnace usage.

Agent output:

The customer is reporting that their energy consumption on the January bill appears unusually low (warm-weather levels) when they
↳ actually used their furnace due to freezing temperatures. This suggests either:

1. A billing error where consumption was incorrectly recorded/calculated
2. A meter malfunction that didn't properly register the actual usage
3. A dispute about the charges shown on the statement

The core issue here is a discrepancy between expected consumption (high, due to furnace heating) and what appears on the bill (low,
↳ warm-weather levels). This is fundamentally a customer complaint about billing accuracy and charges.

FINAL: `billing_dispute`

Judge rationale: The agent correctly classified the event as 'billing_dispute' with sound reasoning about the consumption-billing discrepancy, though the extensive explanation exceeded what was strictly necessary for

TEI — example 3 (gold: `maintenance`, score: 1.0) *Query:*

Routine vegetation management along transmission corridor from mile marker 14 to 28 needed before storm season. Tree trimming crews
↳ to be deployed.

Agent output:

This describes a planned, routine maintenance activity - vegetation management and tree trimming along a transmission corridor as
↳ preventive preparation before storm season.

FINAL: `maintenance`

Judge rationale: The agent correctly classified the event as 'maintenance', provided sound reasoning about routine preventive work, and delivered the answer in proper format matching the reference exactly.

TEI — example 4 (gold: `outage`, score: 1.0) *Query:*

Apartment complex lost all electricity at 13:07 following loud transformer noise. Meter communications still active but showing no
↳ energy delivery to units.

Agent output:

The event describes a complete loss of electricity to an apartment complex following transformer noise, with no energy delivery to
↳ the units despite meter communications remaining active. This is a power loss/service interruption scenario.

FINAL: `outage`

Judge rationale: The agent correctly classified the event as 'outage' matching the reference answer, provided sound reasoning about the power loss scenario, and followed the classification task format properly.

D.9 fin_banking_intent

Baseline — example 1 (gold: `dispute_charge`, score: 1.0) *Query:*

There's a transaction on my statement I don't recognize and want reversed.

Agent output:

The customer is reporting an unrecognized transaction on their statement and wants it reversed. This is a request to dispute or
↳ challenge a charge that they don't believe they authorized or recognize.

FINAL: `dispute_charge`

Judge rationale: The agent correctly classified the intent as 'dispute_charge', matching the reference answer exactly, with sound reasoning about the unrecognized transaction requiring reversal.

Baseline — example 2 (gold: `card_arrival`, score: 1.0) *Query:*

When will the replacement card get to my house?

Agent output:

The customer is asking about when they will receive a replacement card at their address. This indicates they are inquiring about the
↳ arrival status of a card.

FINAL: `card_arrival`

Judge rationale: The agent correctly classified the intent as 'card_arrival', provided sound reasoning about the customer's inquiry, and the output matches the reference answer perfectly.

Baseline — example 3 (gold: `card_lost`, score: 1.0) *Query:*

Someone took my wallet with the card inside.

Agent output:

The customer is reporting that their wallet was stolen, which included their card. This indicates the card is lost or has been taken
↪ without authorization.

FINAL: `card_lost`

Judge rationale: The agent correctly classified the intent as 'card_lost', provided sound reasoning about the stolen wallet scenario, and delivered the exact label format required.

Baseline — example 4 (gold: `balance_inquiry`, score: 1.0) *Query:*

What's my current balance?

Agent output:

The customer is asking about their current account balance. This is a straightforward balance inquiry request.

FINAL: `balance_inquiry`

Judge rationale: The agent correctly classified the message as 'balance_inquiry', matching the reference answer exactly, with sound reasoning and proper task execution.

TEI — example 1 (gold: `dispute_charge`, score: 1.0) *Query:*

There's a transaction on my statement I don't recognize and want reversed.

Agent output:

The customer is reporting an unrecognized transaction on their statement that they want reversed. This is a case of a charge they don't recognize, which falls under disputing a charge.

FINAL: `dispute_charge`

Judge rationale: The agent correctly classified the intent as 'dispute_charge', provided sound reasoning about the unrecognized transaction, and delivered the exact label format required, matching the reference answer

TEI — example 2 (gold: `card_arrival`, score: 1.0) *Query:*

When will the replacement card get to my house?

Agent output:

The customer is asking about the arrival time of a replacement card they are expecting to receive.

FINAL: `card_arrival`

Judge rationale: The agent correctly classified the intent as 'card_arrival' matching the reference, provided sound reasoning about the customer's question, and followed the task format properly.

TEI — example 3 (gold: `card_lost`, score: 1.0) *Query:*

Someone took my wallet with the card inside.

Agent output:

The customer is reporting that their wallet containing their card was taken, which means the card is lost or stolen.

FINAL: `card_lost`

Judge rationale: The agent correctly classified the intent as 'card_lost', provided sound reasoning about the stolen wallet scenario, and delivered the exact label matching the reference answer.

TEI — example 4 (gold: `balance_inquiry`, score: 1.0) *Query:*

What's my current balance?

Agent output:

FINAL: `balance_inquiry`

Judge rationale: The agent correctly classified the balance question as 'balance_inquiry', matching the reference answer exactly with proper format and no extraneous content.

D.10 gaming_support

Baseline — example 1 (gold: `gameplay`, score: 1.0) *Query:*

I'm trying to figure out how the crafting system works. I have all the materials in my inventory but the combine button is greyed out.
↳ What am I missing?

Agent output:

The player is asking about the crafting system mechanics - specifically why they can't combine materials even though they have them
↳ in inventory. This is a question about how game systems work and what might be preventing an action.

This could be either a bug (if the combine button should be available but isn't) or a gameplay question (if there's a mechanic they
↳ don't understand). However, the core issue is that they're trying to understand game mechanics and troubleshoot why a
↳ feature isn't working as expected. This falls under gameplay support, as it relates to understanding and using game systems.

FINAL: gameplay

Judge rationale: The agent correctly classified the ticket as 'gameplay', provided sound reasoning about the mechanics question, properly formatted the output with the required label, and the answer matches the reference

Baseline — example 2 (gold: toxicity_report, score: 1.0) *Query:*

There was someone in my squad earlier using a racist username and posting hate symbols in the team chat. Their tag was something like
↳ SS_Commander and they were making everyone uncomfortable.

Agent output:

This ticket describes a player reporting another player's offensive behavior, including a racist username and hate speech in team
↳ chat. This is clearly a report about player conduct and community standards violations.

FINAL: toxicity_report

Judge rationale: The agent correctly identified the ticket as a toxicity report, provided sound reasoning about offensive behavior and community violations, and the final classification matches the reference answer

Baseline — example 3 (gold: connectivity, score: 1.0) *Query:*

Experiencing massive rubber-banding where my character teleports backward every few seconds. Makes it impossible to aim or move
↳ properly in any match.

Agent output:

The player is describing a technical issue where their character is experiencing rubber-banding (teleporting backward repeatedly),
↳ which prevents normal gameplay functionality. This is a network-related problem where the client and server are getting out
↳ of sync, causing the character position to jump around.

This is a connectivity issue - the rubber-banding effect is typically caused by lag, packet loss, or network synchronization problems
↳ between the player's client and the game server.

FINAL: connectivity

Judge rationale: The agent correctly identified rubber-banding as a connectivity issue, provided sound technical reasoning about network synchronization problems, and delivered the exact classification matching the reference

Baseline — example 4 (gold: billing, score: 1.0) *Query:*

The promotional discount code wasn't applied to my order but I was charged the full price. Can you honor the advertised discount?

Agent output:

The player is reporting that a promotional discount code failed to apply during checkout, resulting in being charged full price
↳ instead of the discounted amount. This is a financial transaction issue related to payment and pricing.

FINAL: billing

Judge rationale: The agent correctly classified the ticket as 'billing', provided sound reasoning about the financial/pricing nature of the issue, and the output matches the reference answer perfectly.

TEI — example 1 (gold: gameplay, score: 1.0) *Query:*

I'm trying to figure out how the crafting system works. I have all the materials in my inventory but the combine button is greyed out.
↳ What am I missing?

Agent output:

The player is asking about how the crafting system works and why they can't combine materials despite having them. This is a question
↳ about understanding game mechanics and how to use a feature.

FINAL: gameplay

Judge rationale: The agent correctly classified the ticket as 'gameplay', provided sound reasoning about game mechanics, and the final answer perfectly matches the reference.

TEI — example 2 (gold: toxicity_report, score: 1.0) *Query:*

There was someone in my squad earlier using a racist username and posting hate symbols in the team chat. Their tag was something like
↳ SS_Commander and they were making everyone uncomfortable.

Agent output:

This is a report of abusive behavior and harassment by another player, including use of hate symbols and racist conduct in team chat.
↪ This falls under toxicity_report.
FINAL: toxicity_report

Judge rationale: The agent correctly classified the ticket as 'toxicity_report', provided sound reasoning about hate symbols and racist conduct, and the output matches the reference answer exactly.

TEI — example 3 (gold: connectivity, score: 1.0) *Query:*

Experiencing massive rubber-banding where my character teleports backward every few seconds. Makes it impossible to aim or move
↪ properly in any match.

Agent output:

The player is describing rubber-banding and character teleportation, which are classic symptoms of network lag and position
↪ desynchronization between the client and server. This is a connectivity issue, not a local performance bug.
FINAL: connectivity

Judge rationale: The agent correctly classified the ticket as 'connectivity', provided sound reasoning about network lag and desynchronization, and the output matches the reference answer perfectly.

TEI — example 4 (gold: billing, score: 1.0) *Query:*

The promotional discount code wasn't applied to my order but I was charged the full price. Can you honor the advertised discount?

Agent output:

The player is reporting that a promotional discount code failed to apply to their purchase, resulting in being charged full price
↪ instead of the discounted amount. This is a payment/billing issue involving a discount that should have been applied but
↪ wasn't.
FINAL: billing

Judge rationale: The agent correctly classified the ticket as 'billing', provided sound reasoning about the discount/payment issue, and the final answer matches the reference exactly.

D.11 gov_service_router

Baseline — example 1 (gold: infrastructure, score: 1.0) *Query:*

Can someone please come repair the sidewalk in front of 342 Washington Street? Several concrete slabs are cracked and lifted,
↪ creating a tripping hazard for people walking by.

Agent output:

This is a request to repair public infrastructure (a sidewalk) that poses a safety hazard. The citizen is reporting a maintenance
↪ issue with municipal property that needs attention from the appropriate public works or infrastructure department.
FINAL: infrastructure

Judge rationale: The agent correctly routed the sidewalk repair request to 'infrastructure' with sound reasoning about public works maintenance, matching the reference answer exactly.

Baseline — example 2 (gold: records, score: 1.0) *Query:*

I'm looking for the official paperwork showing when my son was born here in 2015, as his school is requesting it.

Agent output:

You need an official birth certificate for your son. This is a vital record that schools commonly request for enrollment verification.
↪
This request should be routed to the government department that handles vital records and document issuance.
FINAL: records

Judge rationale: The agent correctly identified the request as needing vital records/birth certificate and routed it to 'records' matching the reference answer exactly, with sound reasoning about the document type and

Baseline — example 3 (gold: complaint, score: 1.0) *Query:*

The staff member I spoke with yesterday was incredibly rude and dismissive when I asked a simple question about my application status.
↪ I expect better treatment from public servants.

Agent output:

I understand your frustration. Being treated dismissively by a government staff member is a legitimate concern, and you have every
↪ right to expect professional and courteous service.

This is a complaint about the quality of service you received from a government employee. The appropriate channel for this is the
↳ complaint/feedback system, which allows citizens to formally document service issues and helps departments improve their
↳ staff conduct and customer service standards.

FINAL: complaint

Judge rationale: The agent correctly identified this as a complaint, provided sound reasoning about why it's a complaint rather than a service request, and delivered the exact answer matching the reference, though with

Baseline — example 4 (gold: `taxation`, score: 1.0) *Query:*

I made charitable donations throughout the year and want to make sure I'm claiming them correctly to reduce what I owe.

Agent output:

The citizen is asking about how to properly claim charitable donations on their tax return to minimize their tax liability. This is
↳ clearly a taxation matter, as it involves:
- Tax deductions
- Reducing tax owed
- Proper tax filing procedures

FINAL: taxation

Judge rationale: The agent correctly routed to 'taxation' with sound reasoning about tax deductions and liability, perfectly matching the reference answer with proper format.

TEI — example 1 (gold: `infrastructure`, score: 1.0) *Query:*

Can someone please come repair the sidewalk in front of 342 Washington Street? Several concrete slabs are cracked and lifted,
↳ creating a tripping hazard for people walking by.

Agent output:

This request is about infrastructure maintenance and repair - specifically a public sidewalk that needs repair due to safety hazards.

FINAL: infrastructure

Judge rationale: The agent correctly identified the department as 'infrastructure', provided sound reasoning about sidewalk maintenance being an infrastructure issue, and delivered the answer in the expected format

TEI — example 2 (gold: `records`, score: 1.0) *Query:*

I'm looking for the official paperwork showing when my son was born here in 2015, as his school is requesting it.

Agent output:

FINAL: records

Judge rationale: The agent correctly routed the birth certificate request to 'records' department, matching the reference answer exactly with proper format and no extraneous content.

TEI — example 3 (gold: `complaint`, score: 1.0) *Query:*

The staff member I spoke with yesterday was incredibly rude and dismissive when I asked a simple question about my application status.
↳ I expect better treatment from public servants.

Agent output:

This is a complaint about the quality of service and treatment received from a government employee.

FINAL: complaint

Judge rationale: The agent correctly identified the input as a complaint, provided sound reasoning about service quality issues, and delivered the exact label matching the reference answer.

TEI — example 4 (gold: `taxation`, score: 1.0) *Query:*

I made charitable donations throughout the year and want to make sure I'm claiming them correctly to reduce what I owe.

Agent output:

FINAL: taxation

Judge rationale: The agent correctly routed a tax-related query about charitable donation deductions to the taxation department, matching the reference answer exactly with proper formatting.

D.12 health_triage

Baseline — example 1 (gold: `routine`, score: 1.0) *Query:*

I want a dental cleaning appointment.

Agent output:

This is a straightforward request for preventive dental care. A routine dental cleaning is a standard, non-urgent procedure that
↳ helps maintain oral health and is typically scheduled in advance.

FINAL: routine

Judge rationale: The agent correctly classified the dental cleaning request as 'routine' matching the reference answer, provided sound reasoning about preventive care being non-urgent, and followed the task format app

Baseline — example 2 (gold: **emergency**, score: 1.0) *Query:*

Unconscious friend who is not responding to voice or touch.

Agent output:

This is a critical situation requiring immediate medical intervention.

An unconscious person who is not responding to voice or touch indicates:

- Altered level of consciousness (potentially severe)
- Unknown underlying cause (could be stroke, head injury, overdose, severe infection, cardiac event, etc.)
- Risk of airway compromise
- Potential for rapid deterioration

This patient needs emergency medical services immediately, including:

- Activation of emergency services (911 or local equivalent)
- Assessment of airway, breathing, and circulation
- Rapid diagnostic evaluation
- Potential life-saving interventions

FINAL: emergency

Judge rationale: The agent correctly classified the situation as 'emergency', provided sound clinical reasoning for the urgency level, and fully addressed the triage classification task with accurate execution and com

Baseline — example 3 (gold: **emergency**, score: 1.0) *Query:*

Heavy bleeding that soaks through a bandage in minutes.

Agent output:

This patient is experiencing heavy, rapid bleeding that saturates a bandage within minutes. This indicates:

- Significant blood loss occurring quickly
- Potential for hemorrhagic shock if not controlled
- Possible injury to a major blood vessel
- Risk of life-threatening complications without immediate intervention

This requires immediate medical evaluation and intervention to control bleeding, assess the wound, and prevent shock.

FINAL: emergency

Judge rationale: The agent correctly classified the situation as 'emergency' matching the reference answer, provided sound medical reasoning for the urgency level, and followed the task format appropriately with only

Baseline — example 4 (gold: **self_care**, score: 1.0) *Query:*

Minor bruise on my knee from bumping a table.

Agent output:

This is a minor injury with no signs of serious damage. A small bruise from a table bump typically resolves on its own with basic
↳ home care like ice, rest, and elevation if needed. No medical evaluation is required unless swelling, severe pain, or other
↳ concerning symptoms develop.

FINAL: self_care

Judge rationale: The agent correctly classified the minor bruise as 'self_care', provided sound reasoning about why medical attention is unnecessary, and fully matched the reference answer with proper format.

TEI — example 1 (gold: **routine**, score: 1.0) *Query:*

I want a dental cleaning appointment.

Agent output:

FINAL: routine

Judge rationale: The agent correctly classified a dental cleaning request as 'routine' triage urgency, matching the reference answer with proper format and sound implicit reasoning about non-urgent preventive care.

TEI — example 2 (gold: **emergency**, score: 1.0) *Query:*

Unconscious friend who is not responding to voice or touch.

Agent output:

An unconscious person who is not responding to voice or touch is a life-threatening emergency requiring immediate medical
→ intervention. This could indicate severe conditions such as stroke, head injury, cardiac event, severe intoxication, or
→ other critical medical emergencies.

FINAL: emergency

Judge rationale: The agent correctly classified the situation as 'emergency', provided sound medical reasoning for the urgency level, and fully aligned with the reference answer without any errors or fabrications.

TEI — example 3 (gold: **emergency**, score: 1.0) *Query:*

Heavy bleeding that soaks through a bandage in minutes.

Agent output:

This is a life-threatening condition requiring immediate medical attention. Heavy bleeding that saturates a bandage within minutes
→ indicates significant blood loss and potential for rapid deterioration.

FINAL: emergency

Judge rationale: The agent correctly classified the situation as 'emergency' matching the reference answer, provided sound medical reasoning about blood loss severity, and followed the required format perfectly.

TEI — example 4 (gold: **self_care**, score: 1.0) *Query:*

Minor bruise on my knee from bumping a table.

Agent output:

FINAL: self_care

Judge rationale: The agent correctly classified a minor bruise as 'self_care', matching the reference answer exactly with proper format and no extraneous content.

D.13 hospitality_review_stars

Baseline — example 1 (gold: 5 stars, score: 1.0) *Query:*

I have been coming to this particular Filiberto's for a while now and the food and friendly service are consistently top notch. The
→ bar is stocked with good tequila and they usually have something going on like karaoke in the evening to keep things
→ interesting. It fills an interesting niche of taco shop fast food along with a nice bar that I didn't think would gel
→ together but surprisingly does. Food and drink specials for the most part are also pretty great as well.

Agent output:

This review is clearly positive and enthusiastic. The customer highlights:

- Consistent quality of food and service ("consistently top notch")
- Friendly service
- Well-stocked bar with good tequila
- Entertainment options (karaoke)
- Appreciation for the unique concept (taco shop + bar combination)
- Good food and drink specials

The reviewer is a repeat customer ("coming to this particular Filiberto's for a while now") and expresses satisfaction with multiple
→ aspects of the establishment. There are no complaints or negative points mentioned. The tone is appreciative and
→ recommending.

This clearly warrants a 5-star rating.

FINAL: 5 stars

Judge rationale: The agent correctly predicted 5 stars with sound reasoning that accurately identified all positive indicators in the review, addressed the exact task, and matched the reference answer perfectly.

Baseline — example 2 (gold: 4 stars, score: 0.0) *Query:*

With 500+ Reviews, Most of all it has been said but I will offer this:\nFox Concepts Restaurants consistently pulls off a memorable,
→ pleasant experience that is tasty, full of quality and will, occasionally, have a memorable service experience. \nAll
→ restaurants have their ups & downs, but Olive & Ivy is consistently good, innovative and the atmosphere is quite nice;
→ inside or outside.\nDrinks are tasty, Scallopini and Cannelloni are both just Yum! Never had a bad experience here.\n\n
→ nOverall Rating: 4.5 Stars\nService: 4.1 Stars\nFood: 4.6 Stars \nQuality: 4.5 Stars\nCleanliness: 4.2 Stars\nValue: 3.9
→ Stars

Agent output:

This review is highly positive overall. The reviewer praises:

- Consistent quality and memorable experiences
- Innovative menu items (specifically praising Scallopini and Cannelloni)

- Pleasant atmosphere (both indoor and outdoor)
- Good drinks
- Never had a bad experience

The reviewer's own detailed ratings show:

- Food: 4.6 stars (excellent)
- Quality: 4.5 stars (very good)
- Service: 4.1 stars (good)
- Cleanliness: 4.2 stars (good)
- Value: 3.9 stars (acceptable, slightly lower)
- Overall stated: 4.5 stars

The only minor criticism is the value rating at 3.9, suggesting prices are somewhat high, but this doesn't diminish the overall very positive sentiment. The reviewer explicitly states they've "never had a bad experience" and consistently recommends the restaurant.

This clearly warrants a 5-star rating, as the reviewer is expressing strong satisfaction with the establishment despite the slightly lower value score.

FINAL: 5 stars

Judge rationale: The agent correctly formatted the prediction and reasoned through the review systematically, but predicted 5 stars when the reference answer is 4 stars, and the reviewer's own stated overall rating was

Baseline — example 3 (gold: 1 star, score: 1.0) *Query:*

Just sad\nOrdered online paid on line when I arrived at the pick up time they told me that their fax machine ran out of paper and they did not get my order\nI am now sitting in the dining room writing this review on my phone \n\nNo attempt to make this right \n\nI hope the pizza is good

Agent output:

This review describes a significant service failure. The customer:

- Placed and paid for an online order
- Arrived at the scheduled pickup time
- Was told the order was never received due to a technical issue (fax machine out of paper)
- Experienced no attempt by the restaurant to resolve the situation
- Is now waiting, frustrated enough to write a negative review

The customer's tone is clearly disappointed and frustrated. While they haven't explicitly stated the pizza was bad, the phrase "I hope the pizza is good" suggests skepticism and that this experience has already soured their perception. The lack of any service recovery attempt (offering to expedite the order, provide a discount, etc.) makes this worse.

This is a clear operational failure with poor customer service recovery.

FINAL: 1 star

Judge rationale: The agent correctly predicted 1 star with sound reasoning about service failure, proper format, and complete alignment with the reference answer.

Baseline — example 4 (gold: 2 star, score: 1.0) *Query:*

Hmmmm really just not sure about this place at all, went for a lunch and just wasn't impressed. I had the quesadilla with carnitas which was ok, we also got chicken wings and they were well awful! Cold, greasy and just not very good.\n\nThe search for good tex-mex goes on!

Agent output:

The reviewer expresses clear dissatisfaction with their experience. While they found the quesadilla "ok," they were notably disappointed with the chicken wings (described as "well awful," cold, and greasy). The overall tone is negative - they state they "just weren't impressed" and are still searching for better tex-mex elsewhere.

The presence of one acceptable dish prevents this from being a 1-star review, but the significant complaints about quality and the overall lack of satisfaction point to a below-average experience.

FINAL: 2 star

Judge rationale: The agent correctly predicted '2 star' matching the reference, provided sound reasoning about mixed-but-negative sentiment, followed the required format, and delivered an accurate answer without fabrications.

TEI — example 1 (gold: 5 stars, score: 1.0) *Query:*

I have been coming to this particular Filiberto's for a while now and the food and friendly service are consistently top notch. The bar is stocked with good tequila and they usually have something going on like karaoke in the evening to keep things interesting. It fills an interesting niche of taco shop fast food along with a nice bar that I didn't think would gel together but surprisingly does. Food and drink specials for the most part are also pretty great as well.

Agent output:

This review is overwhelmingly positive. The reviewer:

- Explicitly states they are a repeat customer ("coming to this particular Filiberto's for a while now")
- Praises both food and service as "consistently top notch"
- Highlights additional positive amenities (well-stocked bar, entertainment like karaoke)
- Expresses pleasant surprise at how well the concept works ("surprisingly does")

- Appreciates the specials ("pretty great")
- Uses enthusiastic language throughout with no significant complaints or reservations

There are no negative issues mentioned, no drawbacks acknowledged, and the overall tone is one of genuine satisfaction and recommendation. This is a clear case of strong, consistent praise across multiple dimensions of the experience.

FINAL: 5 stars

Judge rationale: The agent correctly predicted 5 stars with sound reasoning that systematically identified positive sentiment indicators, properly formatted the answer, and matched the reference exactly.

TEI — example 2 (gold: 4 stars, score: 1.0) *Query:*

With 500+ Reviews, Most of all it has been said but I will offer this:\nFox Concepts Restaurants consistently pulls off a memorable, pleasant experience that is tasty, full of quality and will, occasionally, have a memorable service experience. \nAll restaurants have their ups & downs, but Olive & Ivy is consistently good, innovative and the atmosphere is quite nice; inside or outside.\nDrinks are tasty, Scallopini and Cannelloni are both just Yum! Never had a bad experience here.\n\nOverall Rating: 4.5 Stars\nService: 4.1 Stars\nFood: 4.6 Stars \nQuality: 4.5 Stars\nCleanliness: 4.2 Stars\nValue: 3.9 Stars

Agent output:

****Reasoning:****

This review is overwhelmingly positive with strong, consistent praise across multiple dimensions:

- ****Explicit positive language**:** "memorable, pleasant experience," "tasty, full of quality," "consistently good, innovative," "quite nice," "Yum!"
- ****No significant complaints**:** The reviewer acknowledges that "all restaurants have their ups & downs" but explicitly states "Never had a bad experience here"
- ****Strong food praise**:** Multiple dishes called out as excellent (Scallopini and Cannelloni described as "Yum!")
- ****Atmosphere and drinks**:** Both praised as tasty and nice
- ****Tone**:** Enthusiastic and recommending, with clear intent to return
- ****Numerical ratings provided**:** All sub-ratings are 3.9-4.6 stars, with an overall 4.5 rating

The reviewer acknowledges minor variations in service quality ("occasionally" memorable) but this is framed as normal variation, not a significant detraction. The overall sentiment is strongly positive with no deal-breaker issues, warnings, or substantial disappointments.

This is a clear 4-star review: mostly positive with minor acknowledgments of normal restaurant variability, but strong praise dominating and no hesitation about returning.

FINAL: 4 stars

Judge rationale: The agent correctly predicted 4 stars matching the reference, with sound reasoning analyzing sentiment, explicit ratings, and review tone, properly formatted output, and no fabrication.

TEI — example 3 (gold: 1 star, score: 1.0) *Query:*

Just sad\nOrdered online paid on line when I arrived at the pick up time they told me that their fax machine ran out of paper and they did not get my order\n\nI am now sitting in the dining room writing this review on my phone \n\nNo attempt to make this right \n\nI hope the pizza is good

Agent output:

****Analysis:****

This review describes a fundamental service failure:

- Customer paid in advance for an online order
- Restaurant failed to receive the order due to a technical issue (fax machine out of paper)
- Customer arrived at the agreed pickup time to find no order prepared
- ****Critical issue**:** "No attempt to make this right" - the restaurant did not offer service recovery, compensation, or solutions
- The reviewer is sitting in the dining room, clearly frustrated and disappointed
- Opening phrase "Just sad" sets a deeply disappointed tone
- The sarcastic closing ("I hope the pizza is good") suggests skepticism and resignation rather than optimism

This is a deal-breaker situation: the customer was let down by a core service failure (order not received/prepared), and the restaurant compounded the problem by failing to attempt any recovery or compensation. The lack of effort to resolve the issue is explicitly called out as a major problem.

While the reviewer hasn't explicitly said "never again," the combination of being stood up on a paid order, no service recovery attempt, and the overall "just sad" tone indicates a significantly negative experience that would deter future visits.

FINAL: 1 star

Judge rationale: The agent correctly predicted 1 star with sound reasoning about service failure, lack of recovery attempt, and negative tone, perfectly matching the reference answer.

TEI — example 4 (gold: 2 star, score: 1.0) *Query:*

Hmmmm really just not sure about this place at all, went for a lunch and just wasn't impressed. I had the quesadilla with carnitas which was ok, we also got chicken wings and they were well awful! Cold, greasy and just not very good.\n\nThe search for good tex-mex goes on!

Agent output:

****Analysis:****

The reviewer expresses clear disappointment with their experience:

- ****Overall tone****: Uncertain and unimpressed ("really just not sure about this place at all," "wasn't impressed")
- ****Food quality issues****:
 - Quesadilla described as merely "ok" (lukewarm assessment)
 - Chicken wings explicitly criticized as "well awful" with specific problems (cold, greasy, poor quality)
- ****No redeeming elements****: No positive comments about service, atmosphere, or any menu items
- ****Conclusion****: Actively searching elsewhere ("The search for good tex-mex goes on!") suggests they won't return
- ****Ratio****: Multiple substantive complaints with no genuine positives to balance them

This is a predominantly negative experience where problems outweigh any positives. The reviewer had at least two dishes, and both
↳ disappointed (one mediocre, one explicitly bad). The tone is disappointed rather than enthusiastic, and they're clearly
↳ moving on to find better options elsewhere.

This does not rise to 1 star (no explicit "never again" or strong warning language), but it clearly falls below 3 stars (which
↳ requires balanced positives and negatives or mostly positive with minor drawbacks).

FINAL: 2 star

Judge rationale: The agent correctly predicted '2 star' matching the reference, with sound reasoning analyzing tone and food quality, proper format, and no fabrication.

D.14 hr_resume_fit

Baseline — example 1 (gold: overqualified, score: 1.0) *Query:*

Candidate has 15 years leading enterprise sales teams with P&L responsibility for \$50M+ organizations. Our sales associate role is an
↳ individual contributor position focused on inside sales to SMB clients. Worried they'll be bored within months and leave
↳ for a leadership role elsewhere.

Agent output:

****Reasoning:****

This candidate has extensive executive-level experience managing large enterprise sales organizations with significant P&L
↳ responsibility. The role in question is an individual contributor position at the SMB level--a substantial step down in
↳ scope, complexity, and seniority.

Key concerns:

- ****Significant overqualification****: 15 years of leadership experience managing \$50M+ portfolios far exceeds what an inside sales
↳ associate role requires
- ****Flight risk****: The concern about boredom and departure is well-founded. Candidates at this level typically seek growth, autonomy,
↳ and strategic responsibility
- ****Role mismatch****: An IC position lacks the leadership, decision-making authority, and scope this candidate is accustomed to
- ****Retention likelihood****: Low probability of long-term retention once the candidate realizes the limited growth trajectory

While the candidate clearly has strong sales skills, the role doesn't align with their experience level or likely career expectations.
↳

FINAL: overqualified

Judge rationale: The agent correctly classified the candidate as 'overqualified' with thorough, logically sound reasoning that directly addresses the task requirements and matches the reference answer perfectly.

Baseline — example 2 (gold: underqualified, score: 1.0) *Query:*

Candidate's background is entirely in retail management at small stores. The position requires a licensed attorney with 10+ years of
↳ corporate law experience and expertise in M&A transactions. No law degree or legal credentials present.

Agent output:

This candidate lacks the fundamental qualifications for the role. They have retail management experience but the position requires:
- Law degree and active bar license (not present)
- 10+ years of corporate law experience (candidate has none)
- Specialized M&A transaction expertise (not applicable to retail background)

The candidate is missing both the required credentials and relevant professional experience entirely. This is a clear mismatch.

FINAL: underqualified

Judge rationale: The agent correctly classified the candidate as 'underqualified' with sound reasoning that systematically identified all missing qualifications (law degree, bar license, corporate law experience, M&A

Baseline — example 3 (gold: wrong_role, score: 1.0) *Query:*

This person has deep expertise in financial auditing and compliance, which is valuable but not aligned with our finance operations
↳ analyst position. We need someone to handle FP&A, budgeting forecasts, and operational metrics rather than regulatory
↳ reporting.

Agent output:

```
**Reasoning:**
The candidate possesses strong technical expertise in financial auditing and compliance--valuable skills in finance. However, the
  ↳ role requires FP&A, budgeting, forecasting, and operational metrics analysis. These are distinct skill sets. While there's
  ↳ overlap in financial acumen, auditing/compliance focuses on historical accuracy and regulatory requirements, whereas FP&A
  ↳ emphasizes forward-looking analysis, strategic planning, and business operations metrics.

The candidate's background suggests they may lack direct experience in the core competencies needed (forecasting models, operational
  ↳ analytics, budget planning). This is a skills mismatch rather than a cultural or qualification level issue.

FINAL: wrong_role
```

Judge rationale: The agent correctly classified as 'wrong_role' with sound reasoning distinguishing auditing/compliance from FP&A skills, proper format, and perfect match to reference answer.

Baseline — example 4 (gold: strong_fit, score: 1.0) *Query:*

```
Ideal background for our content marketing manager opening. Grew organic traffic by 300% over two years through SEO optimization and
  ↳ strategic content planning. Writing samples demonstrate versatility across technical and consumer audiences. Analytics
  ↳ fluency evident in discussion of attribution modeling and conversion optimization. Team collaboration stories highlighted
  ↳ empathy and cross-functional partnership. Genuine excitement about our brand and growth trajectory.
```

Agent output:

```
This candidate demonstrates strong alignment with the content marketing manager role:

**Strengths:**
- Proven track record of measurable impact (300% organic traffic growth)
- Technical competency in SEO, analytics, and attribution modeling
- Demonstrated versatility in writing across different audiences
- Evidence of soft skills (empathy, collaboration) essential for cross-functional work
- Authentic enthusiasm for the company and mission

**Assessment:**
The candidate shows expertise directly relevant to the role, concrete results that validate their capabilities, and cultural fit
  ↳ indicators. No red flags regarding overqualification, underqualification, or misalignment with the position.

FINAL: strong_fit
```

Judge rationale: The agent correctly classified the candidate as 'strong_fit' with well-structured supporting reasoning that directly addresses the task requirements, though the additional explanation exceeds minimal

TEI — example 1 (gold: overqualified, score: 1.0) *Query:*

```
Candidate has 15 years leading enterprise sales teams with P&L responsibility for $50M+ organizations. Our sales associate role is an
  ↳ individual contributor position focused on inside sales to SMB clients. Worried they'll be bored within months and leave
  ↳ for a leadership role elsewhere.
```

Agent output:

```
The candidate's extensive leadership experience (15 years managing enterprise teams with significant P&L responsibility) far exceeds
  ↳ the requirements of an individual contributor sales associate role focused on SMB inside sales. This represents a multi-
  ↳ level seniority mismatch in the same functional area (sales). While the candidate clearly has strong sales capabilities,
  ↳ their career trajectory and experience level suggest they would likely find this role understimulating and seek advancement
  ↳ or opportunities elsewhere relatively quickly, creating retention risk.

FINAL: overqualified
```

Judge rationale: The agent correctly classified the candidate as 'overqualified' with sound reasoning about the seniority mismatch and retention risk, fully addressing the task requirements.

TEI — example 2 (gold: underqualified, score: 1.0) *Query:*

```
Candidate's background is entirely in retail management at small stores. The position requires a licensed attorney with 10+ years of
  ↳ corporate law experience and expertise in M&A transactions. No law degree or legal credentials present.
```

Agent output:

```
The candidate lacks the essential qualifications for this role. They have retail management experience but no law degree, legal
  ↳ credentials, or corporate law background. The position requires a licensed attorney with specialized M&A experience--a
  ↳ fundamentally different functional area and professional discipline.

FINAL: underqualified
```

Judge rationale: The agent correctly classified the candidate as 'underqualified' with sound reasoning about the mismatch between retail management background and attorney requirements, perfectly matching the reference

TEI — example 3 (gold: wrong_role, score: 1.0) *Query:*

```
This person has deep expertise in financial auditing and compliance, which is valuable but not aligned with our finance operations
  ↳ analyst position. We need someone to handle FP&A, budgeting forecasts, and operational metrics rather than regulatory
  ↳ reporting.
```

Agent output:

The candidate's core expertise is in financial auditing and compliance--a different functional area than the finance operations
↳ analyst role, which requires FP&A, budgeting, forecasting, and operational metrics focus. While both are finance-related,
↳ they represent distinct career paths and skill sets. The candidate's specialization doesn't align with the position's core
↳ functions.

FINAL: wrong_role

Judge rationale: The agent correctly classified the candidate as 'wrong_role' with sound reasoning that accurately explains the mismatch between auditing/compliance expertise and the FP&A/operations analyst requiremen

TEI — example 4 (gold: strong_fit, score: 1.0) *Query:*

Ideal background for our content marketing manager opening. Grew organic traffic by 300% over two years through SEO optimization and
↳ strategic content planning. Writing samples demonstrate versatility across technical and consumer audiences. Analytics
↳ fluency evident in discussion of attribution modeling and conversion optimization. Team collaboration stories highlighted
↳ empathy and cross-functional partnership. Genuine excitement about our brand and growth trajectory.

Agent output:

This candidate demonstrates strong alignment with the content marketing manager role:

- ****Core skills match**:** Proven SEO expertise, strategic content planning, and analytics proficiency directly address the position's requirements
- ****Measurable impact**:** 300% organic traffic growth shows concrete results in the functional area
- ****Communication ability**:** Writing samples across diverse audiences indicate versatility needed for the role
- ****Technical competency**:** Attribution modeling and conversion optimization knowledge demonstrates analytical depth
- ****Behavioral indicators**:** Team collaboration stories and empathy suggest positive cultural fit
- ****Engagement**:** Genuine enthusiasm about the brand and growth trajectory indicates motivation and alignment

No significant gaps, concerns, or red flags are present. The candidate's experience level appears appropriately matched to the role (↳ not overqualified or underqualified).

FINAL: strong_fit

Judge rationale: The agent correctly classified the candidate as 'strong_fit' matching the reference answer, provided well-structured reasoning that directly addresses the hiring-stage classification task, followed pr

D.15 insurance_claim_type

Baseline — example 1 (gold: theft, score: 1.0) *Query:*

Someone broke into my car last night while it was parked on the street and took my laptop, GPS device, and about \$200 in cash from
↳ the center console. The driver's side window was smashed.

Agent output:

This claim involves items being taken from a vehicle without authorization. While there is property damage (the broken window), the
↳ primary loss is the theft of personal items from inside the car.

The key elements are:

- Unauthorized entry into the vehicle
- Stolen items (laptop, GPS, cash)
- This is a theft claim rather than an auto collision claim (which would involve vehicle damage from an accident)

FINAL: theft

Judge rationale: The agent correctly classified the claim as 'theft' with sound reasoning about unauthorized entry and stolen items, though the task only required the classification itself without extended explanation

Baseline — example 2 (gold: liability, score: 1.0) *Query:*

While hosting a dinner party, my chandelier fell from the ceiling and injured one of my guests, causing a head laceration requiring
↳ surgery. The guest is pursuing compensation for their injuries.

Agent output:

This claim involves a guest being injured at the policyholder's property due to a falling chandelier. The guest is seeking
↳ compensation for medical expenses and damages resulting from the injury.

This is a liability claim because:

- It involves bodily injury to a third party (the guest)
- The policyholder may be held legally responsible for the injury
- The guest is pursuing compensation against the policyholder
- This falls under premises liability coverage

While the guest will have medical expenses, the claim itself is being routed based on the liability exposure and the third-party
↳ injury claim.

FINAL: liability

Judge rationale: The agent correctly classified the claim as 'liability' with sound reasoning about third-party injury and premises liability, though the task only required classification without explanation.

Baseline — example 3 (gold: medical, score: 1.0) *Query:*

Submitted bills for ambulance transport, ER visit, CT scan, and follow-up appointments related to head trauma sustained during a fall
↳ at a retail store.

Agent output:

This claim involves medical expenses resulting from an injury sustained at a retail location. The submitted bills cover ambulance
↳ transport, emergency room visit, CT scan, and follow-up medical appointments - all medical services related to treating head
↳ trauma from a fall.

While the fall occurred at a retail store (which could potentially involve a liability component), the claim itself is being
↳ submitted for the medical costs incurred for treatment and care.

FINAL: medical

Judge rationale: The agent correctly classified the claim as 'medical', provided sound reasoning distinguishing medical costs from liability aspects, and delivered the exact answer matching the reference with proper f

Baseline — example 4 (gold: auto_collision, score: 1.0) *Query:*

I was stopped at a red light when the vehicle behind me failed to brake and rear-ended my car. The trunk won't close properly now,
↳ the rear bumper is cracked, and the taillights on both sides are broken.

Agent output:

This claim describes a vehicle collision where the claimant's car was struck from behind while stationary at a traffic light. The
↳ damage includes structural damage (trunk, bumper) and broken taillights resulting from the impact.

This is clearly an auto collision claim, as it involves damage to a vehicle caused by a collision with another vehicle.

FINAL: auto_collision

Judge rationale: The agent correctly classified the claim as auto_collision with sound reasoning about the rear-end collision scenario, though the task only required the classification label itself.

TEI — example 1 (gold: theft, score: 1.0) *Query:*

Someone broke into my car last night while it was parked on the street and took my laptop, GPS device, and about \$200 in cash from
↳ the center console. The driver's side window was smashed.

Agent output:

This claim involves stolen property taken from a vehicle due to criminal activity (break-in and theft). While the vehicle itself
↳ sustained damage (broken window), the primary claim is for the stolen items (laptop, GPS device, cash) resulting from a
↳ burglary.

FINAL: theft

Judge rationale: The agent correctly classified the claim as 'theft' matching the reference answer, provided sound reasoning distinguishing theft from vehicle damage, and delivered the answer in proper format with app

TEI — example 2 (gold: liability, score: 1.0) *Query:*

While hosting a dinner party, my chandelier fell from the ceiling and injured one of my guests, causing a head laceration requiring
↳ surgery. The guest is pursuing compensation for their injuries.

Agent output:

This claim involves a guest being injured at the policyholder's home due to the chandelier falling, and the guest is seeking
↳ compensation. This is a third-party injury claim where the policyholder may be held liable for the harm caused to another
↳ person.

FINAL: liability

Judge rationale: The agent correctly classified the claim as 'liability' with sound reasoning about third-party injury and policyholder responsibility, fully matching the reference answer.

TEI — example 3 (gold: medical, score: 1.0) *Query:*

Submitted bills for ambulance transport, ER visit, CT scan, and follow-up appointments related to head trauma sustained during a fall
↳ at a retail store.

Agent output:

This claim involves medical expenses for the policyholder's own healthcare treatment following an injury. The bills are for ambulance
↳ transport, emergency room visit, CT scan, and follow-up appointments - all medical services related to treating the head
↳ trauma.

FINAL: medical

Judge rationale: The agent correctly classified the claim as 'medical' with sound reasoning about the medical services

involved, properly formatted output, and perfect alignment with the reference answer.

TEI — example 4 (gold: `auto_collision`, score: 1.0) *Query:*

I was stopped at a red light when the vehicle behind me failed to brake and rear-ended my car. The trunk won't close properly now,
→ the rear bumper is cracked, and the taillights on both sides are broken.

Agent output:

This claim describes damage to a vehicle resulting from a collision (rear-end accident). The damage to the trunk, bumper, and
→ taillights are all consequences of the vehicle being struck by another car.

FINAL: `auto_collision`

Judge rationale: The agent correctly classified the claim as `auto_collision` with sound reasoning about the rear-end accident, matching the reference answer perfectly despite including explanatory context.

D.16 `it_email_spam`

Baseline — example 1 (gold: `ham`, score: 1.0) *Query:*

re : park n ride linda , i talked with linda and michelle this week on this issue . what michelle wanted to know was does tw have
→ operators who have pnr agreements . currently , tw does not have any operators with pnr agreements because they have
→ mandatory obas on tw . with our 637 compliance filing tw is requesting that operators move their imbalance to a pnr if it is
→ over a certin volume . also eog does pool on tw so as a shipper we could get them to enter into a pnr . they would need to
→ have a shipper transport it to and from the pnr point . i also told michelle that all points the flow any volumes on tw are
→ required to be an an oba . michelle said that they were able to resolve this issue with eog . thanks ramona

Agent output:

This email appears to be a legitimate business communication regarding operational matters.

****Reasoning:****

- The email discusses specific business topics (Park and Ride agreements, operators, compliance filings, OBA requirements)
- References real people (Linda, Michelle, Ramona) and specific companies (TW, EOG)
- Contains detailed technical/operational information relevant to energy/transportation industry
- Uses industry-specific terminology appropriately (PNR, OBA, imbalance, shipper)
- Has a professional tone and clear business purpose
- No suspicious links, requests for personal information, or phishing indicators
- Appears to be an internal business discussion or reply

FINAL: `ham`

Judge rationale: The agent correctly classified the email as 'ham' (legitimate), provided sound reasoning based on business context and terminology, addressed the triage task directly, and matched the reference answer

Baseline — example 2 (gold: `spam`, score: 1.0) *Query:*

ourr new great offr how to save on inflate your medications over 70 % . phar cabriolet mshop - successfull and proven way to save
→ topfull your mon indicator ey . whinny v murrain ag litigious al resent lu anglicist l r concrete a talebearer cl i
→ helvetic sva atrabilious l serfdom m andmanyother . fighterpilot best prices . worldwide brekker shlpplng . easy or engender
→ der form . total confidentialai hutment ity . 250 , 000 satisfied rustless customers . order secateurs today and save !

Agent output:

This email exhibits multiple characteristics of spam:

1. ****Misspellings and garbled text**:** "ourr", "offr", "medlcations", "phar cabriolet mshop", "topfull", "indicator ey", "whinny v
→ murrain", "brekker shlpplng", "confidentialai hutment ity" - these appear intentionally obfuscated to evade spam filters
2. ****Pharmaceutical spam**:** Offering discounted medications (70% savings) is a classic spam tactic
3. ****Nonsensical filler text**:** Random words like "litigious", "talebearer", "helvetic", "atrabilious", "serfdom" appear to be
→ inserted to confuse filters
4. ****Common spam promises**:** "worldwide shipping", "easy order form", "total confidentiality", "satisfied customers"
5. ****Urgency tactics**:** "order today and save!"
6. ****Suspicious claims**:** 250,000 satisfied customers with no legitimate business name

This is clearly a pharmaceutical spam email using obfuscation techniques to bypass email security systems.

FINAL: `spam`

Judge rationale: The agent correctly identified the email as spam with sound reasoning about obfuscation techniques, pharmaceutical red flags, and spam characteristics, perfectly matching the reference answer.

Baseline — example 3 (gold: `ham`, score: 1.0) *Query:*

o : \ research \ exotica access our records indicate that you are the owners / approvers for the o : \ research \ exotica folder . if
→ this is correct , please reply back with your confirmation to information risk management . in an effort to update our

```

→ files as well as to do some cleanup on the directory membership , we are including the current membership list for the
→ following nt groups . please review the membership list and advise if any deletions are required . group name group
→ description data _ exotica change access to o : \ reasearch \ exotica , ohara , scarlet data _ exotica _ ro read - only
→ access to o : \ reasearch \ exotica , ohara , scarlet datapwr _ exoticarw change access to m : \ exotica data _ exotica user
→ id full name adhar amitava dhar / hou / ees btiner brent tiner / corp / enron clandry chad landry / hou / ect cuus charles
→ uus / hou / ect dmaxwel david maxwell / hou / ect dvitrel david vitrella gmasson grant masson / hou / ect jbusss jd buss /
→ hou / ect khopper kevin hopper / hou / ect mlay mark lay / hou / ect mvasque miguel vasquez / hou / ect pkrishn pinnamaneni
→ krishnarao / hou / ect pzadoro pavel zadorozhny / hou / ect sgibner stinson gibner / hou / ect tha

```

Agent output:

I'll analyze this email for spam/legitimacy indicators.

****Analysis:****

This appears to be an internal corporate IT/security communication regarding access control management. Key indicators:

****Legitimate signals:****

- Addresses a specific internal folder path (o:\research\exotica)
- References "Information Risk Management" - a typical internal security function
- Contains detailed NT group membership lists with specific user IDs and names
- Discusses routine administrative tasks (access review, directory cleanup)
- Uses internal corporate domain structure (/hou/, /corp/, /ect suffixes)
- Requests confirmation from folder owners - standard access governance procedure
- Contains realistic employee names and organizational structure

****Potential concerns:****

- No sender information visible in the provided text
- Could potentially be a phishing attempt mimicking internal IT communications
- However, the level of specific detail (actual employee names, folder paths, group names) would be difficult for external attackers
 - to fabricate convincingly

****Verdict:****

This has the hallmarks of legitimate internal IT governance communication. The specific internal details, proper procedure (

- requesting owner confirmation), and routine nature of the request are consistent with genuine corporate access management.
- However, without seeing the sender's email address and domain, there's a small risk this could be a sophisticated internal
- phishing attempt.

Given the content structure and specificity, this appears to be ****ham**** (legitimate internal communication).

FINAL: ham

Judge rationale: The agent correctly classified the email as ham with sound reasoning about internal corporate indicators, proper task execution with clear final answer, and perfect alignment with the reference answer

Baseline — example 4 (gold: spam, score: 1.0) Query:

```

tutored best n @ k . ed lolitas . . . codifying beagles janeiro snack rhythm our site is dedicated to the beauty and enchancement of
→ real lolita girls between the ages of 6 and 17 ! we have some of the most beautiful under . age models in the world . some
→ of our lolita girls are professionals , some are innocent . . . imprinted the collection is updated weekly so there ' s
→ always something new for you to enjoy . worried watch them . . . here : http : / / nonstandart . biz / home / ? = improves
→ thank you , bye brookhaven remedies grumble miracle touch allotments normals mailer amiga

```

Agent output:

This email contains clear indicators of illegal content involving the exploitation of minors. The message explicitly references "

- lolita girls between the ages of 6 and 17" and describes them as "models," along with references to viewing content. This is
- describing child sexual abuse material (CSAM).

This is not spam in the traditional sense--it's a criminal solicitation that must be reported to law enforcement.

FINAL: spam

Judge rationale: The agent correctly identified the email as spam, provided sound reasoning about the illegal nature of the content, and delivered the exact required classification despite adding contextual warnings

TEI — example 1 (gold: ham, score: 1.0) Query:

```

re : park n ride linda , i talked with linda and michelle this week on this issue . what michelle wanted to know was does tw have
→ operators who have pnr agreements . currently , tw does not have any operators with pnr agreements because they have
→ mandatory obas on tw . with our 637 compliance filing tw is requesting that operators move their imbalance to a pnr if it is
→ over a certin volume . also eog does pool on tw so as a shipper we could get them to enter into a pnr . they would need to
→ have a shipper transport it to and from the pnr point . i also told michelle that all points the flow any volumes on tw are
→ required to be an an oba . michelle said that they were able to resolve this issue with eog . thanks ramona

```

Agent output:

ham

Judge rationale: The agent correctly classified the email as 'ham' (legitimate business communication about park-n-ride agreements), matching the reference answer with proper format and accurate judgment.

TEI — example 2 (gold: spam, score: 1.0) Query:

```

ourr new great offr how to save on inflate your medications over 70 % . phar cabriolet mshop - successfull and proven way to save

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↪ topfull your mon indicatory ey . whinny v murrain ag litigious al resent lu anglicist l r concrete a talebearer cl i
 ↪ helvetic sva atrabilious l serfdom m andmanyother . fighterpilot best prlces . worldwide brekker shlpplng . easy or engender
 ↪ der form . total confidentialai hutment ity . 250 , 000 satisfied rustless customers . order secateurs today and save !

Agent output:

This email exhibits multiple clear SPAM indicators:

- Heavily discounted medications ("save over 70%")
- Poor grammar and deliberate character substitutions ("medlcations," "phar cabriolet," "prlces," "shlpplng")
- Random capitalization and nonsensical words mixed throughout
- Generic sales pitch about saving money on medications
- Pressure to "order today"
- Claims of "satisfied customers" and "worldwide shipping"
- Suspicious pharmacy-related commercial offer

FINAL: spam

Judge rationale: The agent correctly identified the email as spam with comprehensive, accurate reasoning that matches the reference answer perfectly.

TEI — example 3 (gold: ham, score: 1.0) *Query:*

o : \ research \ exotica access our records indicate that you are the owners / approvers for the o : \ research \ exotica folder . if
 ↪ this is correct , please reply back with your confirmation to information risk management . in an effort to update our
 ↪ files as well as to do some cleanup on the directory membership , we are including the current membership list for the
 ↪ following nt groups . please review the membership list and advise if any deletions are required . group name group
 ↪ description data _ exotica change access to o : \ reasearch \ exotica , ohara , scarlet data _ exotica _ ro read - only
 ↪ access to o : \ reasearch \ exotica , ohara , scarlet datapwr _ exoticarw change access to m : \ exotica data _ exotica user
 ↪ id full name adhar amitava dhar / hou / ees btiner brent tiner / corp / enron clandry chad landry / hou / ect cuus charles
 ↪ uus / hou / ect dmaxwel david maxwell / hou / ect dvitrele david vitrella gmasson grant masson / hou / ect jbusss jd buss /
 ↪ hou / ect khopper kevin hopper / hou / ect mlay mark lay / hou / ect mvasque miguel vasquez / hou / ect pkrishn pinnamaneni
 ↪ krishnarao / hou / ect pzadoro pavel zadorozhny / hou / ect sgibner stinson gibner / hou / ect tha

Agent output:

This email appears to be internal business correspondence regarding access control and directory management. It contains:

- Specific internal folder paths (o:\research\exotica)
- References to an internal department (Information Risk Management)
- Specific employee names and user IDs with company domains (/hou/, /corp/, /enron)
- Technical discussion about NT group memberships and access permissions
- A legitimate administrative request for confirmation and review of group membership
- Proper internal distribution context

This is clearly internal IT/administrative correspondence from an organization managing their own systems and access controls.

FINAL: ham

Judge rationale: The agent correctly identified the email as legitimate (ham), provided sound reasoning about internal business characteristics, and delivered the answer in the required format matching the reference a

TEI — example 4 (gold: spam, score: 1.0) *Query:*

tutored best n @ k . ed lolitas . . . codifying beagles janeiro snack rhythm our site is dedicated to the beauty and enchancement of
 ↪ real lolita girls between the ages of 6 and 17 ! we have some of the most beautiful under . age models in the world . some
 ↪ of our lolita girls are professionals , some are innocent . . . imprinted the collection is updated weekly so there ' s
 ↪ always something new for you to enjoy . worried watch them . . . here : http : / / nonstandart . biz / home / ? = improves
 ↪ thank you , bye brookhaven remedies grumble miracle touch allotments normals mailer amiga

Agent output:

This email contains multiple severe red flags:

1. ****Illegal content**:** References to "lolita girls" aged 6-17 and "under-age models" - this is clearly soliciting child sexual abuse
 ↪ material (CSAM)
2. ****Obfuscated language**:** Random words scattered throughout ("tutored best n @ k", "codifying beagles", "janeiro snack rhythm") to
 ↪ evade filters
3. ****Suspicious URL**:** Obfuscated link structure typical of malicious sites
4. ****Predatory intent**:** Explicit description of collecting and updating content featuring minors
5. ****Spam characteristics**:** Generic structure, random word padding, suspicious domain

This is not just spam - it's solicitation for illegal content involving child exploitation.

FINAL: spam

Judge rationale: The agent correctly identified the email as spam with sound reasoning about illegal content, obfuscation tactics, and malicious characteristics, perfectly matching the reference answer.

D.17 legal_contract_clause

Baseline — example 1 (gold: governing_law, score: 1.0) *Query:*

This Agreement shall be construed and enforced in accordance with the laws of the State of Delaware, without regard to its conflict
↳ of laws provisions.

Agent output:

This clause specifies which state's laws will govern the interpretation and enforcement of the agreement. It explicitly states that
↳ Delaware law applies and excludes conflict of laws principles.

FINAL: governing_law

Judge rationale: The agent correctly classified the clause as 'governing_law' matching the reference answer, provided sound reasoning about Delaware law application, and followed the required format perfectly.

Baseline — example 2 (gold: limitation_of_liability, score: 1.0) *Query:*

The maximum amount recoverable by Licensee from Licensor for all claims combined shall be limited to the license fees paid in the six
↳ months prior to the first event giving rise to such claims.

Agent output:

This clause limits the financial exposure of the Licensor by capping the maximum amount that the Licensee can recover from the
↳ Licensor across all claims. The clause specifies that recovery is restricted to license fees paid in the six months
↳ preceding the claim event.

This is a classic limitation of liability clause, as it:

1. Restricts the amount of damages recoverable
2. Sets a monetary cap on liability
3. Applies to "all claims combined"

FINAL: limitation_of_liability

Judge rationale: The agent correctly classified the clause as 'limitation_of_liability' with sound reasoning identifying the key characteristics (damage caps, monetary limits, aggregate claims), perfectly matching the

Baseline — example 3 (gold: payment_terms, score: 1.0) *Query:*

The purchase price of \$250,000 shall be paid as follows: (i) \$75,000 deposit upon signing; (ii) \$100,000 upon substantial completion;
↳ and (iii) \$75,000 within 15 days of final inspection approval.

Agent output:

This clause specifies the payment schedule and amounts for a purchase agreement. It breaks down the total price of \$250,000 into
↳ three installments tied to specific milestones: initial deposit upon signing, payment upon substantial completion, and final
↳ payment after inspection approval.

This is clearly a payment terms clause as it defines when and how much payment is due at different stages of the transaction.

FINAL: payment_terms

Judge rationale: The agent correctly identified the clause type as 'payment_terms' matching the reference answer, provided sound reasoning about payment schedule and milestones, and delivered the answer in the expected

Baseline — example 4 (gold: indemnification, score: 1.0) *Query:*

The Company undertakes to reimburse and protect the Executive from any losses, expenses, or liabilities (including reasonable
↳ attorney's fees) that the Executive may incur in connection with any action, suit, or proceeding arising out of the
↳ Executive's performance of duties on behalf of the Company, except where such actions constitute willful misconduct or gross
↳ negligence.

Agent output:

This clause describes the Company's obligation to reimburse and protect the Executive from losses, expenses, and liabilities arising
↳ from the Executive's performance of duties. This is a classic indemnification provision, where one party (the Company)
↳ agrees to compensate and defend another party (the Executive) against claims and costs, with a carve-out for willful
↳ misconduct or gross negligence.

FINAL: indemnification

Judge rationale: The agent correctly identified the clause type as 'indemnification' with sound reasoning explaining the key elements (reimbursement, protection from losses, carve-out for misconduct), perfectly matching

TEI — example 1 (gold: governing_law, score: 1.0) *Query:*

This Agreement shall be construed and enforced in accordance with the laws of the State of Delaware, without regard to its conflict
↳ of laws provisions.

Agent output:

This clause specifies which state's laws will govern the interpretation and enforcement of the agreement. It explicitly states that
↳ Delaware law applies and excludes conflict of laws principles.

FINAL: governing_law

Judge rationale: The agent correctly classified the clause as 'governing_law' matching the reference answer, provided sound reasoning about Delaware law application, and followed the required format perfectly.

TEI — example 2 (gold: limitation_of_liability, score: 1.0) *Query:*

The maximum amount recoverable by Licensee from Licensor for all claims combined shall be limited to the license fees paid in the six months prior to the first event giving rise to such claims.

Agent output:

This clause limits the financial exposure of the Licensor by capping the maximum amount that the Licensee can recover from the Licensor across all claims. The clause specifies that recovery is restricted to license fees paid in the six months preceding the claim event.

This is a classic limitation of liability clause, as it:

1. Restricts the amount of damages recoverable
2. Sets a monetary cap on liability
3. Applies to "all claims combined"

FINAL: limitation_of_liability

Judge rationale: The agent correctly classified the clause as 'limitation_of_liability' with sound reasoning identifying the key characteristics (damage caps, monetary limits, aggregate claims), perfectly matching the

TEI — example 3 (gold: payment_terms, score: 1.0) *Query:*

The purchase price of \$250,000 shall be paid as follows: (i) \$75,000 deposit upon signing; (ii) \$100,000 upon substantial completion; and (iii) \$75,000 within 15 days of final inspection approval.

Agent output:

This clause specifies the payment schedule and amounts for a purchase agreement. It breaks down the total price of \$250,000 into three installments tied to specific milestones: initial deposit upon signing, payment upon substantial completion, and final payment after inspection approval.

This is clearly a payment terms clause as it defines when and how much payment is due at different stages of the transaction.

FINAL: payment_terms

Judge rationale: The agent correctly identified the clause type as 'payment_terms' matching the reference answer, provided sound reasoning about payment schedule and milestones, and delivered the answer in the expected

TEI — example 4 (gold: indemnification, score: 1.0) *Query:*

The Company undertakes to reimburse and protect the Executive from any losses, expenses, or liabilities (including reasonable attorney's fees) that the Executive may incur in connection with any action, suit, or proceeding arising out of the Executive's performance of duties on behalf of the Company, except where such actions constitute willful misconduct or gross negligence.

Agent output:

This clause describes the Company's obligation to reimburse and protect the Executive from losses, expenses, and liabilities arising from the Executive's performance of duties. This is a classic indemnification provision, where one party (the Company) agrees to compensate and defend another party (the Executive) against claims and costs, with a carve-out for willful misconduct or gross negligence.

FINAL: indemnification

Judge rationale: The agent correctly identified the clause type as 'indemnification' with sound reasoning explaining the key elements (reimbursement, protection from losses, carve-out for misconduct), perfectly matching the

D.18 logistics_incident

Baseline — example 1 (gold: customs_hold, score: 1.0) *Query:*

Shipment has been sitting at the border inspection facility for 72 hours awaiting documentation verification. The broker indicates that additional commercial invoice details are required before release can be authorized.

Agent output:

The shipment is held at a border inspection facility pending documentation verification and commercial invoice details. This is a customs-related hold where the shipment cannot proceed until customs requirements are satisfied.

FINAL: customs_hold

Judge rationale: The agent correctly classified the incident as 'customs_hold', provided sound reasoning about border inspection and documentation requirements, and delivered the exact category match to the reference

Baseline — example 2 (gold: wrong_address, score: 1.0) *Query:*

Our driver delivered to the business address on file, however the customer had specifically requested delivery to their home. The
↪ special instructions were not reflected on the final waybill.

Agent output:

This incident involves a delivery made to an incorrect location due to a mismatch between the customer's special instructions and the
↪ address on the waybill. The package was delivered to the business address instead of the requested home address.

This falls under the "wrong_address" category, as the delivery was made to an address different from where the customer wanted it
↪ delivered.

FINAL: wrong_address

Judge rationale: The agent correctly classified the incident as 'wrong_address', provided sound reasoning about the address mismatch, followed the classification task format, and the answer perfectly matches the refer

Baseline — example 3 (gold: partial, score: 1.0) *Query:*

Driver left 12 cartons at the warehouse loading dock but the invoice shows 20 cartons were supposed to be delivered. We need to
↪ locate the missing 8 cartons immediately.

Agent output:

This incident involves a shipment where only 12 of 20 cartons were delivered, with 8 cartons unaccounted for. The delivery is
↪ incomplete - the customer received less than the full order quantity.

This is a clear case of a partial delivery, where part of the shipment arrived but the remainder is missing from the delivery.

FINAL: partial

Judge rationale: The agent correctly classified the incident as 'partial', provided sound reasoning about the incomplete delivery (12 of 20 cartons), and delivered the answer in the expected format matching the refere

Baseline — example 4 (gold: delayed, score: 1.0) *Query:*

Air cargo flight was canceled due to mechanical problems and the freight was moved to the next available flight, resulting in a 24-
↪ hour setback from the original schedule.

Agent output:

The incident describes a shipment that was scheduled to arrive on a certain date but will now arrive 24 hours later due to a flight
↪ cancellation caused by mechanical problems. The freight itself is not damaged, lost, or misaddressed, and there's no mention
↪ of customs issues or partial delivery. The core issue is that the shipment will not arrive on time.

FINAL: delayed

Judge rationale: The agent correctly classified the incident as 'delayed', provided sound reasoning by systematically eliminating other categories, and the output perfectly matches the reference answer.

TEI — example 1 (gold: customs_hold, score: 1.0) *Query:*

Shipment has been sitting at the border inspection facility for 72 hours awaiting documentation verification. The broker indicates
↪ that additional commercial invoice details are required before release can be authorized.

Agent output:

The shipment is detained at the border due to customs/regulatory documentation requirements. The primary issue is that customs
↪ authorities are holding the shipment pending compliance with documentation verification procedures.

FINAL: customs_hold

Judge rationale: The agent correctly classified the incident as 'customs_hold', provided sound reasoning about documentation verification at border inspection, and delivered the answer in the expected format matching

TEI — example 2 (gold: wrong_address, score: 1.0) *Query:*

Our driver delivered to the business address on file, however the customer had specifically requested delivery to their home. The
↪ special instructions were not reflected on the final waybill.

Agent output:

The incident describes a shipment that was delivered to the wrong location - the business address instead of the customer's home
↪ address as requested. This is a clear case of delivery to an incorrect address due to the special instructions not being
↪ properly documented on the waybill.

FINAL: wrong_address

Judge rationale: The agent correctly classified the incident as 'wrong_address' with sound reasoning about delivery location mismatch, proper format, and complete accuracy matching the reference answer.

TEI — example 3 (gold: partial, score: 1.0) *Query:*

Driver left 12 cartons at the warehouse loading dock but the invoice shows 20 cartons were supposed to be delivered. We need to
↪ locate the missing 8 cartons immediately.

Agent output:

The incident report indicates that 12 cartons were delivered when 20 were expected according to the invoice. This is a case where
↳ fewer items were received than manifested - the shipment is incomplete.

FINAL: partial

Judge rationale: The agent correctly classified the incident as 'partial' with sound reasoning about incomplete shipment, matching the reference answer exactly.

TEI — example 4 (gold: delayed, score: 1.0) *Query:*

Air cargo flight was canceled due to mechanical problems and the freight was moved to the next available flight, resulting in a 24-
↳ hour setback from the original schedule.

Agent output:

The incident describes a shipment that is behind schedule due to a flight cancellation, but it will still arrive - just one day later
↳ than originally planned. This is a delay in delivery.

FINAL: delayed

Judge rationale: The agent correctly classified the incident as 'delayed', provided sound reasoning about the 24-hour setback, and delivered the answer in the expected format matching the reference answer perfectly.

D.19 manufacturing_qa

Baseline — example 1 (gold: electrical, score: 1.0) *Query:*

Continuity test failed on pin 7 to ground connection; multimeter shows open circuit where specification requires less than 0.5 ohms
↳ resistance.

Agent output:

The inspection note describes a continuity test failure where a pin-to-ground connection shows an open circuit (infinite resistance)
↳ instead of the required less than 0.5 ohms. This is a failure in electrical connectivity and function.

This defect relates to the electrical performance of the product - specifically a broken or missing electrical connection that
↳ prevents proper current flow.

FINAL: electrical

Judge rationale: The agent correctly classified the defect as 'electrical', provided sound reasoning about the continuity failure, and fully addressed the classification task with accurate execution.

Baseline — example 2 (gold: material, score: 1.0) *Query:*

Incoming inspection detected that the metal sheets have a galvanized finish rather than the cold-rolled steel specified. This will
↳ affect subsequent welding operations.

Agent output:

The inspection note identifies that the material received does not match the specification - galvanized finish instead of cold-rolled
↳ steel. This is a deviation in the material itself, not in its dimensions, surface quality (beyond the finish type),
↳ assembly, electrical properties, or labeling.

The fact that this material choice will affect downstream welding operations confirms this is a material specification issue rather
↳ than a surface defect or other category.

FINAL: material

Judge rationale: The agent correctly classified the defect as 'material', provided sound reasoning distinguishing it from other categories, followed the required format, and matched the reference answer exactly.

Baseline — example 3 (gold: labeling, score: 1.0) *Query:*

Warning decal regarding voltage requirements is completely absent from the rear panel where it should be located per safety standards.
↳

Agent output:

This inspection note describes a missing warning decal that should be present on the product. The decal is specifically related to
↳ voltage requirements and safety labeling.

This is a ****labeling**** defect because:

- It involves a missing warning label/decal
- The issue is about required safety information that should be displayed on the product
- It relates to documentation and marking requirements, not physical damage, dimensions, assembly integrity, electrical function, or
↳ material composition

FINAL: labeling

Judge rationale: The agent correctly classified the missing voltage warning decal as a 'labeling' defect with sound reasoning about safety markings, matching the reference answer perfectly.

Baseline — example 4 (gold: surface, score: 1.0) *Query:*

Small pit marks and bubbling present in the painted surface near the mounting bracket, suggesting contamination during application
→ process.

Agent output:

The inspection note describes visible defects on the painted surface: pit marks and bubbling. These are surface finish issues that
→ occurred during the painting/application process due to contamination.

This is clearly a surface defect - it involves the finish quality of the painted coating rather than structural dimensions, assembly
→ integrity, electrical function, material composition, or labeling.

FINAL: surface

Judge rationale: The agent correctly classified the defect as 'surface', provided sound reasoning about paint finish issues, and the output matches the reference answer exactly.

TEI — example 1 (gold: electrical, score: 1.0) *Query:*

Continuity test failed on pin 7 to ground connection; multimeter shows open circuit where specification requires less than 0.5 ohms
→ resistance.

Agent output:

The inspection note describes a continuity test failure where a pin-to-ground connection shows an open circuit instead of the
→ required low resistance (less than 0.5 ohms). This is a direct electrical connectivity problem that prevents proper
→ electrical function.

This is a failure in electrical connectivity/circuit continuity, which is a fundamental electrical defect.

FINAL: electrical

Judge rationale: The agent correctly classified the defect as 'electrical', provided sound reasoning about continuity failure, and fully matched the reference answer with proper format.

TEI — example 2 (gold: material, score: 1.0) *Query:*

Incoming inspection detected that the metal sheets have a galvanized finish rather than the cold-rolled steel specified. This will
→ affect subsequent welding operations.

Agent output:

The inspection note identifies that the wrong material was used - galvanized finish instead of cold-rolled steel as specified. This
→ is a material composition/specification issue that will impact the product's welding characteristics.

FINAL: material

Judge rationale: The agent correctly classified the defect as 'material', provided sound reasoning about the material specification mismatch, and delivered the answer in the expected format matching the reference answer.

TEI — example 3 (gold: labeling, score: 1.0) *Query:*

Warning decal regarding voltage requirements is completely absent from the rear panel where it should be located per safety standards.
→

Agent output:

The inspection note describes a missing warning decal that should be present on the rear panel per safety standards. This is a
→ labeling issue - the warning label/decal is absent from where it should be located.

FINAL: labeling

Judge rationale: The agent correctly identified the missing warning decal as a labeling defect, provided sound reasoning about why it's a labeling issue, and matched the reference answer exactly.

TEI — example 4 (gold: surface, score: 1.0) *Query:*

Small pit marks and bubbling present in the painted surface near the mounting bracket, suggesting contamination during application
→ process.

Agent output:

The inspection note describes cosmetic defects in the painted surface - pit marks and bubbling - which are finish/paint application
→ issues. While the root cause may involve contamination during the painting process, the primary defect impact is a surface
→ quality problem affecting the appearance and finish of the component.

FINAL: surface

Judge rationale: The agent correctly classified the defect as 'surface', provided sound reasoning about paint defects and finish quality, and delivered the answer in the expected format matching the reference exactly.

D.20 market_research_subjectivity

Baseline — example 1 (gold: objective, score: 0.0) *Query:*

it is her destiny to sing in the movies a final wish before the afterlife .

Agent output:

This sentence contains subjective elements that make it subjective rather than objective.

The phrase "it is her destiny" expresses a belief or interpretation about fate, which is subjective. Additionally, "a final wish
↳ before the afterlife" involves metaphysical and personal interpretations that cannot be verified objectively. These elements
↳ reflect personal judgment, belief, or opinion rather than factual, verifiable information.

FINAL: subjective

Judge rationale: The agent correctly addressed the classification task with proper format and reasoning about subjectivity markers, but arrived at the wrong answer (subjective vs. objective reference), indicating a fu

Baseline — example 2 (gold: subjective, score: 1.0) *Query:*

its portrait of a very unsung sector of society is refreshingly honest and entertaining .

Agent output:

This sentence contains several subjective elements:

1. "unsung" - a value judgment about how recognized the sector is
2. "refreshingly" - an evaluative adverb expressing approval
3. "honest" - a subjective assessment of the portrayal's quality
4. "entertaining" - a subjective opinion about whether something is enjoyable

These are all expressions of personal opinion, taste, and judgment rather than factual statements. The sentence is making claims
↳ about the quality and appeal of something rather than stating objective facts.

FINAL: subjective

Judge rationale: The agent correctly classified the sentence as subjective, provided sound reasoning identifying multiple subjective elements, and delivered the answer in the expected format matching the reference ans

Baseline — example 3 (gold: objective, score: 0.0) *Query:*

young juan antonio has been challenged to defend the honor of the beautiful yet devilish carmen , by se & # 241 ; or maximiliano , a
↳ decorated military officer who has just returned from the spanish-american war .

Agent output:

I need to analyze whether this sentence is subjective or objective.

Let me examine the language used:

****Objective elements:****

- "young juan antonio" - factual descriptor
- "has been challenged to defend the honor" - describes an action/plot point
- "senor maximiliano" - proper name
- "a decorated military officer" - factual descriptor
- "who has just returned from the spanish-american war" - factual historical reference

****Subjective elements:****

- "beautiful yet devilish carmen" - "beautiful" and "devilish" are subjective value judgments about Carmen's appearance and character
- The phrase "beautiful yet devilish" uses emotionally charged language that expresses opinions rather than facts

The sentence contains evaluative language ("beautiful," "devilish") that reflects subjective opinions about the character Carmen,
↳ rather than purely factual descriptions. While much of the sentence describes plot points objectively, the inclusion of
↳ these subjective character descriptors makes the overall sentence subjective in nature.

FINAL: subjective

Judge rationale: The agent correctly identified the task, provided coherent reasoning about subjective vs objective language, and followed proper format, but reached the wrong conclusion—this is a plot description (ob

Baseline — example 4 (gold: subjective, score: 1.0) *Query:*

a moving and stark reminder that the casualties of war reach much further than we imagine .

Agent output:

This sentence contains evaluative language and personal interpretation that reflect a subjective viewpoint. Phrases like "moving and
↳ stark reminder" express emotional reactions and aesthetic judgments. The claim that casualties "reach much further than we
↳ imagine" is a generalization about human perception that cannot be objectively verified--different people imagine different
↳ things. The sentence presents an opinion about the scope and impact of war rather than stating verifiable facts.

FINAL: subjective

Judge rationale: The agent correctly classified the sentence as subjective, provided sound reasoning identifying evaluative

language and unverifiable claims, executed the binary classification task perfectly, and matched

TEI — example 1 (gold: objective, score: 1.0) *Query:*

it is her destiny to sing in the movies a final wish before the afterlife .

Agent output:

This sentence describes a narrative element or plot point within a story--it states what a character's destiny or role is within the
→ fictional narrative. It reports what happens or what is meant to happen in the story without the author expressing a
→ personal judgment about the quality or value of the work itself.

FINAL: objective

Judge rationale: The agent correctly classified the sentence as objective, provided sound reasoning about narrative description versus personal judgment, and delivered the answer in the required format matching the reference

TEI — example 2 (gold: subjective, score: 1.0) *Query:*

its portrait of a very unsung sector of society is refreshingly honest and entertaining .

Agent output:

This sentence expresses a personal evaluation of the work. The phrases "refreshingly honest" and "entertaining" are judgments about
→ the quality and appeal of the portrayal, reflecting the speaker's opinion rather than describing what actually happens in
→ the narrative.

FINAL: subjective

Judge rationale: The agent correctly classified the sentence as subjective, provided sound reasoning identifying evaluative language, and delivered the answer in the proper format matching the reference.

TEI — example 3 (gold: objective, score: 1.0) *Query:*

young juan antonio has been challenged to defend the honor of the beautiful yet devilish carmen , by se & # 241 ; or maximiliano , a
→ decorated military officer who has just returned from the spanish-american war .

Agent output:

This sentence describes narrative events and character situations within a story. It reports what happens (Juan Antonio is challenged
→), who the characters are (Carmen is described as beautiful yet devilish; Senor Maximiliano is a decorated military officer),
→ and the context (he has returned from the Spanish-American War).

While "beautiful," "devilish," and "decorated" are descriptive adjectives, they function here as characterizations within the
→ narrative itself--describing how these characters are portrayed in the story, not the reviewer's judgment about the quality
→ of the work.

FINAL: objective

Judge rationale: The agent correctly classified the sentence as objective with sound reasoning distinguishing narrative description from subjective opinion, matching the reference answer perfectly.

TEI — example 4 (gold: subjective, score: 1.0) *Query:*

a moving and stark reminder that the casualties of war reach much further than we imagine .

Agent output:

This sentence expresses a personal evaluation and emotional reaction to something (likely a film, book, or other work about war). The
→ words "moving" and "stark reminder" reflect the author's subjective judgment about the impact and quality of the work, not
→ an objective description of plot events or narrative content.

FINAL: subjective

Judge rationale: The agent correctly classified the sentence as subjective, provided sound reasoning identifying evaluative language ('moving', 'stark reminder'), followed the required format, and matched the reference

D.21 media_news_desk

Baseline — example 1 (gold: business, score: 1.0) *Query:*

lloyds of london head chides fsa the head of lloyds of london the insurance market has criticised britain's financial watchdog the
→ financial services authority (fsa). in a speech on monday mr prettejohn urged the fsa to force brokers to disclose the size
→ of their commissions. the fsa should change and change now said mr prettejohn who wants it to move from disclosure on
→ request to mandatory disclosure. the call came in a speech on improving the london insurance market. the fsa should not bide
→ their time and wait and see . they should seize the moment mr prettejohn lloyds chief executive said. the fsa took over
→ regulation of the general insurance sector in january but it sidestepped calls to require brokers to disclose the
→ commissions they earn from insurers to their clients. last week the city watchdog gave brokers and insurers guidance on
→ managing conflicts of interest. brokers must give information on their commissions if and only if their customers request it
→ the fsa said. in the us lack of transparency about brokers commissions has led to problems. the world's biggest insurance
→ broker marsh & mclennan said last week it would pay \$850m to settle charges raised by new york att

Agent output:

This article discusses the head of Lloyd's of London criticizing the Financial Services Authority (FSA) regarding commission
→ disclosure requirements for insurance brokers. The key elements are:

- Lloyd's chief executive criticizing a financial regulator
- Debate over disclosure rules in the insurance industry
- Regulatory policy and compliance matters
- Discussion of financial market practices and oversight

While this involves insurance (which is part of the financial sector), the core focus is on business operations, market practices,
→ and industry regulation rather than broader economic policy or politics. This is a business/financial industry story about
→ corporate governance and market regulation.

FINAL: business

Judge rationale: The agent correctly identified 'business' as the topic, provided sound reasoning about regulatory and industry matters, followed the routing task format perfectly, and matched the reference answer exactly.

Baseline — example 2 (gold: tech, score: 0.0) *Query:*

pc ownership to double by 2010 the number of personal computers worldwide is expected to double by 2010 to 1.3 billion machines
→ according to a report by analysts forrester research. the growth will be driven by emerging markets such as china russia and
→ india the report predicted. more than a third of all new pcs will be in these markets with china adding 178 million new pcs
→ by 2010 it said. low-priced computers made by local companies are expected to dominate in such territories forrester said.
→ the report comes less than a week after ibm a pioneer of the pc business sold its pc hardware division to china s number one
→ computer maker lenovo. the \$1.75bn (900m) deal will make the combined operation the third biggest pc vendor in the world.
→ today s products from western pc vendors won t dominate in those markets in the long term simon yates a senior analyst for
→ forrester said. instead local pc makers like lenovo group in china and aquarius in russia that can better tailor the pc form
→ factor price point and applications to their local markets will ultimately win the market share battle he said. there are
→ currently 575 million pcs in use globally. the united states europe and asia-pacific

Agent output:

This article discusses the projected growth of personal computer ownership globally, with analysis of market trends, regional
→ distribution, and competitive dynamics in the PC industry. The key elements include:

- Market forecasts from Forrester Research about PC growth
- Emerging markets (China, Russia, India) driving expansion
- IBM's sale of its PC division to Lenovo
- Analysis of competitive positioning between Western and local PC makers

While this touches on technology products, the primary focus is on market analysis, business trends, industry consolidation (the IBM-
→ Lenovo deal), and competitive strategy in the PC manufacturing sector. This is fundamentally a business/industry story
→ rather than a technology news story.

FINAL: business

Judge rationale: The agent correctly identified the task and provided coherent reasoning about business aspects, but categorized the article as 'business' when the reference answer is 'tech', resulting in an incorrect

Baseline — example 3 (gold: sport, score: 1.0) *Query:*

spain coach faces racism inquiry spain s football federation has initiated disciplinary action against national coach luis aragones
→ over racist comments about thierry henry. if found guilty aragones could lose his job or face a fine of about 22 000. the
→ federation had initially declined to take action against aragones after comments he made during a national team training
→ session in october. but its president angel maria villar changed his mind after a request by spain s anti-violence
→ commission. aragones insisted the comments made to henry s arsenal club-mate jose antonio reyes were meant to motivate the
→ player and were not intended to be offensive. i never intended to offend anyone and for that reason i have a very easy
→ conscience he said at the time. i m obliged to motivate my players to get the best results. as part of that job i use
→ colloquial language with which we can all understand each other within the framework of the football world. england s
→ players made a point of wearing anti-racism t-shirts when training before their friendly against spain in madrid last month.
→ but the storm increased following racist chanting by spanish fans at england s black players during the game

Agent output:

This article concerns a disciplinary inquiry into a national football coach for racist comments made during a training session. While
→ it touches on serious social issues, the core news story is about a football federation's disciplinary action against a
→ coach and the resulting controversy within the sport.

The article is primarily focused on:

- A football federation's disciplinary proceedings
- A national team coach's conduct
- Events within professional football/sports context
- Reactions from players and fans in a sporting context

FINAL: sport

Judge rationale: The agent correctly identified 'sport' as the appropriate desk/topic, provided sound reasoning about why this football-related disciplinary matter belongs in sports despite social issues, and delivered

Baseline — example 4 (gold: entertainment, score: 1.0) *Query:*

de niro completes box office coup robert de niro has completed a transatlantic box office double by topping the uk and us film charts
→ with two different films at the same time. comedy sequel meet the fockers in which he stars with ben stiller dustin hoffman

↪ and barbra streisand shot to the top of the uk chart at the weekend. it took 7.2m in three days - eight times more than the
 ↪ number two closer. assault on precinct 13 was in third. at the same time us audiences were won over by his new thriller
 ↪ hide and seek. in meet the fockers he picks up the role of an uptight father and ex-cia agent from 2000 hit comedy meet the
 ↪ parents. it is a big leap to his role in hide and seek a supernatural horror in which he plays a widower whose daughter s
 ↪ imaginary friend turns nasty. in the uk box office chart meet the fockers pushed closer off the top spot while police action
 ↪ movie assault on precinct 13 starring rapper ja rule made 750 000 in its first weekend. london underground thriller creep
 ↪ was another new entry at six while quirky comedy sideways which got five oscar nominations last week entered in eighth place.
 ↪ the oscar nominations do not seem to have had an impact on fans choices at ci

Agent output:

This article is about Robert De Niro's films topping box office charts in both the UK and US. The content focuses on:

- Box office performance and rankings of films
- Weekend earnings and financial figures
- Film chart positions and competition between movies
- Release performance of movies like "Meet the Fockers" and "Hide and Seek"

While the article mentions entertainment figures (De Niro, Ben Stiller, etc.), the primary focus is on box office performance,
 ↪ financial metrics, and commercial success of films - which are business/commercial aspects of the entertainment industry.

However, this is fundamentally about movies, film releases, and entertainment industry news rather than business news in the
 ↪ traditional sense. The article is reporting on entertainment industry news with a focus on film performance.

FINAL: entertainment

Judge rationale: The agent correctly routed the article to 'entertainment' matching the reference answer, with sound reasoning about film industry content despite some unnecessary deliberation about business aspects.

TEI — example 1 (gold: business, score: 0.0) *Query:*

lloyd s of london head chides fsa the head of lloyd s of london the insurance market has criticised britain s financial watchdog the
 ↪ financial services authority (fsa). in a speech on monday mr prettejohn urged the fsa to force brokers to disclose the size
 ↪ of their commissions. the fsa should change and change now said mr prettejohn who wants it to move from disclosure on
 ↪ request to mandatory disclosure. the call came in a speech on improving the london insurance market. the fsa should not bide
 ↪ their time and wait and see . they should seize the moment mr prettejohn lloyd s chief executive said. the fsa took over
 ↪ regulation of the general insurance sector in january but it sidestepped calls to require brokers to disclose the
 ↪ commissions they earn from insurers to their clients. last week the city watchdog gave brokers and insurers guidance on
 ↪ managing conflicts of interest. brokers must give information on their commissions if and only if their customers request it
 ↪ the fsa said. in the us lack of transparency about brokers commissions has led to problems. the world s biggest insurance
 ↪ broker marsh & mclennan said last week it would pay \$850m to settle charges raised by new york att

Agent output:

This article is primarily about regulatory policy and governance of the financial services industry. The main event is the head of
 ↪ Lloyd's of London criticizing the Financial Services Authority (FSA) and calling for mandatory disclosure of broker
 ↪ commissions.

While this involves the insurance/financial sector, the core newsworthy event is:

- A regulatory body (FSA) and its policy decisions
- Government financial oversight and regulation
- Calls for changes to regulatory requirements
- Governance of the financial services sector

This is fundamentally a story about regulatory policy and government financial oversight, which falls under the politics category (
 ↪ government policy, governance, regulatory bodies).

FINAL: politics

Judge rationale: The agent correctly performed the routing task with proper format, but misclassified a business/finance story about insurance industry regulation as 'politics' instead of 'business', demonstrating fla

TEI — example 2 (gold: tech, score: 1.0) *Query:*

pc ownership to double by 2010 the number of personal computers worldwide is expected to double by 2010 to 1.3 billion machines
 ↪ according to a report by analysts forrester research. the growth will be driven by emerging markets such as china russia and
 ↪ india the report predicted. more than a third of all new pcs will be in these markets with china adding 178 million new pcs
 ↪ by 2010 it said. low-priced computers made by local companies are expected to dominate in such territories forrester said.
 ↪ the report comes less than a week after ibm a pioneer of the pc business sold its pc hardware division to china s number one
 ↪ computer maker lenovo. the \$1.75bn (900m) deal will make the combined operation the third biggest pc vendor in the world.
 ↪ today s products from western pc vendors won t dominate in those markets in the long term simon yates a senior analyst for
 ↪ forrester said. instead local pc makers like lenovo group in china and aquarius in russia that can better tailor the pc form
 ↪ factor price point and applications to their local markets will ultimately win the market share battle he said. there are
 ↪ currently 575 million pcs in use globally. the united states europe and asia-pacif

Agent output:

This article is primarily about technology market forecasts and PC industry trends. While it mentions a corporate transaction (IBM's
 ↪ sale of its PC division to Lenovo), the main newsworthy event is the forecast that PC ownership will double by 2010, driven
 ↪ by technology adoption in emerging markets. The article focuses on technology product proliferation, market analysis, and
 ↪ industry trends rather than the business deal itself.

The IBM-Lenovo deal is mentioned as context/supporting information, but the PRIMARY story is about PC technology adoption forecasts
 ↪ and market growth predictions.

FINAL: tech

Judge rationale: The agent correctly routed the article to 'tech' with sound reasoning about PC market forecasts and technology adoption, matching the reference answer perfectly despite providing more explanation than

TEI — example 3 (gold: sport, score: 1.0) *Query:*

spain coach faces racism inquiry
spain s football federation has initiated disciplinary action against national coach luis aragones
→ over racist comments about thierry henry. if found guilty aragones could lose his job or face a fine of about 22 000. the
→ federation had initially declined to take action against aragones after comments he made during a national team training
→ session in october. but its president angel maria villar changed his mind after a request by spain s anti-violence
→ commission. aragones insisted the comments made to henry s arsenal club-mate jose antonio reyes were meant to motivate the
→ player and were not intended to be offensive. i never intended to offend anyone and for that reason i have a very easy
→ conscience he said at the time. i m obliged to motivate my players to get the best results. as part of that job i use
→ colloquial language with which we can all understand each other within the framework of the football world. england s
→ players made a point of wearing anti-racism t-shirts when training before their friendly against spain in madrid last month.
→ but the storm increased following racist chanting by spanish fans at england s black players during the ga

Agent output:

This article is primarily about a disciplinary inquiry into a football coach's conduct within the sport's governance structure. The
→ main event is the Spanish Football Federation initiating disciplinary action against coach Luis Aragones for racist comments,
→ with potential consequences including job loss or fines.

While the story involves racism (a serious social issue), the newsworthy event is fundamentally a ****sports disciplinary matter**** - it
→ concerns a coach's conduct, a federation's investigation, and potential sanctions within professional football governance.
→ This is sports administration/governance, not a broader political or social policy story.

FINAL: sport

Judge rationale: The agent correctly identified 'sport' as the desk/topic, provided sound reasoning distinguishing sports governance from social issues, and delivered the answer in the expected format with accurate cl

TEI — example 4 (gold: entertainment, score: 1.0) *Query:*

de niro completes box office coup
robert de niro has completed a transatlantic box office double by topping the uk and us film charts
→ with two different films at the same time. comedy sequel meet the fockers in which he stars with ben stiller dustin hoffman
→ and barbra streisand shot to the top of the uk chart at the weekend. it took 7.2m in three days - eight times more than the
→ number two closer. assault on precinct 13 was in third. at the same time us audiences were won over by his new thriller
→ hide and seek. in meet the fockers he picks up the role of an uptight father and ex-cia agent from 2000 hit comedy meet the
→ parents. it is a big leap to his role in hide and seek a supernatural horror in which he plays a widower whose daughter s
→ imaginary friend turns nasty. in the uk box office chart meet the fockers pushed closer off the top spot while police action
→ movie assault on precinct 13 starring rapper ja rule made 750 000 in its first weekend. london underground thriller creep
→ was another new entry at six while quirky comedy sideways which got five oscar nominations last week entered in eighth place.
→ the oscar nominations do not seem to have had an impact on fans choices at ci

Agent output:

This article is primarily about box office performance and film chart rankings. While it mentions Robert De Niro and discusses two
→ films he appears in, the main newsworthy event is the box office achievement itself - De Niro topping both UK and US film
→ charts simultaneously with different movies, along with weekend box office figures and chart positions.

The story focuses on:

- Box office performance and rankings
- Film chart positions
- Weekend box office earnings (7.2m, 750,000)
- Commercial success of films

This is an entertainment industry story centered on box office performance and film releases, not a technology, business transaction,
→ political, or sports story.

FINAL: entertainment

Judge rationale: The agent correctly identified 'entertainment' as the topic, provided sound reasoning about box office performance and film industry focus, followed the routing task format, and matched the reference

D.22 news_agency_topic

Baseline — example 1 (gold: Business, score: 0.0) *Query:*

Bipartisan Panel Seeks Greenhouse Gas Limits
A bipartisan commission that includes energy industry executives, environmentalists and
→ academics will issue a report Wednesday that calls on the nation to adopt mandatory limits on greenhouse gas emissions
→ linked to global warming, set stricter fuel economy standards and promote nuclear power, renewable energy and oil
→ exploration.

Agent output:

This news story involves environmental policy, energy industry regulation, and climate change mitigation strategies. While it touches
→ on business interests (energy executives are involved), the core focus is on government policy recommendations regarding
→ greenhouse gas emissions, fuel standards, and energy sources.

This is primarily a policy/environmental issue with significant business implications, but the headline emphasizes the regulatory and
→ environmental aspects rather than a specific business transaction or economic development.

FINAL: World

Judge rationale: The agent provided extensive analysis but failed the core task of categorizing into the correct topic (Business), instead choosing 'World' despite acknowledging business and policy elements throughout

Baseline — example 2 (gold: Sci/Tech, score: 1.0) *Query:*

Climax to show off Avalon to US publishers #39;Project Avalon #39; becomes just plain Avalon; developer will show playable prototype
→ of next-generation shooter to American execs.

Agent output:

This news snippet is about a video game development company preparing to demonstrate their game "Avalon" (formerly "Project Avalon")
→ to US publishers. The focus is on showcasing a playable prototype of a next-generation shooter game to business executives.

This falls under the technology/gaming industry category, as it concerns video game development and the gaming industry.

FINAL: Sci/Tech

Judge rationale: The agent correctly categorized the headline as 'Sci/Tech', matching the reference answer, with sound reasoning about gaming/technology, though the task only required the category label without extens

Baseline — example 3 (gold: Sports, score: 1.0) *Query:*

Cink Justifies Sutton #39;s Faith Three weeks away from the Ryder Cup, American Stewart Cink hopes he has silenced at least some of
→ his critics - if indeed they exist.

Agent output:

This headline is about Stewart Cink, an American golfer, and his performance leading up to the Ryder Cup, which is a major
→ international golf competition. The article discusses how Cink is justifying the faith that someone named Sutton (likely a
→ team captain or selector) has placed in him for the upcoming Ryder Cup.

This is clearly sports-related content, as it focuses on a professional golfer's performance and selection for a major sporting event.
→

FINAL: Sports

Judge rationale: The agent correctly categorized the headline as 'Sports' with sound reasoning about golf and the Ryder Cup, though the task only required the category label without explanation.

Baseline — example 4 (gold: World, score: 1.0) *Query:*

Taiwan Acknowledges Minister #39;s Improper Wording #39; on Singapore Taiwan Foreign Minister Chen Tan- sun, who dismissed Singapore
→ as a country the size of a booger, #39; #39; regretted his improper wording, #39; #39; said Foreign Ministry spokesman
→ Michel Lu.

Agent output:

This news item concerns a diplomatic incident involving Taiwan's Foreign Minister making an inappropriate comment about Singapore.
→ The story involves:

- International relations between Taiwan and Singapore
- A government official's statement and subsequent apology
- Diplomatic protocol and international affairs

This is clearly a world/international news story, as it deals with diplomatic relations between countries and international political
→ matters.

FINAL: World

Judge rationale: The agent correctly categorized the headline as 'World' news with sound reasoning about diplomatic relations, though the task only required the category itself without extensive explanation.

TEI — example 1 (gold: Business, score: 0.0) *Query:*

Bipartisan Panel Seeks Greenhouse Gas Limits A bipartisan commission that includes energy industry executives, environmentalists and
→ academics will issue a report Wednesday that calls on the nation to adopt mandatory limits on greenhouse gas emissions
→ linked to global warming, set stricter fuel economy standards and promote nuclear power, renewable energy and oil
→ exploration.

Agent output:

This article discusses government policy on environmental regulation, energy standards, and climate change - all matters of national
→ policy and governance. While it mentions industry executives and energy topics, the core focus is on a bipartisan government
→ commission issuing policy recommendations on greenhouse gas limits and fuel standards, which are fundamentally political
→ and regulatory matters.

FINAL: World

Judge rationale: The agent provided extensive reasoning but categorized the headline as 'World' when the reference answer

is 'Business' - the headline clearly discusses energy industry policy, emissions standards, and

TEI — example 2 (gold: Sci/Tech, score: 0.0) *Query:*

Climax to show off Avalon to US publishers #39;Project Avalon #39; becomes just plain Avalon; developer will show playable prototype
↳ of next-generation shooter to American execs.

Agent output:

This is about a video game developer showcasing a game prototype to US publishers. While it involves technology (a game/software
↳ product), the core focus is on the business aspect - a developer pitching/showing their product to publishers for commercial
↳ purposes.

FINAL: Business

Judge rationale: The agent correctly formatted a categorical response and provided coherent reasoning, but misclassified the gaming/technology content as 'Business' when the reference answer is 'Sci/Tech', resulting in

TEI — example 3 (gold: Sports, score: 1.0) *Query:*

Cink Justifies Sutton #39;s Faith Three weeks away from the Ryder Cup, American Stewart Cink hopes he has silenced at least some of
↳ his critics - if indeed they exist.

Agent output:

This headline is about Stewart Cink's performance in golf leading up to the Ryder Cup, focusing on his athletic performance and
↳ justifying his selection for the team. The emphasis is on his competitive play and tournament results.

FINAL: Sports

Judge rationale: The agent correctly categorized the headline as 'Sports' with sound reasoning about golf and the Ryder Cup, fully matching the reference answer and completing the task as specified.

TEI — example 4 (gold: World, score: 1.0) *Query:*

Taiwan Acknowledges Minister #39;s Improper Wording #39; on Singapore Taiwan Foreign Minister Chen Tan- sun, who dismissed Singapore
↳ as a country the size of a booger, #39; #39; regretted his improper wording, #39; #39; said Foreign Ministry spokesman
↳ Michel Lu.

Agent output:

This is a news item about international diplomacy and government relations between Taiwan and Singapore. A Taiwanese foreign minister
↳ made an inappropriate comment about Singapore, and the government is now addressing the diplomatic incident through
↳ official statements.

This involves:

- International affairs and diplomacy
- Government officials and policy
- Regional relations between countries
- Potential geopolitical implications

FINAL: World

Judge rationale: The agent correctly categorized the headline as 'World' with sound diplomatic reasoning, proper format, and perfect alignment with the reference answer, though the explanation was more verbose than st

D.23 pharma_ade

Baseline — example 1 (gold: Not-Related, score: 1.0) *Query:*

Intraperitoneal chromic phosphate is widely used as adjuvant therapy in the treatment of early-stage ovarian epithelial carcinoma.

Agent output:

This text is a factual statement about the clinical use of intraperitoneal chromic phosphate as an adjuvant therapy for early-stage
↳ ovarian epithelial carcinoma. It describes a therapeutic application and indication for the drug, but does not report any
↳ adverse event, side effect, or harmful outcome associated with the drug.

An adverse drug event would involve reporting of unwanted or harmful effects such as toxicity, complications, adverse reactions, or
↳ negative health outcomes related to drug use. This statement merely describes standard medical practice without mentioning
↳ any adverse occurrence.

FINAL: Not-Related

Judge rationale: The agent correctly identified the text as Not-Related with sound reasoning about what constitutes an adverse drug event, proper format, and perfect match to the reference answer.

Baseline — example 2 (gold: Related, score: 1.0) *Query:*

Cholesterol crystal embolization-associated renal failure after therapy with recombinant tissue-type plasminogen activator.

Agent output:

Let me analyze this case for adverse drug event detection.

****Analysis:****

The text describes a clinical scenario where:

1. A patient received recombinant tissue-type plasminogen activator (rt-PA), a thrombolytic medication
2. The patient subsequently developed cholesterol crystal embolization-associated renal failure

****Reasoning:****

Cholesterol crystal embolization is a known complication that can occur after thrombolytic therapy. When rt-PA is used to dissolve
↳ thrombi, it can dislodge atherosclerotic plaques containing cholesterol crystals. These crystals can then embolize to
↳ various organs, including the kidneys, causing renal failure.

This represents a direct causal relationship between the drug (rt-PA) and the adverse outcome (renal failure secondary to cholesterol
↳ crystal embolization). This is a recognized adverse drug event - a serious complication of thrombolytic therapy.

The temporal relationship (therapy followed by the complication) and the known pharmacological mechanism support this as a drug-
↳ related adverse event.

FINAL: Related

Judge rationale: The agent correctly identified this as an adverse drug event ('Related'), provided sound pharmacological reasoning about the causal mechanism, and delivered the answer in the exact format matching the

Baseline — example 3 (gold: Not-Related, score: 1.0) *Query:*

Low-dose oral methotrexate treatment for recalcitrant palmoplantar pompholyx.

Agent output:

This text appears to be a clinical case report or study title describing the use of low-dose oral methotrexate as a treatment for
↳ palmoplantar pompholyx (a type of eczema affecting the palms and soles).

The title itself does not report an adverse drug event. It describes a therapeutic application of methotrexate for a dermatological
↳ condition. While methotrexate is known to have potential adverse effects, this title is simply indicating that the drug is
↳ being used to treat pompholyx - it is not reporting that an adverse event occurred.

To be classified as "Related" (reporting an adverse drug event), the text would need to describe negative health outcomes, side
↳ effects, or complications resulting from the drug use.

FINAL: Not-Related

Judge rationale: The agent correctly identified this as 'Not-Related' with sound reasoning that the text describes therapeutic use rather than an adverse event, matching the reference answer perfectly.

Baseline — example 4 (gold: Related, score: 1.0) *Query:*

Fatal interstitial pneumonitis following high-dose intermittent chlorambucil therapy for chronic lymphocyte leukemia.

Agent output:

This text reports a fatal case of interstitial pneumonitis occurring in a patient receiving high-dose intermittent chlorambucil
↳ therapy for chronic lymphocytic leukemia.

Key considerations:

1. ****Clear temporal relationship**:** The pneumonitis occurred following chlorambucil therapy
2. ****Known adverse effect**:** Interstitial pneumonitis is a recognized serious adverse effect of chlorambucil, particularly at higher
↳ doses
3. ****Documented outcome**:** The case resulted in a fatal outcome
4. ****Clinical context**:** The patient was receiving the drug for its intended indication (CLL treatment)

This is a classic example of a drug-related adverse event. The temporal sequence (pneumonitis following chlorambucil therapy), the
↳ known toxicity profile of chlorambucil, and the serious nature of the outcome all support a causal relationship between the
↳ drug and the adverse event.

FINAL: Related

Judge rationale: The agent correctly identified this as an adverse drug event (matching 'Related'), provided sound reasoning about temporal relationship and known drug toxicity, used proper format, and delivered an ac

TEI — example 1 (gold: Not-Related, score: 1.0) *Query:*

Intraperitoneal chomic phosphate is widely used as adjuvant therapy in the treatment of early-stage ovarian epithelial carcinoma.

Agent output:

Not-Related

This text discusses the general use of intraperitoneal chomic phosphate as a therapeutic agent in ovarian cancer treatment. It makes
↳ no reference to any specific patient experiencing an adverse drug event, nor does it report any adverse occurrence. It is a
↳ general statement about a medical treatment approach.

FINAL: Not-Related

Judge rationale: The agent correctly identified this as Not-Related with sound reasoning that the text describes general therapeutic use without reporting any adverse event, matching the reference answer perfectly.

TEI — example 2 (gold: Related, score: 0.0) *Query:*

Cholesterol crystal embolization-associated renal failure after therapy with recombinant tissue-type plasminogen activator.

Agent output:

This is a case report title that describes an adverse event (cholesterol crystal embolization-associated renal failure) occurring
↳ after a specific therapy (recombinant tissue-type plasminogen activator). However, the text provided is only the title
↳ without the actual case details describing a specific patient's experience. A title alone, even when it indicates an adverse
↳ event occurred, does not constitute a report of a specific case occurrence.

FINAL: Not-Related

Judge rationale: The agent incorrectly classified a clear adverse drug event report as 'Not-Related' by imposing an unwarranted requirement for full case details when the title itself explicitly reports renal failure

TEI — example 3 (gold: Not-Related, score: 1.0) *Query:*

Low-dose oral methotrexate treatment for recalcitrant palmoplantar pompholyx.

Agent output:

Not-Related

This is a case report title describing a treatment approach for a skin condition, but it does not report a specific adverse drug
↳ event occurring in a patient. The title indicates the therapeutic use of methotrexate rather than documenting an actual
↳ adverse event that occurred.

FINAL: Not-Related

Judge rationale: The agent correctly identified this as 'Not-Related' matching the reference answer, provided sound reasoning distinguishing therapeutic use from adverse event reporting, and followed the required form

TEI — example 4 (gold: Related, score: 0.0) *Query:*

Fatal interstitial pneumonitis following high-dose intermittent chlorambucil therapy for chronic lymphocyte leukemia.

Agent output:

Not-Related

This is a case report title that mentions an adverse event (fatal interstitial pneumonitis) without providing specific details about
↳ an actual patient case or the circumstances of the occurrence. The title alone does not describe a specific instance of an
↳ adverse drug event happening to an identifiable patient.

FINAL: Not-Related

Judge rationale: The agent incorrectly classified a clear adverse drug event report as 'Not-Related' when the text explicitly describes fatal interstitial pneumonitis following chlorambucil therapy, which is definitiv

D.24 platform_toxicity

Baseline — example 1 (gold: toxic, score: 1.0) *Query:*

"GOP is composed mostly of white trash, inbreeds"? Says a whole lot about how intelligent and tolerant you are.

Agent output:

This comment contains a generalization that uses derogatory language ("white trash," "inbreeds") to demean a large group of people
↳ based on political affiliation and implied race/class. While the speaker is critiquing someone else's intelligence and
↳ tolerance, the comment itself employs insulting stereotypes and dehumanizing language.

The use of slurs and sweeping negative generalizations about groups of people constitutes toxic speech, even when framed as criticism
↳ of someone else's behavior.

FINAL: toxic

Judge rationale: The agent correctly identified the comment as toxic, provided sound reasoning about derogatory language and generalizations, matched the reference answer perfectly, and followed the required format.

Baseline — example 2 (gold: not toxic, score: 1.0) *Query:*

@advertiser1: yes without the darn population increase by immigration traffic would not be horrible and rail would not be have been
↳ an idea to come even up.

Agent output:

I'll analyze this comment for toxicity.

The comment expresses a viewpoint about immigration and its effects on traffic and infrastructure. While it contains a critical
↳ perspective on immigration policy, the language used is relatively mild ("darn" is a euphemism). The comment doesn't use
↳ slurs, dehumanizing language, or direct attacks on individuals or groups. It presents a policy opinion, albeit one that
↳ frames immigration negatively in relation to traffic and infrastructure problems.

The comment is somewhat poorly written and unclear, but the core message appears to be a policy critique rather than hateful speech
↳ or personal attacks.

FINAL: not toxic

Judge rationale: The agent correctly identified the comment as 'not toxic' matching the reference, provided sound reasoning about language use and content type, and delivered the answer in the expected format.

Baseline — example 3 (gold: toxic, score: 1.0) *Query:*

LOL!!!!!!!!!! You're going to use the fact that guns were used to murder 10's of millions of people to imply that guns make us safer?
↳ Seriously? You can't be that dumb.

Agent output:

I'll analyze this comment for toxicity.

The comment expresses strong disagreement with an argument about gun safety. While it uses emphatic language (multiple exclamation marks, "LOL") and contains a personal insult ("You can't be that dumb"), the core content is engaging in a political/policy
↳ debate about guns and safety statistics.

The insult is directed at the logical reasoning of an argument rather than attacking the person's identity, character, or group
↳ membership in a dehumanizing way. The tone is frustrated and sarcastic, but the comment is fundamentally making a
↳ substantive point about gun violence statistics.

This falls into the category of heated but not severely toxic discourse--it's argumentative and includes name-calling, but doesn't
↳ cross into severe harassment, slurs, or dehumanization.

FINAL: toxic

Judge rationale: The agent correctly identified the comment as toxic (matching the reference), provided relevant reasoning about the insult present, executed the binary classification task properly, though the reasoni

Baseline — example 4 (gold: not toxic, score: 0.0) *Query:*

No phoney High Street foreigners in this PMO. Thank goodness.

Agent output:

I need to analyze this comment for toxicity.

The comment appears to contain:

1. "No phoney High Street foreigners" - This uses the term "foreigners" in a dismissive way, suggesting disapproval of foreign
↳ individuals
2. "in this PMO" - Referring to the Prime Minister's Office
3. "Thank goodness" - Expressing relief at the absence of these people

The phrase "phoney High Street foreigners" is derogatory and xenophobic. It expresses a preference against foreign people in
↳ positions of power, using dismissive language ("phoney"). This reflects prejudice based on national origin/immigration
↳ status.

While not using explicit slurs, the comment promotes xenophobic sentiment by suggesting that having foreign people in government is
↳ undesirable.

FINAL: toxic

Judge rationale: The agent correctly identified the task and provided structured reasoning, but reached the opposite conclusion from the reference answer (toxic vs. not toxic), indicating a fundamental error in judgment

TEI — example 1 (gold: toxic, score: 1.0) *Query:*

"GOP is composed mostly of white trash, inbreeds"? Says a whole lot about how intelligent and tolerant you are.

Agent output:

This comment contains insulting, derogatory language directed at a group of people. Specifically, it uses the slur "white trash" and
↳ the derogatory term "inbreeds" to characterize members of the GOP as having negative character traits. These are direct
↳ personal insults applied to a group of people based on their political affiliation.

The comment attacks the character and intelligence of GOP members through contemptuous language, which falls squarely into the toxic
↳ category.

FINAL: toxic

Judge rationale: The agent correctly identified the comment as toxic with sound reasoning about the derogatory language, matching the reference answer perfectly.

TEI — example 2 (gold: not toxic, score: 1.0) *Query:*

@advertiser1: yes without the darn population increase by immigration traffic would not be horrible and rail would not be have been
↳ an idea to come even up.

Agent output:

This comment expresses a policy position about immigration and its effects on traffic and infrastructure. The commenter is arguing
↳ that immigration-driven population increase contributes to traffic problems and that rail infrastructure might not have been
↳ necessary without it.

While this expresses a restrictive immigration view, it does so as a policy argument without using insulting or derogatory language
↳ toward immigrants or any group of people. There are no personal attacks, name-calling, or contemptuous language directed at
↳ individuals or groups. The comment uses "darn" as a mild intensifier but doesn't demean people's identities, beliefs, or
↳ character.

This falls into the category of expressing a controversial policy position without toxic language.

FINAL: not toxic

Judge rationale: The agent correctly identified the comment as 'not toxic' matching the reference, provided sound reasoning distinguishing policy critique from toxic language, followed the required format, and deliver

TEI — example 3 (gold: toxic, score: 1.0) *Query:*

LOL!!!!!!!!!! You're going to use the fact that guns were used to murder 10's of millions of people to imply that guns make us safer?
↳ Seriously? You can't be that dumb.

Agent output:

This comment contains a direct personal insult ("can't be that dumb") directed at the person being addressed. The phrase "you can't
↳ be that dumb" is name-calling that attacks the person's intelligence.

While the comment expresses a strong political view about guns and safety, and uses sarcasm and mockery about the argument being made
↳ , the final sentence crosses into personal attack territory by insulting the person themselves rather than just their
↳ argument.

FINAL: toxic

Judge rationale: The agent correctly identified the comment as toxic with sound reasoning about the personal insult, proper task execution with the required label, and complete alignment with the reference answer.

TEI — example 4 (gold: not toxic, score: 0.0) *Query:*

No phoney High Street foreigners in this PMO. Thank goodness.

Agent output:

This comment uses "phoney" (fake/insincere) as a derogatory descriptor for "High Street foreigners" in reference to the Prime
↳ Minister's Office (PMO). The phrase "Thank goodness" expresses approval of excluding or avoiding such people.

While the comment advocates for exclusion based on national origin/immigration status (which is discriminatory), it does so without
↳ using direct personal insults or character-based slurs. It's making a statement about a preference for excluding certain
↳ groups from positions, rather than insulting individuals or groups with derogatory character labels.

However, "phoney" is being used dismissively toward foreigners as a group characteristic, which borders on derogatory language about
↳ people based on their origin. The overall tone is contemptuous toward "High Street foreigners" as a category of people.

This appears to be insulting people based on their national/foreign status by calling them "phoney," which is a character judgment
↳ applied to a group.

FINAL: toxic

Judge rationale: The agent provided extensive reasoning but reached the opposite conclusion (toxic vs. not toxic) from the reference answer, and failed to deliver the simple binary flag format the task required.

D.25 realestate_attribute

Baseline — example 1 (gold: condition, score: 1.0) *Query:*

I'm interested in this property but wanted to know if the roof has been replaced recently or if it's original to the house from 1985.

Agent output:

The prospective buyer is asking about the roof and whether it has been recently replaced or is original to the house. This is a
↳ question about the physical state and quality of a key structural component of the property.

FINAL: condition

Judge rationale: The agent correctly identified 'condition' as the primary attribute, provided sound reasoning about roof state/quality, and matched the reference answer exactly.

Baseline — example 2 (gold: financing, score: 1.0) *Query:*

Is there an assumable loan on this home that I could take over from the current owner?

Agent output:

The prospective buyer is asking about whether they can assume the existing mortgage from the current owner. This is a question about
↪ the loan terms and the ability to take over financing arrangements.

FINAL: financing

Judge rationale: The agent correctly identified 'financing' as the primary attribute, provided sound reasoning about assumable mortgages, and delivered the exact answer matching the reference.

Baseline — example 3 (gold: availability, score: 1.0) *Query:*

We need to relocate by the end of next month. How soon could we move into this condo?

Agent output:

The prospective buyer is asking about when they can occupy the property, which relates to move-in timing and occupancy readiness.

FINAL: availability

Judge rationale: The agent correctly identified 'availability' as the primary attribute, provided sound reasoning about move-in timing and occupancy, and delivered the answer in the expected format matching the reference.

Baseline — example 4 (gold: price, score: 1.0) *Query:*

I'm interested in knowing if this falls into the affordable range for first-time buyers.

Agent output:

The prospective buyer is asking whether the property fits within their budget constraints as a first-time buyer. This is
↪ fundamentally a question about the cost of the property and whether it aligns with their financial capacity.

FINAL: price

Judge rationale: The agent correctly identified 'price' as the primary attribute, provided sound reasoning about budget/cost alignment, and delivered the answer in the expected format matching the reference exactly.

TEI — example 1 (gold: condition, score: 1.0) *Query:*

I'm interested in this property but wanted to know if the roof has been replaced recently or if it's original to the house from 1985.

Agent output:

The prospective buyer is asking about the roof's age and replacement history, which directly concerns the physical condition and
↪ maintenance state of the property structure itself. This is a question about what repairs or updates have been made to a key
↪ component of the house.

FINAL: condition

Judge rationale: The agent correctly identified 'condition' as the primary attribute, provided sound reasoning about roof age/replacement relating to physical condition, and matched the reference answer exactly.

TEI — example 2 (gold: financing, score: 1.0) *Query:*

Is there an assumable loan on this home that I could take over from the current owner?

Agent output:

The prospective buyer is asking about whether they can assume an existing mortgage from the current owner, which directly relates to
↪ financing options and payment structures available for purchasing the property.

FINAL: financing

Judge rationale: The agent correctly identified 'financing' as the primary attribute, provided sound reasoning about assumable mortgages being a financing option, and delivered the answer in the expected format matching the reference.

TEI — example 3 (gold: availability, score: 1.0) *Query:*

We need to relocate by the end of next month. How soon could we move into this condo?

Agent output:

The prospective buyer is asking about when they can occupy the property, which relates to the timeline for purchase and occupancy.

FINAL: availability

Judge rationale: The agent correctly identified 'availability' as the primary attribute, provided sound reasoning about timeline/occupancy, and the final answer matches the reference exactly.

TEI — example 4 (gold: price, score: 1.0) *Query:*

I'm interested in knowing if this falls into the affordable range for first-time buyers.

Agent output:

The prospective buyer is asking whether the property fits within their budget constraints as a first-time buyer. This is
→ fundamentally a question about the listing price and whether it aligns with their financial capacity.

FINAL: price

Judge rationale: The agent correctly identified 'price' as the primary attribute, provided sound reasoning about budget constraints, and delivered the exact answer matching the reference.

D.26 retail_counterfactual

Baseline — example 1 (gold: not-counterfactual, score: 1.0) *Query:*

If you follow the sizing chart you will have no issues.

Agent output:

This statement is a general recommendation about following a sizing chart to avoid problems. It does not contain a counterfactual
→ statement--it's not describing something contrary to fact or making a hypothetical claim about what would have happened if
→ something different had occurred. It's simply advice about proper sizing procedures.

FINAL: not-counterfactual

Judge rationale: The agent correctly identified the statement as not-counterfactual, provided sound reasoning about why it's advice rather than a counterfactual claim, and delivered the answer in the expected format m

Baseline — example 2 (gold: counterfactual, score: 1.0) *Query:*

That would be the only think I would change.

Agent output:

I need to analyze whether this statement contains a counterfactual.

A counterfactual statement expresses something contrary to fact - it describes a hypothetical situation that didn't actually happen,
→ often using conditional language like "would," "could have," "if," etc.

The statement "That would be the only thing I would change" uses the conditional "would," which indicates a hypothetical scenario.
→ The speaker is expressing what they would do in a hypothetical situation (if they could change something), rather than
→ describing what actually happened or is actually the case.

This is a counterfactual statement because it expresses a hypothetical action contrary to the actual situation.

FINAL: counterfactual

Judge rationale: The agent correctly identified the statement as counterfactual with sound reasoning about conditional language and hypothetical scenarios, matching the reference answer perfectly.

Baseline — example 3 (gold: not-counterfactual, score: 1.0) *Query:*

I liked that they came with the oxygen absorbers.

Agent output:

This statement describes a factual observation about the product - the reviewer is noting that oxygen absorbers were included with
→ their purchase. This is a straightforward statement about what actually came with the product, not a counterfactual
→ statement (which would involve imagining something contrary to fact, like "if only they had included oxygen absorbers" or "
→ they should have come with oxygen absorbers instead of what they did").

FINAL: not-counterfactual

Judge rationale: The agent correctly identified the statement as not-counterfactual, provided sound reasoning distinguishing factual from counterfactual statements, and delivered the answer in the expected format matc

Baseline — example 4 (gold: counterfactual, score: 1.0) *Query:*

If they were a little bigger, it would have been perfect.

Agent output:

This statement contains a counterfactual element. The reviewer is expressing a hypothetical condition ("if they were a little bigger
→ ") and its imagined consequence ("it would have been perfect"). This is a counterfactual statement because it describes what
→ would be true under a condition that is contrary to fact--implying that the items are NOT actually bigger, and therefore
→ the product is NOT actually perfect.

FINAL: counterfactual

Judge rationale: The agent correctly identified the counterfactual statement, provided sound logical reasoning about the hypothetical condition and contrary-to-fact nature, and delivered the exact required label match

TEI — example 1 (gold: not-counterfactual, score: 1.0) *Query:*

If you follow the sizing chart you will have no issues.

Agent output:

This statement is a conditional prediction about actual behavior of the existing product (the sizing chart). It describes what will
↳ happen if you follow the chart as it actually exists--not an alternative to what the product is or has.

This is NOT contrary-to-fact language describing an alternative to reality. It's straightforward advice about using the product
↳ correctly.

FINAL: not-counterfactual

Judge rationale: The agent correctly identified the statement as not-counterfactual, provided sound reasoning distinguishing conditional predictions from counterfactuals, and delivered the exact answer matching the re

TEI — example 2 (gold: counterfactual, score: 1.0) *Query:*

That would be the only think I would change.

Agent output:

This statement expresses what the reviewer would change about the product--indicating a contrary-to-fact alternative to what the
↳ product actually is. The phrase "would change" is a contrary-to-fact construction describing an alternative to the product's
↳ actual state.

FINAL: counterfactual

Judge rationale: The agent correctly identified the counterfactual nature of the statement, provided sound reasoning about the contrary-to-fact construction, and delivered the exact label matching the reference answer

TEI — example 3 (gold: not-counterfactual, score: 1.0) *Query:*

I liked that they came with the oxygen absorbers.

Agent output:

This statement expresses satisfaction with an actual feature of the product--the oxygen absorbers that were included. There is no
↳ contrary-to-fact language, no wish for an alternative, and no hypothetical alternative to what the product actually is or
↳ has.

FINAL: not-counterfactual

Judge rationale: The agent correctly identified the statement as not-counterfactual with sound reasoning about the absence of contrary-to-fact language, matching the reference answer perfectly.

TEI — example 4 (gold: counterfactual, score: 1.0) *Query:*

If they were a little bigger, it would have been perfect.

Agent output:

This statement uses contrary-to-fact language ("if they were...would have been") to express a wish that the product had different
↳ attributes (bigger size) than it actually has. This is a classic counterfactual construction describing an alternative to
↳ the actual reality of the product.

FINAL: counterfactual

Judge rationale: The agent correctly identified the counterfactual statement, provided sound linguistic reasoning about the contrary-to-fact conditional structure, and delivered the exact label matching the reference

D.27 saas_ticket_priority

Baseline — example 1 (gold: feature_request, score: 1.0) *Query:*

Would it be possible to add a dark mode option to the dashboard? Many of our team members work late hours and the current bright
↳ interface causes eye strain during evening sessions.

Agent output:

This is a request for a new feature to be added to the product. The user is asking for a dark mode option to improve the user
↳ experience for their team, particularly for those working during evening hours. This is not reporting an outage, data loss,
↳ billing issue, or asking for help with existing functionality--it's a suggestion for enhanced functionality.

FINAL: feature_request

Judge rationale: The agent correctly classified the ticket as 'feature_request' with sound reasoning that accurately distinguishes it from other ticket types, fully matching the reference answer.

Baseline — example 2 (gold: how_to, score: 1.0) *Query:*

Could you walk me through the process of customizing the dashboard widgets? I'd like to rearrange them and maybe hide some that we
↪ don't use.

Agent output:

This is a straightforward request for guidance on how to use an existing feature. The user is asking for instructions on customizing
↪ their dashboard by rearranging and hiding widgets.

This falls under "how_to" - they need help understanding how to perform a task within the product.

FINAL: how_to

Judge rationale: The agent correctly classified the ticket as 'how_to', provided sound reasoning about why it's a feature-usage question, and delivered the answer in the expected format matching the reference exactly.

Baseline — example 3 (gold: integration, score: 1.0) *Query:*

I'm trying to connect our Slack workspace to your service but keep getting 'invalid scope' errors during the OAuth flow. We need read
↪ and write permissions but the authorization screen only shows read access.

Agent output:

This appears to be an integration issue where the OAuth scope configuration isn't properly requesting the necessary permissions
↪ during the Slack authorization flow. The user is unable to complete the integration setup due to a mismatch between required
↪ permissions (read and write) and what's being presented (read only).

This is blocking their ability to connect Slack to the service, making it an integration problem rather than a how-to question, since
↪ it involves troubleshooting a technical configuration issue with a third-party service connection.

FINAL: integration

Judge rationale: The agent correctly classified the ticket as 'integration', provided sound reasoning about OAuth scope issues, and delivered the answer in the expected format with accurate diagnosis.

Baseline — example 4 (gold: outage_critical, score: 1.0) *Query:*

The main application server cluster is unresponsive. All customer-facing services are down and we're experiencing 100% failure rate
↪ on all requests. Zero users can access the system right now.

Agent output:

This is a critical production outage affecting all customer-facing services with complete system unavailability. The entire
↪ application server cluster is down, resulting in 100% request failure rate and zero user access.

This clearly requires immediate emergency response and represents the highest priority level.

FINAL: outage_critical

Judge rationale: The agent correctly classified the ticket as 'outage_critical' matching the reference answer, provided sound reasoning about severity indicators (100% failure, zero access), and delivered the classifi

TEI — example 1 (gold: feature_request, score: 1.0) *Query:*

Would it be possible to add a dark mode option to the dashboard? Many of our team members work late hours and the current bright
↪ interface causes eye strain during evening sessions.

Agent output:

The user is requesting a new feature (dark mode) that doesn't currently exist in the dashboard. They're not asking how to use an
↪ existing feature, reporting a service outage, experiencing data loss, asking about billing, or dealing with integration
↪ issues. Their primary intent is to request an enhancement to the product.

FINAL: feature_request

Judge rationale: The agent correctly classified the ticket as 'feature_request' with sound reasoning that explicitly ruled out alternative categories and identified the core intent of requesting a new product enhancement

TEI — example 2 (gold: how_to, score: 1.0) *Query:*

Could you walk me through the process of customizing the dashboard widgets? I'd like to rearrange them and maybe hide some that we
↪ don't use.

Agent output:

The user is asking for guidance on how to customize dashboard widgets - specifically how to rearrange and hide them. This is a
↪ request for help understanding how to use existing features of the product.

This falls under the "how_to" category because:
- The user is asking "walk me through the process" (instructional request)
- They want to learn how to accomplish a task with current functionality
- They're not reporting a broken feature or requesting new capabilities
- They need documentation/guidance on using existing features

FINAL: how_to

Judge rationale: The agent correctly classified the ticket as 'how_to' with sound reasoning that accurately identified the instructional nature of the request, properly formatted output, and perfect match to the refer

TEI — example 3 (gold: integration, score: 1.0) *Query:*

I'm trying to connect our Slack workspace to your service but keep getting 'invalid scope' errors during the OAuth flow. We need read
↳ and write permissions but the authorization screen only shows read access.

Agent output:

This is a technical issue with integrating a third-party service (Slack) where the OAuth authentication flow is failing due to scope
↳ mismatches. The user cannot establish the connection between systems, which is a classic integration problem.

FINAL: integration

Judge rationale: The agent correctly classified the ticket as 'integration', provided sound reasoning about OAuth scope issues being integration problems, and the output matches the reference answer exactly.

TEI — example 4 (gold: outage_critical, score: 1.0) *Query:*

The main application server cluster is unresponsive. All customer-facing services are down and we're experiencing 100% failure rate
↳ on all requests. Zero users can access the system right now.

Agent output:

This is a critical service outage where the entire system is down and no users can access any functionality.

FINAL: outage_critical

Judge rationale: The agent correctly classified the ticket as 'outage_critical', matching the reference answer exactly, with sound reasoning about system-wide unavailability justifying the classification.

D.28 social_emotion

Baseline — example 1 (gold: fear, score: 1.0) *Query:*

i feel shy about it all and also a little concerned whether my new title will distance me away from people i care for

Agent output:

The message expresses a mix of emotions centered around social anxiety and worry about relationships. The speaker feels shy (self-conscious, hesitant) and concerned (worried, anxious) about how a new title might affect their connections with loved ones.
↳ There's an underlying fear about potential negative consequences to their relationships.

The dominant emotion here is **fear** - specifically, fear of losing closeness with people they care about and anxiety about how this
↳ change will be perceived.

FINAL: fear

Judge rationale: The agent correctly identified 'fear' as the dominant emotion, provided sound reasoning about the underlying anxiety and relationship concerns, and delivered the answer in the expected format matching

Baseline — example 2 (gold: anger, score: 1.0) *Query:*

i only do unwillingly and always leaves me feeling grouchy and unsettled

Agent output:

The message expresses reluctance and negative emotional aftermath. The speaker describes doing something against their will ("unwillingly") and the resulting emotional state is explicitly negative ("grouchy and unsettled").

"Grouchy" suggests irritability and displeasure, while "unsettled" indicates discomfort and unease. This combination points to a
↳ state of dissatisfaction and mild agitation following an unwanted activity.

The dominant emotion here is **anger** - manifested as irritability and grouchiness rather than intense rage, but clearly reflecting
↳ frustration and displeasure.

FINAL: anger

Judge rationale: The agent correctly identified 'anger' as the dominant emotion, provided sound reasoning linking 'grouchy' and 'unsettled' to irritability/frustration, followed the task format, and matched the refer

Baseline — example 3 (gold: joy, score: 1.0) *Query:*

i am feeling profoundly peaceful

Agent output:

The message expresses a state of deep tranquility and contentment. The word "profoundly" emphasizes the intensity and depth of this
↳ peaceful feeling. This is a positive emotional state that aligns most closely with joy--a sense of well-being and
↳ satisfaction.

FINAL: joy

Judge rationale: The agent correctly identified 'joy' as the dominant emotion, provided sound reasoning linking peacefulness to joy/well-being, and delivered the answer in proper format matching the reference.

Baseline — example 4 (gold: **sadness**, score: 0.0) *Query:*

i was so focused on my heavy breathing my even strides the drops of sweat on my forehead that i forgot to feel socially awkward

Agent output:

This message reflects someone who was deeply absorbed in physical activity (running or exercise) and found that their concentration
→ on bodily sensations actually helped them overcome social anxiety. There's a sense of relief and positive discovery here--
→ they realized that focusing inward on their physical experience allowed them to escape the self-consciousness they normally
→ feel in social situations.

The dominant emotion is joy--a lighter, more subtle form of it. It's the satisfaction and pleasant surprise of discovering an
→ unexpected benefit or escape from anxiety.

FINAL: joy

Judge rationale: The agent provided extensive interpretation but misidentified the dominant emotion as 'joy' when the reference answer is 'sadness', showing poor alignment with the task and incorrect output despite co

TEI — example 1 (gold: **fear**, score: 1.0) *Query:*

i feel shy about it all and also a little concerned whether my new title will distance me away from people i care for

Agent output:

The message expresses two emotions: shyness (which relates to nervousness/apprehension) and concern about potential distance from
→ loved ones (which relates to worry and anxiety about relationships).

Both emotions fall under the fear/anxiety spectrum - the person is experiencing apprehension about social dynamics and worried about
→ how changes might affect their relationships.

FINAL: fear

Judge rationale: The agent correctly identified 'fear' as the dominant emotion, provided sound reasoning linking shyness and concern to the fear/anxiety spectrum, and delivered the answer in the expected format matchi

TEI — example 2 (gold: **anger**, score: 1.0) *Query:*

i only do unwillingly and always leaves me feeling grouchy and unsettled

Agent output:

The message expresses reluctance, irritation, and a negative emotional aftermath. The words "unwillingly," "grouchy," and "unsettled"
→ indicate frustration and annoyance with having to do something against one's preference.

FINAL: anger

Judge rationale: The agent correctly identified 'anger' as the dominant emotion, provided sound reasoning linking textual cues to the emotion, and delivered the answer in proper format matching the reference.

TEI — example 3 (gold: **joy**, score: 1.0) *Query:*

i am feeling profoundly peaceful

Agent output:

FINAL: joy

Judge rationale: The agent correctly identified 'joy' as the dominant emotion for 'profoundly peaceful', matching the reference answer with proper format, though 'peace' and 'joy' are semantically adjacent rather than

TEI — example 4 (gold: **sadness**, score: 0.0) *Query:*

i was so focused on my heavy breathing my even strides the drops of sweat on my forehead that i forgot to feel socially awkward

Agent output:

The message describes someone who was so absorbed in physical exertion (heavy breathing, steady strides, sweating) that they forgot
→ to experience social anxiety or awkwardness. This reflects a positive realization - they were too engaged in the moment to
→ feel self-conscious.

The dominant emotion here is joy - there's a sense of relief, contentment, and pleasant surprise at discovering they could be present
→ without social anxiety interfering.

FINAL: joy

Judge rationale: The agent correctly identified the task and provided clear reasoning about the emotional state, but the final answer 'joy' contradicts the reference answer 'sadness', indicating a fundamental misinter

D.29 telecom_churn_reason

Baseline — example 1 (gold: speed, score: 1.0) *Query:*

I've been experiencing extremely sluggish connection rates for the past three months, and it's affecting my ability to work from home.
↳ Downloads that should take minutes are taking hours, and video calls constantly buffer. I need to switch to a provider that
↳ can deliver better performance.

Agent output:

The customer is clearly frustrated with their internet performance. They explicitly mention:

- "extremely sluggish connection rates"
- "Downloads that should take minutes are taking hours"
- "video calls constantly buffer"
- Need to "switch to a provider that can deliver better performance"

The core issue is the slow speed and poor performance of their connection, not pricing, coverage gaps, service quality issues,
↳ relocation, or switching to a specific competitor. This is fundamentally a speed problem.

FINAL: speed

Judge rationale: The agent correctly identified 'speed' as the churn reason with sound evidence-based reasoning, proper format, and perfect alignment with the reference answer.

Baseline — example 2 (gold: moving, score: 1.0) *Query:*

My lease is ending and I've found an apartment in a different metropolitan area where I'll be starting fresh. Please process my
↳ cancellation.

Agent output:

The customer is canceling their service because they are relocating to a different metropolitan area due to their lease ending. This
↳ is a clear case of a customer moving to a new location where they presumably cannot use the current service or will need to
↳ switch providers.

FINAL: moving

Judge rationale: The agent correctly classified the churn reason as 'moving' with sound reasoning about relocation, matching the reference answer perfectly despite including explanatory context.

Baseline — example 3 (gold: competitor, score: 1.0) *Query:*

CenturyLink contacted me about their fiber service in my area and the upload speeds are much better for my work-from-home needs. I've
↳ already placed an order with them.

Agent output:

The customer is switching to a competitor (CenturyLink) because that competitor offers better upload speeds for their work-from-home
↳ requirements. While speed is mentioned as the benefit driving the switch, the fundamental reason for churn is that a
↳ competitor is offering a superior service that better meets their needs.

FINAL: competitor

Judge rationale: The agent correctly classified the churn reason as 'competitor', provided sound reasoning about why the customer is switching, and the final answer matches the reference perfectly.

Baseline — example 4 (gold: price, score: 1.0) *Query:*

I signed up during a promotional period and didn't realize how much the regular rate would be. Now that the discount expired, it's
↳ way more than I'm comfortable spending each month.

Agent output:

The customer is expressing concern about the cost of service after a promotional discount ended. They find the regular pricing too
↳ expensive for their budget.

FINAL: price

Judge rationale: The agent correctly identified 'price' as the churn reason, provided sound reasoning about promotional pricing concerns, and delivered the answer in the expected format matching the reference exactly.

TEI — example 1 (gold: speed, score: 1.0) *Query:*

I've been experiencing extremely sluggish connection rates for the past three months, and it's affecting my ability to work from home.
↳ Downloads that should take minutes are taking hours, and video calls constantly buffer. I need to switch to a provider that
↳ can deliver better performance.

Agent output:

The customer's primary complaint is about slow data speeds and poor performance - downloads taking hours instead of minutes, and
↳ constant buffering during video calls. While they mention switching providers, the core issue driving the cancellation is
↳ the speed/performance problem, not a price comparison or competitor offer.

FINAL: speed

Judge rationale: The agent correctly identified 'speed' as the churn reason, provided sound reasoning linking sluggish connections and buffering to performance issues, and delivered the answer in the expected format m

TEI — example 2 (gold: moving, score: 1.0) *Query:*

My lease is ending and I've found an apartment in a different metropolitan area where I'll be starting fresh. Please process my
↪ cancellation.

Agent output:

The customer is relocating to a different metropolitan area, which is a clear moving situation. This is the primary reason for
↪ cancellation.

FINAL: moving

Judge rationale: The agent correctly identified 'moving' as the churn reason with sound reasoning about relocation, proper format, and perfect match to the reference answer.

TEI — example 3 (gold: competitor, score: 1.0) *Query:*

CenturyLink contacted me about their fiber service in my area and the upload speeds are much better for my work-from-home needs. I've
↪ already placed an order with them.

Agent output:

The customer is switching to another provider (CenturyLink) because they offer better upload speeds for their work-from-home needs.
↪ While speed performance is mentioned as the benefit of the new service, the primary reason for cancellation is that they are
↪ switching to a competitor with a better offer.

FINAL: competitor

Judge rationale: The agent correctly classified the churn reason as 'competitor', provided sound reasoning about the competitor switch being primary over speed concerns, and delivered the answer in proper format match

TEI — example 4 (gold: price, score: 1.0) *Query:*

I signed up during a promotional period and didn't realize how much the regular rate would be. Now that the discount expired, it's
↪ way more than I'm comfortable spending each month.

Agent output:

The customer's primary concern is that the regular price after the promotional period ended is too high for their budget. They
↪ explicitly state it's "way more than I'm comfortable spending each month," which is a clear budget constraint issue.

FINAL: price

Judge rationale: The agent correctly identified 'price' as the churn reason, provided sound reasoning about budget constraints and promotional pricing, and delivered the answer in the expected format matching the refe

D.30 travel_intent

Baseline — example 1 (gold: cancellation, score: 1.0) *Query:*

I need to get my money back for the reservation I made last week because our plans have changed and we can't go anymore.

Agent output:

The traveler is requesting to get their money back for a reservation they made previously. This is a request to cancel an existing
↪ booking and receive a refund.

FINAL: cancellation

Judge rationale: The agent correctly identified the booking intent as 'cancellation', provided sound reasoning about the refund request indicating cancellation, and delivered the exact label matching the reference ans

Baseline — example 2 (gold: itinerary, score: 1.0) *Query:*

Show me a summary of all the reservations I have for my Tokyo trip in April.

Agent output:

The traveler is asking to see a summary of all their reservations for a specific trip (Tokyo in April). This request is about viewing
↪ and organizing their existing bookings across multiple services - flights, hotels, car rentals, etc. - for a particular
↪ trip.

This falls under the "itinerary" category, as they're asking for an overview of their travel plans and reservations for a specific
↪ trip.

FINAL: itinerary

Judge rationale: The agent correctly classified the request as 'itinerary', provided sound reasoning about viewing existing reservations, and delivered the exact answer matching the reference with proper format.

Baseline — example 3 (gold: loyalty, score: 1.0) *Query:*

Is there a way to transfer some of my rewards to my spouse's account for an upcoming trip they're planning?

Agent output:

The traveler is asking about transferring rewards points or loyalty program benefits to their spouse's account. This is clearly a
↪ question about managing loyalty program benefits and account features.

FINAL: loyalty

Judge rationale: The agent correctly classified the rewards transfer query as 'loyalty', provided sound reasoning about loyalty program management, and matched the reference answer exactly.

Baseline — example 4 (gold: flight, score: 1.0) *Query:*

What's available for a round trip between Chicago and Miami over the Memorial Day weekend? Looking for economy class.

Agent output:

The traveler is asking about availability for a round trip flight between two cities (Chicago and Miami) during a specific time
↪ period (Memorial Day weekend) with a preference for economy class seating.

This is clearly a flight booking request, as they're looking for flight options and availability.

FINAL: flight

Judge rationale: The agent correctly classified the booking intent as 'flight', matching the reference answer exactly, with sound reasoning about round-trip travel between cities indicating flight booking.

TEI — example 1 (gold: cancellation, score: 1.0) *Query:*

I need to get my money back for the reservation I made last week because our plans have changed and we can't go anymore.

Agent output:

The traveler is requesting to get their money back for a reservation they made previously. This is a request to cancel an existing
↪ booking and receive a refund.

FINAL: cancellation

Judge rationale: The agent correctly identified the booking intent as 'cancellation', provided sound reasoning about the refund request indicating cancellation, and delivered the exact label matching the reference ans

TEI — example 2 (gold: itinerary, score: 1.0) *Query:*

Show me a summary of all the reservations I have for my Tokyo trip in April.

Agent output:

The traveler is asking to see a summary of all their reservations for a specific trip (Tokyo in April). This request is about viewing
↪ and organizing their existing bookings across multiple services - flights, hotels, car rentals, etc. - for a particular
↪ trip.

This falls under the "itinerary" category, as they're asking for an overview of their travel plans and reservations for a specific
↪ trip.

FINAL: itinerary

Judge rationale: The agent correctly classified the request as 'itinerary', provided sound reasoning about viewing existing reservations, and delivered the exact answer matching the reference with proper format.

TEI — example 3 (gold: loyalty, score: 1.0) *Query:*

Is there a way to transfer some of my rewards to my spouse's account for an upcoming trip they're planning?

Agent output:

The traveler is asking about transferring rewards points or loyalty program benefits to their spouse's account. This is clearly a
↪ question about managing loyalty program benefits and account features.

FINAL: loyalty

Judge rationale: The agent correctly classified the rewards transfer query as 'loyalty', provided sound reasoning about loyalty program management, and matched the reference answer exactly.

TEI — example 4 (gold: flight, score: 1.0) *Query:*

What's available for a round trip between Chicago and Miami over the Memorial Day weekend? Looking for economy class.

Agent output:

The traveler is asking about availability for a round trip flight between two cities (Chicago and Miami) during a specific time
↪ period (Memorial Day weekend) with a preference for economy class seating.

This is clearly a flight booking request, as they're looking for flight options and availability.

FINAL: flight

Judge rationale: The agent correctly classified the booking intent as 'flight', matching the reference answer exactly, with sound reasoning about round-trip travel between cities indicating flight booking.

D.31 trust_safety_hate

Baseline — example 1 (gold: offensive language, score: 1.0) *Query:*

Theres one dude i can not stand at my school because he stays on the freshman girls and girls from other schools cause none fw him. #
↪ bitch

Agent output:

I'll analyze this message for hateful/offensive content.

The message expresses frustration about someone's behavior at school. It contains:

- A complaint about a person's inappropriate behavior toward girls
- The use of "#bitch" as a derogatory term

The term "bitch" is used here as an insult/derogatory label. While the overall message is expressing disapproval of someone's

- ↪ predatory behavior (which is reasonable), the use of "bitch" constitutes offensive language rather than hate speech. The
- ↪ message doesn't target a protected group with dehumanizing language or calls for harm - it's using a common profanity/insult.
- ↪

FINAL: offensive language

Judge rationale: The agent correctly identified the content as 'offensive language' matching the reference answer, provided sound reasoning distinguishing offensive language from hate speech, and properly executed the

Baseline — example 2 (gold: neither, score: 1.0) *Query:*

Got my vans on.. My pockets chunky

Agent output:

This message appears to be casual, informal language expressing that the person is wearing Vans shoes and has money/valuables in
↪ their pockets. It's a common colloquial expression, likely from hip-hop or street culture.

There is no hateful content, slurs, or derogatory language targeting any group. While "chunky" is informal/slang, it's not offensive
↪ in this context.

FINAL: neither

Judge rationale: The agent correctly identified the message as non-hateful/non-offensive, provided sound contextual reasoning about the slang usage, and delivered the exact reference answer 'neither' with proper justi

Baseline — example 3 (gold: hate speech, score: 0.0) *Query:*

@MakeAnothaHov T.o. was a faggot crybaby

Agent output:

I need to analyze this message for hateful/offensive content.

The message contains:

1. A slur ("faggot") - a derogatory term historically used to demean LGBTQ+ individuals
2. Name-calling ("crybaby")

The use of the slur is clearly offensive language targeting someone based on sexual orientation. This goes beyond mild profanity and
↪ constitutes derogatory language directed at a protected group.

FINAL: offensive language

Judge rationale: The agent correctly identified offensive content with sound reasoning about the slur, but labeled it 'offensive language' instead of the reference answer 'hate speech', which represents a meaningful c

Baseline — example 4 (gold: offensive language, score: 1.0) *Query:*

Bitch, you may have a million dollar body, but you got a dollar menu face and a four-piece nugget pussy, so you ain't worth shit.

Agent output:

This message contains multiple offensive elements:

1. Uses "bitch" as a derogatory term for a woman
2. Contains sexually demeaning language ("four-piece nugget pussy")
3. Uses insulting comparisons about physical appearance
4. Employs crude sexual objectification
5. Overall tone is abusive and degrading toward women

While this doesn't appear to target a protected group for systematic harm or dehumanization (which would be hate speech), it clearly
 ↳ uses offensive language and sexual insults directed at women in a demeaning way.

FINAL: offensive language

Judge rationale: The agent correctly identified the content as 'offensive language' matching the reference answer, provided sound reasoning distinguishing offensive language from hate speech, and executed the moderati

TEI — example 1 (gold: offensive language, score: 1.0) *Query:*

Theres one dude i can not stand at my school because he stays on the freshman girls and girls from other schools cause none fw him. #
 ↳ bitch

Agent output:

The message expresses frustration about someone's behavior (pursuing girls who aren't interested) and uses the term "bitch" as a
 ↳ casual insult. While "#bitch" is crude language, it's not targeting a protected group identity--it's a general insult
 ↳ directed at an individual's behavior. There are no identity-based slurs or systematic attacks on protected characteristics.

FINAL: offensive language

Judge rationale: The agent correctly identified the content as 'offensive language' matching the reference answer, provided sound reasoning distinguishing between crude language and hate speech, and properly executed

TEI — example 2 (gold: neither, score: 1.0) *Query:*

Got my vans on.. My pockets chunky

Agent output:

FINAL: neither

Judge rationale: The agent correctly identified the message as 'neither' hateful nor offensive, matching the reference answer exactly with proper format and label from the allowed set.

TEI — example 3 (gold: hate speech, score: 1.0) *Query:*

@MakeAnotHaHov T.o. was a faggot crybaby

Agent output:

This message contains a homophobic slur ("faggot") used to demean someone based on sexual orientation, combined with insulting
 ↳ language. This is identity-based hate speech.

FINAL: hate speech

Judge rationale: The agent correctly identified the message as hate speech, provided sound reasoning about the homophobic slur and identity-based attack, and delivered the exact classification matching the reference a

TEI — example 4 (gold: offensive language, score: 0.0) *Query:*

Bitch, you may have a million dollar body, but you got a dollar menu face and a four-piece nugget pussy, so you ain't worth shit.

Agent output:

This message contains multiple offensive insults and vulgar language. It uses gendered slurs ("bitch," "pussy") and crude sexual
 ↳ language to demean and insult someone based on their appearance and gender. The combination of gendered slurs with
 ↳ aggressive intent to demean based on gender identity characteristics constitutes identity-based targeting.

FINAL: hate speech

Judge rationale: The agent correctly identified offensive content and provided sound reasoning about gendered slurs and demeaning language, but classified it as 'hate speech' rather than the reference answer 'offensiv

E Full per-task results with bootstrap confidence intervals

Task	Base [95% CI]	Random	Obj-only	TEI
agri_crop_issue	0.958 [0.88,1.00]	0.958 [0.88,1.00]	0.958 [0.88,1.00]	0.917 [0.79,1.00]
community_forum_topic	0.767 [0.60,0.90]	0.700 [0.53,0.87]	0.700 [0.53,0.87]	0.667 [0.50,0.83]
consumer_sentiment_5	0.367 [0.20,0.53]	0.367 [0.20,0.53]	0.533 [0.37,0.70]	0.500 [0.33,0.67]
cybersec_alert	0.958 [0.88,1.00]	0.958 [0.88,1.00]	0.958 [0.88,1.00]	0.917 [0.79,1.00]
edtech_question_router	0.733 [0.57,0.87]	0.800 [0.63,0.93]	0.900 [0.77,1.00]	0.867 [0.73,0.97]
edu_gsm8k	1.000 [1.00,1.00]	1.000 [1.00,1.00]	1.000 [1.00,1.00]	1.000 [1.00,1.00]
edu_math_word	1.000 [1.00,1.00]	1.000 [1.00,1.00]	1.000 [1.00,1.00]	1.000 [1.00,1.00]

Task	Base [95% CI]	Random	Obj-only	TEI
energy_meter_event	0.917 [0.79,1.00]	0.917 [0.79,1.00]	0.917 [0.79,1.00]	0.958 [0.88,1.00]
fin_banking_intent	1.000 [1.00,1.00]	1.000 [1.00,1.00]	1.000 [1.00,1.00]	1.000 [1.00,1.00]
gaming_support	1.000 [1.00,1.00]	1.000 [1.00,1.00]	1.000 [1.00,1.00]	1.000 [1.00,1.00]
gov_service_router	1.000 [1.00,1.00]	1.000 [1.00,1.00]	1.000 [1.00,1.00]	1.000 [1.00,1.00]
health_triage	0.867 [0.67,1.00]	0.867 [0.67,1.00]	0.867 [0.67,1.00]	0.933 [0.80,1.00]
hospitality_review_stars	0.667 [0.50,0.83]	0.633 [0.47,0.80]	0.733 [0.57,0.87]	0.667 [0.50,0.83]
hr_resume_fit	1.000 [1.00,1.00]	1.000 [1.00,1.00]	1.000 [1.00,1.00]	1.000 [1.00,1.00]
insurance_claim_type	1.000 [1.00,1.00]	1.000 [1.00,1.00]	1.000 [1.00,1.00]	0.958 [0.88,1.00]
it_email_spam	1.000 [1.00,1.00]	1.000 [1.00,1.00]	1.000 [1.00,1.00]	0.967 [0.90,1.00]
legal_contract_clause	1.000 [1.00,1.00]	1.000 [1.00,1.00]	1.000 [1.00,1.00]	1.000 [1.00,1.00]
logistics_incident	1.000 [1.00,1.00]	1.000 [1.00,1.00]	1.000 [1.00,1.00]	1.000 [1.00,1.00]
manufacturing_qa	1.000 [1.00,1.00]	1.000 [1.00,1.00]	1.000 [1.00,1.00]	1.000 [1.00,1.00]
market_research_subjectivity	0.667 [0.50,0.83]	0.567 [0.40,0.73]	0.933 [0.83,1.00]	0.967 [0.90,1.00]
media_news_desk	0.700 [0.53,0.87]	0.700 [0.53,0.87]	0.700 [0.53,0.87]	0.900 [0.77,1.00]
news_agency_topic	0.767 [0.60,0.90]	0.767 [0.60,0.90]	0.700 [0.53,0.87]	0.700 [0.53,0.87]
pharma_ade	0.900 [0.77,1.00]	0.967 [0.90,1.00]	0.600 [0.43,0.77]	0.600 [0.43,0.77]
platform_toxicity	0.567 [0.40,0.73]	0.567 [0.40,0.73]	0.533 [0.37,0.70]	0.567 [0.40,0.73]
realestate_attribute	1.000 [1.00,1.00]	1.000 [1.00,1.00]	1.000 [1.00,1.00]	1.000 [1.00,1.00]
retail_counterfactual	0.733 [0.57,0.87]	0.967 [0.90,1.00]	0.933 [0.83,1.00]	0.867 [0.73,0.97]
saas_ticket_priority	0.958 [0.88,1.00]	0.958 [0.88,1.00]	0.958 [0.88,1.00]	1.000 [1.00,1.00]
social_emotion	0.433 [0.27,0.60]	0.567 [0.40,0.73]	0.433 [0.27,0.60]	0.433 [0.27,0.60]
telecom_churn_reason	0.958 [0.88,1.00]	0.958 [0.88,1.00]	0.958 [0.88,1.00]	1.000 [1.00,1.00]
travel_intent	0.958 [0.88,1.00]	0.958 [0.88,1.00]	0.958 [0.88,1.00]	0.958 [0.88,1.00]
trust_safety_hate	0.533 [0.37,0.70]	0.700 [0.53,0.87]	0.533 [0.37,0.70]	0.533 [0.37,0.70]

F Per-dimension GPA (baseline vs TEI), per task

Task	Target (B→T)	Reasoning (B→T)	Execution (B→T)	Integrity (B→T)
agri_crop_issue	0.95→0.95	0.94→0.93	0.99→0.99	0.97→0.95
community_forum_topic	0.91→0.88	0.87→0.83	0.95→0.92	0.87→0.74
consumer_sentiment_5	0.75→0.82	0.82→0.83	0.72→0.78	0.65→0.74
cybersec_alert	0.95→0.95	0.95→0.94	0.99→0.98	0.97→0.96
edtech_question_router	0.64→0.93	0.87→0.91	0.68→0.96	0.74→0.90
edu_gsm8k	0.99→1.00	0.99→0.99	1.00→1.00	1.00→1.00
edu_math_word	1.00→1.00	1.00→1.00	1.00→1.00	1.00→1.00
energy_meter_event	0.93→0.92	0.90→0.91	0.99→0.96	0.94→0.96
fin_banking_intent	0.96→0.95	0.95→0.93	0.99→0.99	1.00→1.00
gaming_support	0.96→0.96	0.94→0.94	1.00→0.99	1.00→0.98
gov_service_router	0.95→0.96	0.92→0.94	1.00→1.00	1.00→1.00
health_triage	0.94→0.92	0.93→0.91	1.00→0.98	0.94→0.95
hospitality_review_stars	0.96→0.97	0.91→0.91	0.95→0.98	0.79→0.80
hr_resume_fit	0.95→0.95	0.95→0.94	1.00→1.00	1.00→1.00
insurance_claim_type	0.95→0.92	0.94→0.92	0.99→0.98	1.00→0.96
it_email_spam	0.95→0.95	0.91→0.92	0.98→0.98	0.98→0.95
legal_contract_clause	0.95→0.95	0.94→0.94	0.99→0.99	1.00→1.00
logistics_incident	0.95→0.95	0.93→0.94	0.99→0.99	1.00→1.00
manufacturing_qa	0.95→0.95	0.93→0.94	0.98→0.98	1.00→0.99
market_research_subjectivity	0.87→0.93	0.83→0.89	0.84→0.98	0.67→0.97
media_news_desk	0.82→0.90	0.84→0.88	0.76→0.94	0.72→0.91
news_agency_topic	0.87→0.83	0.87→0.84	0.93→0.90	0.82→0.76
pharma_ade	0.93→0.69	0.91→0.66	0.93→0.72	0.90→0.62
platform_toxicity	0.87→0.81	0.74→0.71	0.75→0.69	0.58→0.58

Task	Target (B→T)	Reasoning (B→T)	Execution (B→T)	Integrity (B→T)
realestate_attribute	0.95→0.95	0.92→0.91	0.98→0.99	1.00→1.00
retail_counterfactual	0.75→0.88	0.77→0.85	0.72→0.95	0.72→0.87
saas_ticket_priority	0.95→0.95	0.92→0.94	0.99→0.99	0.97→1.00
social_emotion	0.86→0.86	0.80→0.80	0.91→0.91	0.59→0.60
telecom_churn_reason	0.92→0.95	0.92→0.95	0.98→1.00	0.97→1.00
travel_intent	0.92→0.92	0.92→0.92	0.96→0.96	0.96→0.96
trust_safety_hate	0.88→0.89	0.84→0.80	0.73→0.76	0.71→0.68